Kinetic Acceleration to Non-Equilibrium Steady States: An Operator-Theoretic Analysis

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We study decay to non-equilibrium steady states for a class of dissipative evolutions generated on $L^2(\sigma)$ by $\mathcal{L} = \gamma\mathcal{R}+\mathcal{V}_{\text{skew}}+\eta\mathcal{V}_{\text{pert}}$, with a fast self-adjoint dissipator $\mathcal{R}$, a skew transport part $\mathcal{V}_{\text{skew}}$, and a non-reversible perturbation $\eta\mathcal{V}_{\text{pert}}$. In the singular limit $\gamma \to \infty$, we recover the reduced slow operator $\mathcal{L}_O$ (adiabatic elimination / Zeno dynamics), while the coercive structure relevant for finite-friction optimization is encoded by the self-adjoint reference operator $Q_O = -P_S\mathcal{V}_{\text{skew}}\mathcal{R}^+\mathcal{V}_{\text{skew}}P_S$. Using a resolvent-corrected Lyapunov functional, we prove explicit quantitative lower bounds on the spectral bound of $\mathcal{L}$ and identify an optimal friction regime with hypocoercive scaling $\Theta(\sqrt{\lambda(Q_O)})$, improving the overdamped diffusive scale $\mathcal{O}(\lambda(Q_O))$. We further derive explicit perturbative stability thresholds: the accelerated regime persists when the symmetric and skew components of $\mathcal{V}_{\text{pert}}$ are controlled in the $(-\mathcal{R})^{1/2}$-energy norm and η lies below a computable friction-dependent bound. Applications are given to forced underdamped Langevin dynamics and to a finite-dimensional boundary-driven toy model, including cases where the full effective slow operator is non-self-adjoint.

Keywords: Hypocoercivity, non-equilibrium steady states, spectral bound, adiabatic elimination, Quantum Zeno limit

I. INTRODUCTION

We consider a semigroup-theoretic relaxation problem for operators on a Hilbert space $\mathcal{H}$ with decomposition

$$\mathcal{L} = \gamma\mathcal{R} + \mathcal{V}_{\text{skew}} + \eta\mathcal{V}_{\text{pert}}. \tag{1}$$

Here $\gamma > 0$ is a scale-separation parameter, $\mathcal{R}$ is self-adjoint non-positive, $\mathcal{V}_{\text{skew}}$ is skew-adjoint, and $\eta\mathcal{V}_{\text{pert}}$ is a non-reversible perturbation. The functional-analytic objective is to quantify the spectral bound of $\mathcal{L}$ on the mean-zero subspace and to determine how this bound depends on γ, on the reduced slow operator, and on perturbative non-self-adjoint effects.

The framework is operator-theoretic and covers both noncommutative and classical realizations. A natural concrete model is the noncommutative space $L^2(\sigma)$ associated with a faithful state σ (finite-dimensional GKSL setting), while the same estimates apply in the weighted classical spaces $L^2(\mu)$ arising in kinetic PDEs. The main assumptions are therefore formulated at Hilbert-space level; the $L^2(\sigma)$ notation serves only as a concrete realization.

In reversible settings (detailed balance), decay follows from coercivity of the symmetric part and is governed by classical spectral-gap or functional-inequality estimates.[1–3] In the non-reversible regime relevant for driven systems and boundary forcing, one must contend with a non-normal generator for which direct coercivity may be degenerate on large

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subspaces.[4–6] This places the problem squarely in the hypocoercive class: exponential relaxation is produced by coupling between conservative transport directions and dissipative directions, rather than by pointwise coercivity of the symmetric part alone.[7,8]

A second structural ingredient is the singular scaling in γ. For $\mathcal{L}_\gamma = \gamma\mathcal{R} + \mathcal{V}$, adiabatic elimination identifies effective slow dynamics on $\mathcal{H}_S := \ker(\mathcal{R})$, with reduced operator $\mathcal{L}_O = -P_S\mathcal{V}\mathcal{R}^+\mathcal{V}P_S$. Quantitative Zeno-limit control and asymptotic expansions in boundary-driven settings have recently been established in rigorous form.[9] From the viewpoint of long-time rates, however, the overdamped limit typically carries a diffusive γ^{-1} bottleneck; this motivates optimization at finite friction rather than passage to $\gamma = \infty$.

The mechanism under analysis is lifting: the slow dissipative dynamics is embedded into an enlarged phase space where transport and fast relaxation jointly produce faster macroscopic decay.[10,11] The key analytic question is stability of this acceleration under non-reversible perturbations. We address this in an abstract operator framework that accommodates non-self-adjoint effective slow generators while retaining an explicit self-adjoint reference operator $Q_O = -P_S\mathcal{V}_{\mathrm{skew}}\mathcal{R}^+\mathcal{V}_{\mathrm{skew}}P_S$ to measure slow coercivity.

The main estimates are derived from a resolvent-corrected Lyapunov functional

$$\mathcal{E}_{\alpha,\mu}(f) := \frac{1}{2}\|f\|^2 + \alpha\,\Re\langle\mathcal{A}_\mu f, f\rangle, \tag{2}$$

where $\mathcal{A}_\mu$ is a slow-space resolvent corrector. This corrector converts transport-induced cross terms into an effective coercive contribution on slow variables, yielding an explicit lower bound for the decay rate, a finite-friction optimizer with scaling $\gamma_{\mathrm{opt}} \asymp \sqrt{\lambda(Q_O)}$, and a perturbative admissibility threshold for η.

Specifically, the paper establishes: (i) explicit quantitative lower bounds valid for all $\gamma > 0$ under structural assumptions (Theorem V.1); (ii) an overdamped upper ceiling from a singular-value-gap argument, confirming the intrinsic diffusive γ^{-1} regime at large friction (Theorem V.4); and (iii) concrete realizations in nonequilibrium Langevin dynamics and a finite-dimensional boundary-driven toy model.

The remainder of the paper is organized as follows. Section II introduces the abstract Hilbert-space setting and records the $L^2(\sigma)/L^2(\mu)$ realizations used in the examples. Section III revisits adiabatic elimination and the Zeno limit, connecting the full dynamics to the effective slow generator $\mathcal{L}_O$. Section IV formulates the abstract non-equilibrium lifting framework and states the key structural assumptions. Section V develops the operator-modified Lyapunov approach and proves explicit decay-rate bounds, including the optimal friction scaling and quantitative perturbative stability. Section VI provides representative examples and numerical verification. Section VII collects concluding remarks.

II. FUNCTIONAL-ANALYTIC SETUP AND L^2 REALIZATIONS

We fix the Hilbert-space setting, projections, and spectral quantities used throughout the paper. The emphasis is on operator-theoretic structure (adjoints, kernels, spectral bounds), with $L^2(\sigma)$ and $L^2(\mu)$ serving as concrete realizations.

A. Abstract Hilbert setup and concrete realizations

We work on a complex Hilbert space $\mathcal{H}$ with inner product $\langle\cdot,\cdot\rangle$ and norm $\|\cdot\|$. The generator $\mathcal{L}$ (possibly unbounded in PDE settings) is assumed to generate a strongly continuous semigroup on $\mathcal{H}$. The analysis below uses only Hilbert-space notions: adjoints, orthogonal projections, and spectral bounds.

For definiteness, all displayed identities are stated in the noncommutative realization $\mathcal{H} = L^2(\sigma)$ built from a faithful state σ on a finite-dimensional physical Hilbert space $\mathfrak{h}$. In that representation, observables lie in $\mathcal{B}(\mathfrak{h})$, the identity observable is $\mathbf{1}$, and the inner

product is the KMS form

$$\langle X, Y\rangle_\sigma := \mathrm{tr}\big(\sigma^{1/2}X^\dagger\sigma^{1/2}Y\big), \qquad X, Y \in \mathcal{B}(\mathfrak{h}), \tag{3}$$

with norm $\|X\|_\sigma := \sqrt{\langle X, X\rangle_\sigma}$.

In this finite-dimensional quantum realization, a quantum Markov semigroup (QMS) in Heisenberg picture has GKSL generator

$$\mathcal{L}(X) = i[H, X] + \sum_\alpha \Big(L_\alpha X L_\alpha^\dagger - \tfrac{1}{2}\{L_\alpha^\dagger L_\alpha, X\}\Big), \tag{4}$$

where $H = H^\dagger$ and $\{L_\alpha\}$ are jump operators. The same abstract formalism also covers PDE realizations in weighted spaces $L^2(\mu)$: one simply replaces the trace pairing by the weighted integral pairing and $\mathbf{1}$ by the constant mode.

Let $(\mathcal{P}_t)_{t\geq 0}$ denote the semigroup generated by $\mathcal{L}$, and let $\mathcal{L}^\dagger$ be the Hilbert-space adjoint (in the concrete quantum realization: the adjoint with respect to $\langle\cdot,\cdot\rangle_\sigma$).

In the quantum realization, we assume primitiveness (unique faithful invariant state σ for the dual dynamics). We define the symmetric and antisymmetric parts of the generator as

$$\mathcal{S} := \tfrac{1}{2}(\mathcal{L} + \mathcal{L}^\dagger), \qquad \mathcal{A} := \tfrac{1}{2}(\mathcal{L} - \mathcal{L}^\dagger). \tag{5}$$

By construction, $\mathcal{S}$ is self-adjoint and negative semidefinite ($\mathcal{S} \preceq 0$), while $\mathcal{A}$ is skew-adjoint ($\mathcal{A}^\dagger = -\mathcal{A}$).

Definition II.1 (Detailed balance vs. non-reversibility)**.** We say that $\mathcal{L}$ satisfies a detailed-balance condition (DBC) if it is self-adjoint on $\mathcal{H}$, i.e. $\mathcal{L} = \mathcal{L}^\dagger$, equivalently $\mathcal{A} = 0$. Otherwise $\mathcal{L}$ is non-reversible ($\mathcal{L} \neq \mathcal{L}^\dagger$, hence $\mathcal{A} \neq 0$). In the concrete noncommutative realization this is exactly self-adjointness on $L^2(\sigma)$.

B. Spectral Gaps and Hypocoercivity

The convergence analysis relies on the spectral properties of $\mathcal{L}$ and its symmetric part $\mathcal{S}$ on the mean-zero subspace. Let Π denote the orthogonal projection onto the stationary direction (identity/constant mode) and define

$$\mathcal{H}_0 := \ker(\Pi). \tag{6}$$

In the noncommutative realization, $\Pi = \mathbb{E}_\sigma$ with $\mathbb{E}_\sigma(X) := \mathrm{tr}(\sigma X)\mathbf{1}$, so $\mathcal{H}_0 = \{X \in L^2(\sigma) : \mathrm{tr}(\sigma X) = 0\}$. In weighted PDE realizations, $\Pi f = (\int f\, d\mu)\mathbf{1}$ and $\mathcal{H}_0 = \{f : \int f\, d\mu = 0\}$.

Definition II.2 (Two gap parameters)**.** On the mean-zero subspace $\mathcal{H}_0$, we will use two standard quantities.

The (symmetric) coercive gap of the dissipative part $-\mathcal{S}$ is

$$\lambda_S := \inf_{X\in\mathcal{H}_0, X\neq 0} \frac{-\langle X, \mathcal{S}X\rangle}{\|X\|^2}. \tag{7}$$

The asymptotic decay rate (spectral bound) of the full generator is

$$\lambda(\mathcal{L}) := \inf\{\Re(\mu) : \mu \in \mathrm{Spec}(-\mathcal{L}) \setminus \{0\}\}. \tag{8}$$

Definition II.2 is standard in semigroup theory: $\lambda(\mathcal{L})$ is the spectral bound of $-\mathcal{L}$ on the mean-zero subspace, and in finite dimensions it coincides with the exponential decay rate of the semigroup on $\mathcal{H}_0$.[12] In reversible systems (DBC), $\mathcal{L}$ is self-adjoint, so $\lambda(\mathcal{L}) = \lambda_S$. In non-equilibrium systems, it can happen that $\lambda_S = 0$ (degenerate dissipation) while $\lambda(\mathcal{L}) > 0$:

the skew-adjoint part couples directions in $\ker(\mathcal{S}) \cap \mathcal{H}_0$ to dissipative directions, yielding exponential relaxation by hypocoercivity.[7,8]

More quantitatively, the standard *hypocoercive estimate*[7,8] asserts that there exist constants $C \geq 1$ and $\lambda > 0$ such that for all $X \in \mathcal{H}_0$ and $t \geq 0$,

$$\|\mathcal{P}_t X\| \leq C\, e^{-\lambda t}\, \|X\|. \tag{9}$$

If the estimate holds with $C = 1$, we call the dynamics *coercive* in the underlying Hilbert norm (in particular, in $L^2(\sigma)$ realizations); otherwise, the need for $C > 1$ reflects the genuinely hypocoercive nature of the relaxation.

Lemma II.3 (Kernel characterization of coercivity vs. strict hypocoercivity)**.** *Assume* $\ker(\mathcal{L}) = \mathbb{C}\mathbf{1}$ *(primitivity). Then:*

1. *In finite dimensions (or, more generally, whenever $-\mathcal{S}|_{\mathcal{H}_0}$ has a positive spectral gap), the semigroup is coercive in the underlying Hilbert norm (i.e., the decay estimate holds with $C = 1$) if and only if*

$$\ker(\mathcal{S}) \cap \mathcal{H}_0 = \{0\} \qquad \textit{(equivalently, under primitivity, } \ker(\mathcal{S}) = \mathbb{C}\mathbf{1}\textit{).} \tag{10}$$

2. *If $\ker(\mathcal{S}) \cap \mathcal{H}_0 \neq \{0\}$, then no exponential decay estimate can hold with prefactor $C = 1$ and some rate $\lambda > 0$ on $\mathcal{H}_0$. In particular, whenever exponential decay does hold (e.g. for primitive finite-dimensional QMS, where the spectrum of $\mathcal{L}$ is discrete and 0 is a simple eigenvalue), any sharp estimate must be genuinely hypocoercive in the sense that it requires some $C > 1$.*

III. ADIABATIC ELIMINATION AND THE ZENO LIMIT

This section recalls the singular-perturbation reduction from the full generator to an effective slow operator on $\ker(\mathcal{R})$ in the abstract Hilbert setting. The same formulas apply in both the noncommutative $L^2(\sigma)$ and the classical weighted $L^2(\mu)$ realizations. We adopt the adiabatic-elimination scaling to identify the reduced operator and to clarify how the overdamped regime enters the later rate analysis.[13,14]

We consider the generator $\mathcal{L}_\gamma$ acting on the full Hilbert space $\mathcal{H}$ (e.g. $L^2(\sigma)$ or $L^2(\mu)$), decomposed as:

$$\mathcal{L}_\gamma = \gamma\mathcal{R} + \mathcal{V}, \tag{11}$$

where $\gamma > 0$ is the timescale separation parameter. The self-adjoint **dissipator** $\mathcal{R}$ generates the fast relaxation, and the **interactor** $\mathcal{V}$ (not necessarily skew-adjoint) couples the fast and slow degrees of freedom.

We introduce the slow/fast decomposition

$$\mathcal{H}_S := \ker(\mathcal{R}), \qquad \mathcal{H}_F := \mathcal{H}_S^\perp. \tag{12}$$

Let P_S denote the orthogonal projection onto $\mathcal{H}_S$ and set $P_F := \mathrm{id} - P_S$. We assume $\mathcal{R}$ admits a spectral gap $\lambda_R > 0$ on $\mathcal{H}_F$.

A. The Micro-Macro Decomposition

We decompose any $f \in \mathcal{H}$ into macroscopic and microscopic parts:

$$f = u + v, \quad u := P_S f \in \mathcal{H}_S, \quad v := P_F f \in \mathcal{H}_F. \tag{13}$$

B. The Centered Coupling Condition

A structural condition for diffusive scaling is the absence of first-order drift inside the slow space. In operator form, we impose the standard centering constraint[9,15]

$$P_S \mathcal{V} P_S = 0. \tag{14}$$

It enforces second-order effective dynamics on $\mathcal{H}_S$: the reduced operator is generated by slow–fast–slow excursions through $\mathcal{H}_F$ rather than by an $\mathcal{O}(1)$ tangential drift on $\mathcal{H}_S$.

In the non-equilibrium lifting setting used later, we decompose

$$\mathcal{V} = \mathcal{V}_{\text{skew}} + \eta \mathcal{V}_{\text{pert}}, \tag{15}$$

where $\mathcal{V}_{\text{skew}}$ is skew-adjoint on $\mathcal{H}$ (in particular in $L^2(\sigma)$ realizations, $\mathcal{V}_{\text{skew}}^\dagger = -\mathcal{V}_{\text{skew}}$) and $\mathcal{V}_{\text{pert}}$ is a (not necessarily normal) non-equilibrium perturbation with strength parameter η. To remain in the *diffusive* Zeno/AE scaling considered here (macroscopic time $\tau = t/\gamma$), we will enforce (14) by requiring separately that $P_S \mathcal{V}_{\text{skew}} P_S = 0$ and $P_S \mathcal{V}_{\text{pert}} P_S = 0$.

C. The Effective Generator

Under condition (14), the limiting dynamics as $\gamma \to \infty$ are governed by an effective operator acting on $\mathcal{H}_S$.

Definition III.1 (Effective Generator)**.** Let $\mathcal{R}^+$ denote the Moore-Penrose pseudoinverse of $\mathcal{R}$, which is a bounded operator on $\mathcal{H}_F$ with norm $\|\mathcal{R}^+\| \leq \lambda_R^{-1}$. We define the **Effective Generator** $\mathcal{L}_O \in \mathcal{B}(\mathcal{H}_S)$ as:

$$\mathcal{L}_O := -P_S \mathcal{V} \mathcal{R}^+ \mathcal{V} P_S. \tag{16}$$

In the context of lifting, $\mathcal{L}_O$ represents the *target* dynamics we wish to accelerate.

D. Convergence in the Zeno Limit

The physical justification for identifying $\mathcal{L}_O$ as the target dynamics lies in quantitative Zeno-limit convergence results. We consider the system on the *macroscopic* timescale τ, related to the microscopic timescale t by $\tau = \gamma^{-1} t$.

Carlen–Huse–Lebowitz[9] analyze boundary-driven quantum systems in the Schrödinger picture and measure errors in the trace norm $\|\cdot\|_1$. This choice is natural because completely positive trace-preserving (CPTP) maps are contractive in the trace norm. Assuming that the fast boundary dissipator is ergodic and gapped, their results quantify the Zeno-limit approximation on time scales up to order γ. In particular, for initial data on the steady-state manifold, the full evolution is approximated by the projected evolution with an error of order $T\gamma^{-2}$ uniformly for $t \in [0, T]$; for general initial data, the error bound is of order $(\log(1+\gamma) + 1 + T)/\gamma$ uniformly for $t \in [\epsilon, \gamma T]$ (for fixed $0 < \epsilon < T$).

Theorem V.4 later makes this asymptotic statement spectral: in the large-γ regime, the decay rate is bounded above by a diffusive γ^{-1} scale (up to explicit perturbative corrections). The optimization problem is therefore inherently finite-friction: one seeks a lifted representation for which the maximal decay rate occurs at intermediate γ, with kinetic-type scaling relative to the slow gap.

IV. THE NON-EQUILIBRIUM LIFTING FRAMEWORK

We now formulate the abstract hypotheses used for quantitative hypocoercivity at finite friction. The framework identifies the analytic ingredients—spectral gap, centering, flux regularity, and perturbation bounds—needed to transfer coercivity from fast variables to slow variables through the resolvent corrector.[16]

A. Abstract Structural Assumptions

Let $\mathcal{H} = \mathcal{H}_S \oplus \mathcal{H}_F$ be an orthogonal slow–fast decomposition. We study generators depending on friction $\gamma > 0$ and perturbation strength $\eta \geq 0$:

$$\mathcal{L} = \gamma\mathcal{R} + \mathcal{V}. \tag{17}$$

The transport term is decomposed as $\mathcal{V} = \mathcal{V}_{\text{skew}} + \eta\mathcal{V}_{\text{pert}}$, and P_S, P_F denote the orthogonal projections onto $\mathcal{H}_S, \mathcal{H}_F$.

The following assumptions encode the closed set of inequalities exploited in Section V: coercivity of the fast operator, centered coupling, control of slow-induced fluxes, and energy-norm control of non-self-adjoint perturbations.

Assumption IV.1 (Structural Conditions on the Non-Equilibrium Generator)**.** The system is defined on $\mathcal{H} = \mathcal{H}_S \oplus \mathcal{H}_F$. The generator is $\mathcal{L} = \gamma\mathcal{R} + \mathcal{V}_{\text{skew}} + \eta\mathcal{V}_{\text{pert}}$. We assume the following structural conditions:

1. **Microscopic Spectral Gap.** The fast dissipator $\mathcal{R}$ is self-adjoint and non-positive on $\mathcal{H}$, with kernel $\ker(\mathcal{R}) = \mathcal{H}_S$. Moreover, it is strictly dissipative on $\mathcal{H}_F$: there exists $\lambda_R > 0$ such that

$$-\langle \mathcal{R}f, f\rangle \geq \lambda_R \|f\|^2, \qquad \forall f \in \mathcal{H}_F. \tag{18}$$

   In particular, $(-\mathcal{R})^{1/2}$ is well-defined by functional calculus. This allows defining the Moore–Penrose pseudoinverse $\mathcal{R}^+$, which is a bounded self-adjoint operator on $\mathcal{H}_F$ with $-\mathcal{R}^+ \geq 0$ and $\|\mathcal{R}^+\| \leq \lambda_R^{-1}$ (the bound follows from the spectral theorem: $\mathcal{R}|_{\mathcal{H}_F} \leq -\lambda_R$ implies $\|(\mathcal{R}|_{\mathcal{H}_F})^{-1}\| \leq \lambda_R^{-1}$, valid for both bounded and unbounded self-adjoint operators without requiring compact resolvent).

2. **Centered Coupling.** We assume that the interaction does not generate an $\mathcal{O}(1)$ drift on the slow manifold in the macroscopic time scaling $\tau = t/\gamma$. Equivalently, the full interaction $\mathcal{V} := \mathcal{V}_{\text{skew}} + \eta\mathcal{V}_{\text{pert}}$ satisfies

$$P_S \mathcal{V} P_S = 0. \tag{19}$$

   Since η is a free parameter in our framework, we enforce this componentwise:

$$P_S \mathcal{V}_{\text{skew}} P_S = 0, \qquad P_S \mathcal{V}_{\text{pert}} P_S = 0. \tag{20}$$

   This matches the centered coupling condition (14) used to define the effective generator $\mathcal{L}_O$.

3. **Fast-to-slow Regularity (Flux conditioning and flux regularity).** For $\psi \in \text{dom}(Q_O) \subset \mathcal{H}_S$ (where Q_O is the reference effective generator defined in (29) below), define its *microscopic flux* induced by transport as

$$g_\psi := P_F \mathcal{V}_{\text{skew}} P_S \psi \in \mathcal{H}_F. \tag{21}$$

   In the bounded-operator (finite-dimensional) setting, Q_O is a bounded self-adjoint operator on $\mathcal{H}_S$ and thus $\text{dom}(Q_O) = \mathcal{H}_S$. In unbounded settings, the identities below are understood on a common invariant core $\mathcal{C}$ for the operators involved (specified in the examples/appendices) and extended to the relevant energy domains by density. We assume a **flux conditioning bound**: there exists a constant $M_R \in (0, \infty)$ such that

$$\|g_\psi\|^2 \leq M_R \langle g_\psi, -\mathcal{R}^+ g_\psi\rangle, \qquad \forall \psi \in \text{dom}(Q_O) \subset \mathcal{H}_S. \tag{22}$$

   Since (22) is monotone in M_R, if it holds for some constant then it also holds for any larger one; throughout, M_R may be taken to be any convenient upper bound (at

the cost of more conservative constants in subsequent estimates). In the bounded-operator (finite-dimensional) setting, since $-\mathcal{R} \geq 0$ is bounded on $\mathcal{H}_F$, one may take $M_R := \|\mathcal{R}\|$ (equivalently $\|-\mathcal{R}\|$ on $\mathcal{H}_F$). In particular, for all $h \in \mathcal{H}_F$,

$$\langle(-\mathcal{R})h, h\rangle \leq M_R \|h\|^2. \tag{23}$$

We also assume a **flux regularity bound**: applying the fast–fast component of the transport to the corrected flux is controlled by the induced slow operator. Concretely, there exists $\kappa_{\text{ell}} > 0$ such that for all $\psi \in \text{dom}(Q_O)$,

$$\|P_F \mathcal{V}_{\text{skew}} P_F \left(-\mathcal{R}^+\right) g_\psi\| \leq \kappa_{\text{ell}} \|P_S \mathcal{V}_{\text{skew}} P_F (-\mathcal{R}^+) g_\psi\| = \kappa_{\text{ell}} \|(-Q_O)\psi\|. \tag{24}$$

4. **Structural Decomposition of the Perturbation.** The perturbation operator $\mathcal{V}_{\text{pert}}$ admits a decomposition into a skew-symmetric part and a symmetric part, each controlled in the $(-\mathcal{R})^{1/2}$-energy norm $\|\cdot\|_{(1)}$:

$$\mathcal{V}_{\text{pert}} = \mathcal{K}_{\text{pert}} + \mathcal{S}_{\text{pert}}, \tag{25}$$

satisfying the following conditions:

(a) **Skew Perturbation Control:** The operator $\mathcal{K}_{\text{pert}}$ is skew-symmetric, i.e., for all $f \in \text{dom}(\mathcal{V}_{\text{pert}})$:

$$\Re\langle \mathcal{K}_{\text{pert}} f, f\rangle = 0. \tag{26}$$

Moreover, $\mathcal{K}_{\text{pert}}$ is relatively bounded with respect to $(-\mathcal{R})^{1/2}$: there exist constants $a_1, b_1 \geq 0$ such that for all $f \in \text{dom}((-\mathcal{R})^{1/2})$,

$$\|\mathcal{K}_{\text{pert}} f\| \leq a_1 \|(-\mathcal{R})^{1/2} f\| + b_1 \|f\|. \tag{27}$$

(b) **Symmetric Perturbation Control:** The operator $\mathcal{S}_{\text{pert}}$ is symmetric and defined at least on the energy domain $\text{dom}((-\mathcal{R})^{1/2})$. There exist constants $a_2, b_2 \geq 0$ such that for all $f \in \text{dom}((-\mathcal{R})^{1/2})$,

$$\|\mathcal{S}_{\text{pert}} f\| \leq a_2 \|(-\mathcal{R})^{1/2} f\| + b_2 \|f\|. \tag{28}$$

Equivalently, $\mathcal{S}_{\text{pert}}$ is bounded as a map from $(\text{dom}((-\mathcal{R})^{1/2}), \|\cdot\|_{(1)})$ to $\mathcal{H}$. (In PDE models, $\mathcal{S}_{\text{pert}}$ is often an unbounded multiplier on plain L^2; (28) is the standard replacement allowing it to be controlled by the fast dissipation.)

Under Assumption IV.1, we recall the effective generator from Section III, $L_O = -P_S \mathcal{V} \mathcal{R}^+ \mathcal{V} P_S$. To construct the Lyapunov functional, we isolate the reversible contribution to the effective diffusion via the **reference effective generator** Q_O:

$$Q_O := -P_S \mathcal{V}_{\text{skew}} \mathcal{R}^+ \mathcal{V}_{\text{skew}} P_S. \tag{29}$$

From the skew-adjointness of $\mathcal{V}_{\text{skew}}$ and the negativity of $\mathcal{R}^+$, it follows that for any $f \in \mathcal{H}_S$,

$$\langle f, Q_O f\rangle = -\langle f, P_S \mathcal{V}_{\text{skew}} \mathcal{R}^+ \mathcal{V}_{\text{skew}} P_S f\rangle = -\langle \mathcal{V}_{\text{skew}}^\dagger f, \mathcal{R}^+ \mathcal{V}_{\text{skew}} f\rangle = \langle \mathcal{V}_{\text{skew}} f, \mathcal{R}^+ \mathcal{V}_{\text{skew}} f\rangle \leq 0. \tag{30}$$

Thus, Q_O is a self-adjoint, dissipative operator on $\mathcal{H}_S$.

Remark IV.2 (Well-definedness of Q_O in unbounded settings). In finite dimensions the composition $Q_O = -P_S \mathcal{V}_{\text{skew}} \mathcal{R}^+ \mathcal{V}_{\text{skew}} P_S$ is automatic, but in infinite-dimensional kinetic models $\mathcal{V}_{\text{skew}}$ is an unbounded differential operator and one must check that $\mathcal{R}^+ \mathcal{V}_{\text{skew}} P_S \psi \in \text{dom}(\mathcal{V}_{\text{skew}})$. This is ensured by Assumption IV.1.3, which stipulates that all compositions are interpreted on a common invariant core $\mathcal{C}$ (e.g. Schwartz functions) preserved by all operators involved; in the Langevin example, the pseudoinverse $\mathcal{R}^+$ is smoothing in the momentum variable, so the intermediate output $\mathcal{R}^+ \mathcal{V}_{\text{skew}} P_S \psi$ falls in the domain of the second application of $\mathcal{V}_{\text{skew}}$, and Q_O extends to a well-defined self-adjoint operator by integration by parts on $\mathcal{C}$.

Assumption IV.3 (Spectral Gap of Effective Dynamics)**.** Let $\mathcal{K} := \ker(Q_O) \subset \mathcal{H}_S$ be the kernel of the reference generator. We assume Q_O admits a **spectral gap** $\lambda(Q_O) > 0$ on the orthogonal complement $\mathcal{K}^\perp := \mathcal{H}_S \ominus \mathcal{K}$. That is, the Poincaré inequality holds:

$$-\langle f, Q_O f\rangle \geq \lambda(Q_O)\|f - P_{\mathcal{K}} f\|^2, \quad \forall f \in \mathcal{H}_S. \tag{31}$$

Consequently, for any shift parameter $\mu > 0$, the operator $(\mu - Q_O)$ is strictly positive definite on $\mathcal{H}_S$ and invertible, with operator norm $\|(\mu - Q_O)^{-1}\| \leq \mu^{-1}$.

In unbounded settings, we additionally assume (as part of the model-dependent analytic hypotheses) that $(\mu - Q_O)^{-1}$ maps $\mathcal{H}_S$ into $\mathrm{dom}(Q_O)$.

Remark IV.4 (Scope: self-adjointness excludes Liouvillian Exceptional Points)*.* Assumption IV.3 postulates $\lambda(Q_O) > 0$, which in particular requires Q_O to have a complete set of eigenfunctions (no defective spectrum). Liouvillian Exceptional Points (LEPs)—where two eigenvalues coalesce and the generator becomes non-diagonalizable—are therefore excluded by hypothesis. In our Hilbert-space setting this is not an additional restriction: the self-adjointness of Q_O established above already guarantees diagonalizability by the spectral theorem, so genuine LEPs can only arise for non-self-adjoint target effective generator $\mathcal{L}_O$,[17] instead of reference effective generator Q_O.

B. The Resolvent Corrector

To close the hypocoercive estimate, we introduce a bounded corrector that solves a shifted slow-space Poisson problem. This operator captures the response of the slow component to fast transport excursions and provides the cross term that restores coercivity at the Lyapunov level.[8,18]

Definition IV.5 (Resolvent Corrector)**.** Let $\mu > 0$ be a shift parameter, to be optimized in Section V; it is distinct from the reference measures μ_0, μ_η appearing in the examples. We define the **resolvent corrector** $\mathcal{A}_\mu : \mathcal{H} \to \mathcal{H}_S$ by

$$\mathcal{A}_\mu := (\mu - Q_O)^{-1} P_S \mathcal{V}_{\mathrm{skew}}(-\mathcal{R}^+) P_F. \tag{32}$$

The key obstruction is the degeneracy of direct dissipation on $\mathcal{H}_S$: a plain $\frac{1}{2}\|f\|^2$ energy estimate cannot produce a spectral gap when $\ker(\mathcal{R})$ is non-trivial.

Proposition IV.6 (Why the plain energy method fails)**.** *Assume $\eta = 0$ and $\mathcal{L} = \gamma\mathcal{R} + \mathcal{V}_{\mathrm{skew}}$ with $\mathcal{V}_{\mathrm{skew}}$ skew-adjoint and $\ker(\mathcal{R}) = \mathcal{H}_S \neq \{0\}$. Then there is no constant $\lambda > 0$ such that*

$$\frac{d}{dt}\frac{1}{2}\|f_t\|^2 \leq -\lambda\,\|f_t\|^2 \qquad \textit{for all solutions } \partial_t f_t = \mathcal{L} f_t. \tag{33}$$

Proof. For any $\phi \in \mathcal{H}_S$ with $\phi \neq 0$, one has $\mathcal{R}\phi = 0$ and $\Re\langle \mathcal{V}_{\mathrm{skew}}\phi, \phi\rangle = 0$, hence

$$\frac{d}{dt}\frac{1}{2}\|f_t\|^2\Big|_{t=0,\, f_0=\phi} = \Re\langle \mathcal{L}\phi, \phi\rangle = \gamma\langle \mathcal{R}\phi, \phi\rangle + \Re\langle \mathcal{V}_{\mathrm{skew}}\phi, \phi\rangle = 0. \tag{34}$$

If such a $\lambda > 0$ existed, we would get $0 \leq -\lambda\|\phi\|^2 < 0$, a contradiction. □

Differentiating $\Re\langle \mathcal{A}_\mu f, f\rangle$ along the flow generates a slow coercive contribution tied to $-Q_O$, together with controlled remainders from the fast and perturbative components.

The following lemma establishes the critical boundedness properties of $\mathcal{A}_\mu$, including the control over non-equilibrium perturbations.

Lemma IV.7 (Structural Estimates for the Resolvent Corrector)**.** *Let $\mu > 0$ be the resolvent shift parameter. We define the resolvent corrector $\mathcal{A}_\mu := (\mu - Q_O)^{-1} P_S \mathcal{V}_{\mathrm{skew}}(-\mathcal{R}^+) P_F$, where Q_O is the reference effective dynamics.*

Under Assumption IV.1, the operator $\mathcal{A}_\mu$ satisfies the following structural estimates uniformly in μ. The constants defined below depend solely on the structural parameters $(M_R, \lambda_R, \kappa_{\mathrm{ell}}, a_1, b_1, a_2, b_2)$:

1. ***Corrector Boundedness:***

$$\|\mathcal{A}_\mu\| \le k_{\mathcal{A}}\mu^{-1/2}, \quad \textit{with } k_{\mathcal{A}} := \frac{1}{2\sqrt{\lambda_R}}. \tag{35}$$

2. ***Regularity Estimates (Forward and Dual):*** *The corrector acts as a bounded regularizer for the skew-symmetric transport in both directions:*

$$\|\mathcal{V}_{\text{skew}}\mathcal{A}_\mu\| \le k_{\text{reg}}, \quad \textit{with } k_{\text{reg}} := \sqrt{M_R/\lambda_R}, \tag{36a}$$
$$\|\mathcal{A}_\mu\mathcal{V}_{\text{skew}}P_F\| \le k_{\text{dual}}, \quad \textit{with } k_{\text{dual}} := \kappa_{\text{ell}}. \tag{36b}$$

The dual estimate (36b) *is a direct consequence of the Macroscopic Flux Regularity (Assumption IV.1.3).*

3. ***Perturbation Control:*** *The interaction with the non-equilibrium perturbation is controlled in both directions:*

$$\|\mathcal{V}^\dagger_{\text{pert}}\mathcal{A}_\mu\| \le k_P(\mu), \tag{37a}$$
$$\|\mathcal{A}_\mu\mathcal{V}_{\text{pert}}\| \le \tilde{k}_P(\mu). \tag{37b}$$

with control constants

$$k_P(\mu) := (b_1 + b_2)k_{\mathcal{A}}\,\mu^{-1/2}, \tag{38}$$
$$\tilde{k}_P(\mu) := \Big((b_1 + b_2)k_{\mathcal{A}} + \tfrac{a_1+a_2}{2}\Big)\mu^{-1/2}. \tag{39}$$

Moreover, the centered perturbation coupling in Assumption IV.1.2 implies

$$P_S\mathcal{V}_{\text{pert}}P_S = 0 \qquad \Longrightarrow \qquad P_S\mathcal{V}^\dagger_{\text{pert}}P_S = 0, \tag{40}$$

and therefore

$$\mathcal{V}_{\text{pert}}\,\mathcal{H}_S \subset \mathcal{H}_F, \qquad \mathcal{V}^\dagger_{\text{pert}}\,\mathcal{H}_S \subset \mathcal{H}_F. \tag{41}$$

Proof. See Appendix B. □

C. The Modified Lyapunov Functional

We define a modified squared norm that explicitly incorporates the correlation between slow and fast degrees of freedom mediated by the resolvent corrector $\mathcal{A}_\mu$.

Definition IV.8 (Operator-Modified Lyapunov Functional)**.** For any $f \in \mathcal{H}$ and a parameter $\alpha \in (0, 1)$, we define the functional:

$$\mathcal{E}_{\alpha,\mu}(f) := \frac{1}{2}\|f\|^2 + \alpha\Re\langle\mathcal{A}_\mu f, f\rangle. \tag{42}$$

This functional defines a norm equivalent to the underlying Hilbert norm, provided the weight α is sufficiently small relative to the bounded solver $\mathcal{A}_\mu$.

Proposition IV.9 (Norm Equivalence)**.** *If the parameter α satisfies $\alpha\|\mathcal{A}_\mu\| \le 1/2$, then for all $f \in \mathcal{H}$:*

$$\frac{1}{2}(1 - 2\alpha\|\mathcal{A}_\mu\|)\|f\|^2 \le \mathcal{E}_{\alpha,\mu}(f) \le \frac{1}{2}(1 + 2\alpha\|\mathcal{A}_\mu\|)\|f\|^2. \tag{43}$$

Proof. The proof follows directly from the Cauchy-Schwarz inequality $|\Re\langle\mathcal{A}_\mu f, f\rangle| \le \|\mathcal{A}_\mu\|\|f\|^2$ and the triangle inequality. □

V. CONVERGENCE BOUNDS FOR THE LIFTED SEMIGROUP

We now derive quantitative decay estimates for the semigroup on the mean-zero subspace. The analysis yields a constructive lower bound (via a 2×2 dissipation matrix associated with the modified Lyapunov functional) and a complementary overdamped upper ceiling (via singular values). Their combination locates the finite-friction acceleration window.

A. Lower Bound via Modified Hypocoercivity

This subsection derives an explicit lower bound for the spectral decay rate by jointly optimizing the two Lyapunov parameters (α, μ). The optimization is fully algebraic and yields both the friction-dependent rate profile and a perturbative admissibility threshold.

The complete matrix estimates and parameter optimization are deferred to Appendix C; the explicit formulas used throughout the sequel are stated below.

a. Main result: explicit convergence rate. The structural properties established in the appendix imply a uniform norm equivalence (Proposition IV.9), which fixes the prefactor $C = \sqrt{3}$ independently of the friction regime.

Theorem V.1 (Explicit Quantitative Hypocoercivity)**.** *Suppose Assumptions IV.1 and IV.3 hold. Let λ_R be the microscopic spectral gap from* (18)*, and let $\lambda(Q_O)$ be the spectral gap of Q_O from Assumption IV.3. Define the structural constants*

$$\Theta := \frac{\sqrt{M_R}}{2}, \qquad K_{\text{geom}} := k_{\text{reg}} + \kappa_{\text{ell}} = \sqrt{M_R/\lambda_R} + \kappa_{\text{ell}}, \tag{44}$$

$$b_{2,\text{eff}} := b_2 + \frac{a_2}{2}(1 + \lambda_R), \tag{45}$$

where M_R is the flux-conditioning constant from (22)*, k_{reg} is defined in* (36a)*, and κ_{ell} is the flux-regularity constant from* (24)*. Let $\mathcal{Z}_*(\gamma) := \gamma\Theta\lambda(Q_O)^{-1/2} + K_{\text{geom}}$. We define the following rate functions of the friction γ:*

1. ***Unperturbed algebraic gap ($\eta = 0$).***

$$\Lambda_{\text{hypo}}(\gamma) := \frac{\frac{1}{4}\gamma\,\lambda_R}{\mathcal{Z}_*(\gamma)^2 + 2k_{\text{reg}} + \frac{1}{2}}. \tag{46}$$

2. ***Perturbation sensitivity bound.*** *Let C_{sing} be the singular constant associated with the perturbation interaction (defined in Appendix C). Define*

$$B_{\text{pert}}(\gamma) := b_{2,\text{eff}} + \underbrace{\frac{\gamma\lambda_R}{\mathcal{Z}_*(\gamma)^2 + 2k_{\text{reg}}}}_{\textit{Mixing weight } \alpha_*(\gamma)} C_{\text{sing}}\,\lambda(Q_O)^{-1/2}. \tag{47}$$

Here $b_{2,\text{eff}} = b_2 + \frac{a_2}{2}(1 + \lambda_R)$ is the effective symmetric-perturbation coefficient that absorbs both the slow-mode contribution $b_2 + \frac{a_2}{2}$ and the fast-mode λ_R-amplification $\frac{a_2\lambda_R}{2}$ arising from the bound $\|(-\mathcal{R})^{1/2}f\|^2 \geq \lambda_R\|f_F\|^2$; the factor $\lambda(Q_O)^{-1/2}$ reflects the singular behavior of the resolvent corrector as $\lambda(Q_O) \downarrow 0$.

There exists a stability threshold $\eta^*(\gamma) := \Lambda_{\text{hypo}}(\gamma)/(2B_{\text{pert}}(\gamma))$. For any friction $\gamma > 0$, if the non-equilibrium perturbation strength satisfies $\eta \leq \eta^*(\gamma)$, then the semigroup $e^{t\mathcal{L}}$ is exponentially stable on the mean-zero subspace $\mathcal{H}_0$:

$$\|e^{t\mathcal{L}}f\| \leq \sqrt{3}\,e^{-\lambda_\eta t}\|f\|, \quad \forall t \geq 0,\ \forall f \in \mathcal{H}_0. \tag{48}$$

The global decay rate λ_η satisfies the strict lower bound:

$$\lambda_\eta \geq \frac{2}{3}\left(\Lambda_{\text{hypo}}(\gamma) - \eta B_{\text{pert}}(\gamma)\right). \tag{49}$$

Sketch of Proof. The proof is detailed in Appendix C. We optimize the parameters of the modified Lyapunov functional to maximize the algebraic dissipation. At the optimal choice, the distortion parameter admits a uniform bound

$$\delta(\gamma) = 2\alpha_*(\gamma)\,\|\mathcal{A}_{\mu_*}\| \le \delta_{\max} := \frac{\sqrt{\lambda_R}}{4\,\Theta\,K_{\text{geom}}} \le \frac{1}{2}, \qquad \forall \gamma > 0, \tag{50}$$

which follows from the explicit formula $\alpha_*(\gamma) = \frac{\gamma\lambda_R}{\mathcal{Z}_*(\gamma)^2+2k_{\text{reg}}}$, the solver bound $\|\mathcal{A}_{\mu_*}\| \le k_{\mathcal{A}}\,\mu_*^{-1/2}$ with $k_{\mathcal{A}} = (2\sqrt{\lambda_R})^{-1}$, and the elementary maximization

$$\sup_{x\ge 0} \frac{x}{(x+K_{\text{geom}})^2} = \frac{1}{4K_{\text{geom}}} \qquad \text{(attained at } x = K_{\text{geom}}).$$

which guarantees the uniform norm equivalence $\frac{1}{4}\|f\|^2 \le \mathcal{E} \le \frac{3}{4}\|f\|^2$, yielding the prefactor $\sqrt{3}$ and the factor $2/3$ in the rate estimate. □

b. Regime summary. In particular, Theorem V.1 implies diffusive scaling $\Lambda_{\text{hypo}}(\gamma) = \Theta(\gamma^{-1})$ as $\gamma \to \infty$ and kinetic scaling at an intermediate friction.

Corollary V.2 (Kinetic Acceleration and Singularity Cancellation)**.** *Assume the setting of Theorem V.1. Assume furthermore that the structural constants Θ, K_{geom}, k_{reg}, λ_R, $b_{2,\text{eff}}$, and C_{sing} are independent of $\lambda(Q_O)$ (in particular, they remain $\mathcal{O}(1)$ as $\lambda(Q_O) \downarrow 0$), and that $b_{2,\text{eff}} > 0$.*

Then the unperturbed rate function $\Lambda_{\text{hypo}}(\gamma)$ defined in (46) *attains its global maximum at*

$$\gamma_{\text{opt}} = \frac{\sqrt{\lambda(Q_O)}}{\Theta}\sqrt{K_{\text{geom}}^2 + \left(2k_{\text{reg}} + \tfrac{1}{2}\right)} \asymp \sqrt{\lambda(Q_O)}. \tag{51}$$

At this friction scale, the maximal lower bound exhibits kinetic scaling:

$$\max_{\gamma>0} \Lambda_{\text{hypo}}(\gamma) \asymp \sqrt{\lambda(Q_O)}. \tag{52}$$

Moreover, the mixing weight in (47) *satisfies*

$$\alpha_*(\gamma_{\text{opt}}) \asymp \sqrt{\lambda(Q_O)}, \qquad \alpha_*(\gamma_{\text{opt}})\,\lambda(Q_O)^{-1/2} = \mathcal{O}(1), \tag{53}$$

so the solver singularity $\lambda(Q_O)^{-1/2}$ in $B_{\text{pert}}(\gamma)$ is cancelled at the optimal friction. Consequently, $B_{\text{pert}}(\gamma_{\text{opt}}) = \mathcal{O}(1)$ and the stability threshold satisfies

$$\eta^*(\gamma_{\text{opt}}) \asymp \sqrt{\lambda(Q_O)}. \tag{54}$$

Proof. Write

$$\Lambda_{\text{hypo}}(\gamma) = \frac{\lambda_R}{4}\,\frac{\gamma}{\left(\gamma\Theta\lambda(Q_O)^{-1/2} + K_{\text{geom}}\right)^2 + \left(2k_{\text{reg}} + \frac{1}{2}\right)}. \tag{55}$$

Set $x := \gamma\Theta\lambda(Q_O)^{-1/2}$ and $C_0 := 2k_{\text{reg}} + \frac{1}{2}$. Then $\Lambda_{\text{hypo}}(\gamma) = \frac{\lambda_R}{4\Theta}\sqrt{\lambda(Q_O)}\,\frac{x}{(x+K_{\text{geom}})^2+C_0}$. A direct derivative computation shows that the function $x \mapsto \frac{x}{(x+K_{\text{geom}})^2+C_0}$ is maximized at $x = \sqrt{K_{\text{geom}}^2 + C_0}$, giving the stated γ_{opt} and the scaling $\max_\gamma \Lambda_{\text{hypo}}(\gamma) \asymp \sqrt{\lambda(Q_O)}$.

For the perturbation term, by (47), $B_{\text{pert}}(\gamma) = b_{2,\text{eff}} + \alpha_*(\gamma)\,C_{\text{sing}}\,\lambda(Q_O)^{-1/2}$. At $\gamma = \gamma_{\text{opt}}$ we have $x = \mathcal{O}(1)$ and $\mathcal{Z}_*(\gamma_{\text{opt}}) = x + K_{\text{geom}} = \mathcal{O}(1)$, hence $\alpha_*(\gamma_{\text{opt}}) = \frac{\gamma_{\text{opt}}\lambda_R}{\mathcal{Z}_*(\gamma_{\text{opt}})^2+2k_{\text{reg}}} = \mathcal{O}(\sqrt{\lambda(Q_O)})$, which implies $\alpha_*(\gamma_{\text{opt}})\,\lambda(Q_O)^{-1/2} = \mathcal{O}(1)$ and therefore $B_{\text{pert}}(\gamma_{\text{opt}}) = \mathcal{O}(1)$. Finally, $\eta^*(\gamma) = \Lambda_{\text{hypo}}(\gamma)/(2B_{\text{pert}}(\gamma))$ yields $\eta^*(\gamma_{\text{opt}}) \asymp \sqrt{\lambda(Q_O)}$. □

Define the (rigorous) L^2 relaxation time as in[19] by

$$t_{\mathrm{rel}}(\mathcal{L}) := \inf\Big\{t \geq 0\,;\ \|e^{t\mathcal{L}} f\| \leq e^{-1}\,\|f\| \text{ for all } f \in \mathcal{H}_0\Big\}. \tag{56}$$

The following holds:

Corollary V.3 (Acceleration in the L^2 relaxation time)**.** *Assume the setting of Theorem V.1. Then for $\eta \leq \eta^*(\gamma)$, the exponential decay estimate from Theorem V.1 implies*

$$t_{\mathrm{rel}}(\mathcal{L}) \leq \frac{1 + \log\sqrt{3}}{\lambda_\eta}. \tag{57}$$

In particular, at the optimal friction $\gamma_{\mathrm{opt}} \asymp \sqrt{\lambda(Q_O)}$ (Corollary V.2) and for perturbations within the stability threshold, one has

$$t_{\mathrm{rel}}(\mathcal{L}) = \mathcal{O}\big(\lambda(Q_O)^{-1/2}\big). \tag{58}$$

Proof. By Theorem V.1, for all $f \in \mathcal{H}_0$ and $t \geq 0$ one has $\|e^{t\mathcal{L}} f\| \leq \sqrt{3}\, e^{-\lambda_\eta t}\,\|f\|$. Choose $t = (1 + \log\sqrt{3})/\lambda_\eta$ so that $\sqrt{3}\, e^{-\lambda_\eta t} = e^{-1}$. □

B. Overdamped diffusive ceiling

To establish that the high-friction γ^{-1} behavior is intrinsic and not an artifact of the lower-bound construction, we derive a universal upper bound on any admissible L^2 exponential rate in terms of the singular-value gap.

Define the *singular-value gap*

$$s(\mathcal{L}) := \inf_{f \in \mathcal{H}_0 \cap \mathrm{dom}(\mathcal{L}) \setminus \{0\}} \frac{\|\mathcal{L} f\|}{\|f\|}, \tag{59}$$

that is, the smallest singular value of $\mathcal{L}|_{\mathcal{H}_0}$ in finite dimensions, remaining meaningful as an infimum over the operator domain in unbounded settings. Quantities of this type are continuous-time analogues of *multiplicative reversiblizations* and the *pseudo spectral gap* used in non-reversible Markov chain theory; see, e.g.,[20]. In the discrete-time finite-state setting, a closely related point of view is developed by Chatterjee, who defines a nonreversible "spectral gap" via singular values of the generator $I - P$.[21]

Theorem V.4 (Overdamped diffusive ceiling via singular value gap)**.** *Assume Assumptions IV.1.1–4 and Assumption IV.3. Assume also* $\ker(\mathcal{L}) \cap \mathcal{H}_0 = \{0\}$*. Assume moreover that $\lambda(Q_O)$ is attained by a normalized eigenfunction $\phi \in \mathrm{dom}(Q_O) \cap \mathcal{H}_S \cap \mathcal{H}_0 \cap \mathcal{K}^\perp$ (automatic in finite dimensions, and in many PDE settings with compact resolvent). Let $C \geq 1$ and $\nu > 0$ be such that*

$$\|e^{t\mathcal{L}} f\| \leq C\, e^{-\nu t}\,\|f\|, \qquad \forall t \geq 0,\ \forall f \in \mathcal{H}_0. \tag{60}$$

Then for every $\gamma > 0$,

$$\nu \leq C\, s(\mathcal{L}) \leq C\left[\frac{(1 + \kappa_{\mathrm{ell}})\,\lambda(Q_O)}{\gamma} + \eta\,(b_1 + b_2) + \eta\,(a_1 + a_2)\,\frac{\sqrt{\lambda(Q_O)}}{\gamma}\right], \tag{61}$$

where κ_{ell} is the flux-regularity constant from Assumption IV.1.3 and a_1, a_2, b_1, b_2 are from Assumption IV.1.4. In particular, for $\eta \leq c/\gamma$ one obtains the diffusive overdamped scaling $\nu = \mathcal{O}(\gamma^{-1})$ as $\gamma \to \infty$.

Sketch of Proof. The proof is given in Appendix D. The key point is that one can test the singular value gap on a *first-order corrector* $f_\gamma := \phi - \gamma^{-1}\psi$ built from a slow eigenfunction $Q_O\phi = -\lambda(Q_O)\phi$ and $\psi := \mathcal{R}^+\mathcal{V}_{\text{skew}}\phi$. The flux regularity bound (24) controls the only nontrivial fast remainder term $P_F\mathcal{V}_{\text{skew}}\psi$ without ever invoking $\|\mathcal{V}_{\text{skew}}\|$. The perturbation contribution is controlled by the $(-\mathcal{R})^{1/2}$-energy bounds in Assumption IV.1.4 and the explicit identity $\|(-\mathcal{R})^{1/2}\psi\|^2 = \lambda(Q_O)$. □

Theorem V.4 shows that in the overdamped regime the best achievable L^2 exponential decay rate (with any finite prefactor C) is bounded above by the macroscopic diffusion scale $\lambda(Q_O)/\gamma$, up to explicit perturbative contributions. Combined with Theorem V.1, this confirms that the accelerated $\sqrt{\lambda(Q_O)}$ regime must occur at *intermediate* friction $\gamma \asymp \sqrt{\lambda(Q_O)}$, rather than in the large-friction limit.

Corollary V.5 (Diffusive ceiling for the L^2 relaxation time)**.** *Assume the setting of Theorem V.4. Define $t_{\text{rel}}(\mathcal{L})$ as in Corollary V.3. Then the singular-value gap satisfies the explicit ceiling*

$$s(\mathcal{L}) \le \frac{(1+\kappa_{\text{ell}})\,\lambda(Q_O)}{\gamma} + \eta\,(b_1+b_2) + \eta\,(a_1+a_2)\,\frac{\sqrt{\lambda(Q_O)}}{\gamma}. \tag{62}$$

Moreover, the relaxation time is bounded below in terms of the singular-value gap (see Lemma D.2, with the same prefactor C as in Theorem V.4):

$$t_{\text{rel}}(\mathcal{L}) \ge \frac{1}{2C\,s(\mathcal{L})}. \tag{63}$$

Consequently, if $\eta \le c/\gamma$, then $s(\mathcal{L}) = \mathcal{O}(\gamma^{-1})$ as $\gamma \to \infty$ and hence

$$t_{\text{rel}}(\mathcal{L}) = \Omega(\gamma) \qquad (\gamma \to \infty). \tag{64}$$

In particular, the kinetic scaling $t_{\text{rel}}(\mathcal{L}) \asymp \lambda(Q_O)^{-1/2}$ is achieved at optimal friction, combined with Corollary V.3.

VI. EXAMPLES

We now verify the abstract assumptions in two concrete models and compare the predicted scaling laws with numerical spectra. The examples are chosen to illustrate (a) the instantiation of the constants in Theorem V.1, and (b) the faithfulness of the lower/upper rate bounds relative to actual spectral behavior.

Section VI A treats forced underdamped Langevin dynamics in weighted L^2 spaces and recovers the expected underdamped/overdamped crossover together with finite-friction optimization.[15,16] Section VI B gives a finite-dimensional boundary-driven construction with non-self-adjoint effective slow dynamics, illustrating that the framework is not restricted to reversible reduced generators.[19]

A. Example 1: Classical Nonequilibrium Langevin Dynamics

We specialize the abstract framework to a kinetic Fokker–Planck operator with nongradient forcing. The objective is to verify Assumption IV.1 on a natural core, identify Q_O explicitly, and extract the scaling consequences of Theorem V.1 in this PDE setting.

Setup.
Phase space and reference measure. Consider a particle in the phase space $\mathcal{X} = \mathbb{T}^d \times \mathbb{R}^d$ ($d \ge 1$), where the position space is the torus $\mathbb{T}^d$. The system is subject to

a smooth conservative potential $U \in C^\infty(\mathbb{T}^d)$ and a non-conservative external force field $F \in C^\infty(\mathbb{T}^d;\mathbb{R}^d)$. Let μ_0 be the Gibbs measure of the unperturbed Langevin dynamics:

$$d\mu_0(q,p) = Z^{-1} e^{-\beta(\frac{1}{2}|p|^2+U(q))} dq\, dp, \tag{65}$$

where $\beta > 0$ is the inverse temperature and Z is the partition function. Write $d\mu_0(q,p) = d\nu(q)\, d\nu_p(p)$, where $d\nu(q) = Z_q^{-1} e^{-\beta U(q)} dq$ and $d\nu_p(p) = Z_p^{-1} e^{-\frac{\beta}{2}|p|^2} dp$.

For $\eta \neq 0$, the invariant law is generally not μ_0; under standard assumptions it admits a density h_η with respect to μ_0, $d\mu_\eta = h_\eta d\mu_0$.[15,16] We keep the analysis in the reference Hilbert space $L^2(\mu_0)$ and formulate convergence to stationarity as decay toward h_η.

Functional-analytic setting. For $\eta \neq 0$ we work in the reference space $L^2(\mu_0)$ and interpret "convergence to the NESS" as $L^2(\mu_0)$-convergence of the forward evolution toward the stationary density h_η, i.e. $\|e^{t\mathcal{L}^\dagger} f - h_\eta\|_{L^2(\mu_0)} \le Ce^{-\lambda t}\, \|f - h_\eta\|_{L^2(\mu_0)}$, where $\dagger$ denotes the $L^2(\mu_0)$-adjoint.

Core and extension principle. Define the dense test domain

$$\mathcal{C}_{\mathrm{Lan}} := C^\infty(\mathbb{T}^d) \otimes \mathcal{S}(\mathbb{R}^d) \subset L^2(\mu_0). \tag{66}$$

All integrations by parts and operator identities in this example are first performed on $\mathcal{C}_{\mathrm{Lan}}$, which is invariant under $\mathcal{V}_{\mathrm{skew}}$, $\mathcal{R}$, and $\mathcal{V}_{\mathrm{pert}}$. The resulting estimates are expressed in terms of $\|f\|_{L^2(\mu_0)}$ and $\|(-\mathcal{R})^{1/2} f\|_{L^2(\mu_0)}$, so they extend by density from $\mathcal{C}_{\mathrm{Lan}}$ to $\mathrm{dom}((-\mathcal{R})^{1/2})$ (and to $\mathrm{dom}(Q_O)$ on the slow side).

PDE realization in weighted spaces (micro–macro form). Let

$$(\Pi_p f)(q) := \int_{\mathbb{R}^d} f(q,p)\, d\nu_p(p) = P_S f, \qquad f = \Pi_p f + (\mathrm{id} - \Pi_p) f. \tag{67}$$

On $\mathcal{C}_{\mathrm{Lan}}$, with respect to the weighted inner product of $L^2(\mu_0)$, the adjoints of the plain gradients are

$$\nabla_q^\dagger = -\nabla_q + \beta \nabla U(q), \qquad \nabla_p^\dagger = -\nabla_p + \beta p, \tag{68}$$

so

$$\mathcal{R} = -\beta^{-1} \nabla_p^\dagger \nabla_p, \qquad \mathcal{V}_{\mathrm{skew}} = \beta^{-1}(\nabla_p^\dagger \nabla_q - \nabla_q^\dagger \nabla_p). \tag{69}$$

Consequently, for every $f \in \mathcal{C}_{\mathrm{Lan}}$,

$$\langle f, -\mathcal{R} f\rangle_{\mu_0} = \beta^{-1} \|\nabla_p f\|^2_{L^2(\mu_0)}, \tag{70}$$

and Gaussian Poincaré in p gives

$$\|(\mathrm{id} - \Pi_p) f\|^2_{L^2(\mu_0)} \le \beta\, \langle f, -\mathcal{R} f\rangle_{\mu_0}. \tag{71}$$

For slow observables $\psi = \psi(q)$ and $Q_O = \beta^{-1}\Delta_q - \nabla U \cdot \nabla_q$,

$$\langle \psi, (-Q_O)\psi\rangle_{L^2(\nu)} = \beta^{-1} \|\nabla_q \psi\|^2_{L^2(\nu)}, \tag{72}$$

which is the slow-space Dirichlet form used below.

Remark VI.1 (Applicability of main theorems). Since h_η is the unique stationary density, it satisfies $\mathcal{L}^\dagger h_\eta = 0$. Therefore the translated variable $g := f - h_\eta$ evolves by the same semigroup, $e^{t\mathcal{L}^\dagger} g = e^{t\mathcal{L}^\dagger} f - h_\eta$. Consequently, exponential convergence to h_η is equivalent to exponential decay of g to 0 on the mean-zero subspace (with respect to μ_η, or equivalently after centering in $L^2(\mu_0)$). Thus, when applying Theorem V.1 in this example, one simply applies it to the centered function $g = f - h_\eta$; the theorem's "$e^{t\mathcal{L}}$ decays to 0" formulation is exactly the same statement after shifting by the stationary state.

Generator decomposition. The evolution is governed by the generator $\mathcal{L}$, which we decompose according to the hypocoercive structure $\mathcal{L} = \mathcal{V}_{\text{skew}} + \gamma\mathcal{R} + \eta\mathcal{V}_{\text{pert}}$:

1. **Hamiltonian Transport ($\mathcal{V}_{\text{skew}}$):** The generator of the Hamiltonian flow for $H(q,p) = \frac{1}{2}|p|^2 + U(q)$:

$$\mathcal{V}_{\text{skew}} := p \cdot \nabla_q - \nabla U(q) \cdot \nabla_p. \tag{73}$$

2. **Dissipation ($\mathcal{R}$):** The Ornstein-Uhlenbeck operator acting on momenta:

$$\mathcal{R} := -p \cdot \nabla_p + \beta^{-1}\Delta_p. \tag{74}$$

3. **Non-Equilibrium Perturbation ($\mathcal{V}_{\text{pert}}$):** We take the standard (unregularized) forcing

$$\mathcal{V}_{\text{pert}} f := F(q) \cdot \nabla_p f. \tag{75}$$

Effective slow generator (Smoluchowski limit). We compute the effective generator on slow observables by the adiabatic-elimination formula $\mathcal{L}_O = -P_S(\mathcal{V}_{\text{skew}} + \eta\mathcal{V}_{\text{pert}})\,\mathcal{R}^+\,(\mathcal{V}_{\text{skew}} + \eta\mathcal{V}_{\text{pert}})P_S$. Let $\phi \in \mathcal{H}_S$ be a smooth slow observable, i.e. $\phi = \phi(q)$. First, $\mathcal{V}_{\text{pert}}\phi = 0$ and $\mathcal{V}_{\text{skew}}\phi = p \cdot \nabla_q\phi \in \mathcal{H}_F$. Since $\mathcal{R}(p_i) = -p_i$, we have $\mathcal{R}^+(p_i) = -p_i$ on $\mathcal{H}_F$, hence

$$\mathcal{R}^+\mathcal{V}_{\text{skew}}\phi = -p \cdot \nabla_q\phi. \tag{76}$$

Writing the combined transport as $\mathcal{V}_{\text{skew}} + \eta\mathcal{V}_{\text{pert}} = p \cdot \nabla_q - (\nabla U - \eta F) \cdot \nabla_p$, we compute

$$(\mathcal{V}_{\text{skew}} + \eta\mathcal{V}_{\text{pert}})\big(-p \cdot \nabla_q\phi\big) = -\sum_{i,j} p_i p_j\, \partial_{q_i q_j}\phi + (\nabla U - \eta F) \cdot \nabla_q\phi. \tag{77}$$

Projecting onto $\mathcal{H}_S$ amounts to averaging in p against the centered Gaussian, for which $\int p_i p_j\, d\nu_p = \beta^{-1}\delta_{ij}$. Therefore the effective slow generator on observables is

$$\mathcal{L}_O\phi = \beta^{-1}\Delta_q\phi - (\nabla U - \eta F) \cdot \nabla_q\phi. \tag{78}$$

In particular, in the reversible case $\eta = 0$ the corresponding reference operator can be taken as

$$Q_O := \mathcal{L}_O\big|_{\eta=0} = \beta^{-1}\Delta_q - \nabla U \cdot \nabla_q, \tag{79}$$

viewed as a self-adjoint operator on $L^2(\nu)$ with $d\nu \propto e^{-\beta U(q)}dq$. Equivalently, setting $D := \nabla_q$ and letting $D^\dagger$ denote the $L^2(\nu)$-adjoint of the plain gradient, $D^\dagger f = -\nabla_q f + \beta(\nabla U)f$, one has the Dirichlet-form representation

$$Q_O = -(1/\beta)\, D^\dagger D, \tag{80}$$

which is implemented numerically in this direct D-form: we discretize D and its weighted adjoint $D^\dagger$ in the Fourier basis and assemble $Q_O = -(1/\beta)D^\dagger D$ without introducing auxiliary transformed operators. (Explicitly, $D^\dagger D\phi = -\Delta_q\phi + \beta(\nabla U) \cdot \nabla_q\phi$, so $-(1/\beta)D^\dagger D = \beta^{-1}\Delta_q - \nabla U \cdot \nabla_q = Q_O$ as required.) Equivalently, the adjoint effective dynamics for the slow density $\rho(q,t)$ is the (forced) Smoluchowski equation

$$\partial_t\rho = \beta^{-1}\Delta_q\rho + \nabla_q \cdot \big(\rho(\nabla U - \eta F)\big). \tag{81}$$

Verification of the structural assumptions.

We verify that the tuple $(\mathcal{R}, \mathcal{V}_{\text{skew}}, \mathcal{V}_{\text{pert}})$ satisfies Assumption IV.1.1–4 on a natural smooth core in $L^2(\mu_0)$.

Proposition VI.2 (Verification for Nonequilibrium Langevin Dynamics)**.** *The operators defined above satisfy Assumption IV.1.1–4 in $L^2(\mu_0)$. The verification is carried out on $\mathcal{C}_{\mathrm{Lan}}$ and extended to the stated domains by density.*

Proof. We address each condition systematically. All computations below are initially on the core $\mathcal{C}_{\mathrm{Lan}}$.

1. Microscopic Spectral Gap ($\mathcal{R}$). The operator $\mathcal{R}$ acts only on the p-variables. It is classical that the Ornstein–Uhlenbeck operator admits a spectral gap $\lambda_R = 1$ in the Gaussian-weighted space $L^2(e^{-\beta p^2/2}dp)$. Specifically, for any $f \perp \ker(\mathcal{R})$ (where $\ker(\mathcal{R})$ consists of functions of q only), we have $-\langle \mathcal{R}f, f\rangle_{\mu_0} \geq \|f\|^2$.

2. Skew-Adjointness and Centered Coupling ($\mathcal{V}_{\mathrm{skew}}$). The Liouville operator $\mathcal{V}_{\mathrm{skew}}$ satisfies $\mathcal{V}_{\mathrm{skew}}^\dagger = -\mathcal{V}_{\mathrm{skew}}$ in $L^2(\mu_0)$ because the Hamiltonian flow preserves the Gibbs measure. The slow subspace $\mathcal{H}_S$ consists of functions $\phi(q)$ independent of p. The projection is $P_S f = \int f d\nu_p$, where $d\nu_p$ is the marginal Gaussian measure on momenta. For any $\phi \in \mathcal{H}_S$, $\mathcal{V}_{\mathrm{skew}}\phi = p \cdot \nabla_q \phi$. Then, $P_S \mathcal{V}_{\mathrm{skew}} P_S \phi = \int (p \cdot \nabla_q \phi) d\nu_p = (\int p\, d\nu_p) \cdot \nabla_q \phi = 0$, since the Gaussian is centered at zero.

Moreover, for any slow observable $\phi(q)$ we have $\mathcal{V}_{\mathrm{pert}}\phi = F(q) \cdot \nabla_p \phi = 0$. Therefore $P_S \mathcal{V}_{\mathrm{pert}} P_S = 0$, and hence $P_S \mathcal{V} P_S = 0$ for $\mathcal{V} = \mathcal{V}_{\mathrm{skew}} + \eta \mathcal{V}_{\mathrm{pert}}$.

3. Fast-to-slow Regularity (Assumption IV.1.3). Let $\psi \in \mathrm{dom}(Q_O) \subset \mathcal{H}_S$ be a smooth function of q only. Then

$$g_\psi = P_F \mathcal{V}_{\mathrm{skew}} P_S \psi = p \cdot \nabla_q \psi, \tag{82}$$

which is centered in p and therefore lies in $\mathcal{H}_F$. Since the Ornstein–Uhlenbeck operator satisfies $\mathcal{R}(p_i) = -p_i$, we have $\mathcal{R} g_\psi = -g_\psi$ and hence $(-\mathcal{R}^+) g_\psi = g_\psi$ (on this first Hermite sector). Therefore

$$\langle g_\psi, -\mathcal{R}^+ g_\psi \rangle_{\mu_0} = \|g_\psi\|_{\mu_0}^2, \tag{83}$$

so the flux conditioning bound (22) holds with $M_R = 1$.

It remains to check the flux regularity bound (24) in a self-contained way. Since $(-\mathcal{R}^+) g_\psi = g_\psi$ and $P_F(p \cdot \nabla_q \psi) = p \cdot \nabla_q \psi$, a direct computation gives

$$P_F \mathcal{V}_{\mathrm{skew}} P_F (-\mathcal{R}^+) g_\psi = P_F \mathcal{V}_{\mathrm{skew}} \left(p \cdot \nabla_q \psi\right) = \sum_{i,j} \left(p_i p_j - \beta^{-1}\delta_{ij}\right) \partial_{q_i q_j} \psi. \tag{84}$$

Step 1: explicit reduction to a second-derivative norm. Define the centered quadratic polynomials

$$Z_{ij}(p) := p_i p_j - \beta^{-1}\delta_{ij}. \tag{85}$$

For each fixed q, Cauchy–Schwarz in the finite sum over i, j yields

$$\left| \sum_{i,j} Z_{ij}(p)\, \partial_{q_i q_j} \psi(q) \right|^2 \leq \left(\sum_{i,j} |Z_{ij}(p)|^2 \right) \left(\sum_{i,j} |\partial_{q_i q_j} \psi(q)|^2 \right). \tag{86}$$

Integrating first in p and then in q gives

$$\left\| P_F \mathcal{V}_{\mathrm{skew}} P_F (-\mathcal{R}^+) g_\psi \right\|_{L^2(\mu_0)} \leq c_{d,\beta} \left\| \nabla_q^2 \psi \right\|_{L^2(\nu)}, \tag{87}$$

where the constant $c_{d,\beta}$ is defined explicitly by

$$c_{d,\beta}^2 := \int_{\mathbb{R}^d} \sum_{i,j} |Z_{ij}(p)|^2\, d\nu_p(p). \tag{88}$$

Since the coordinates of p are i.i.d. $N(0,\beta^{-1})$ under ν_p, we have $\mathbb{E}[p_i^4]=3\beta^{-2}$ and $\mathbb{E}[p_i^2p_j^2]=\beta^{-2}$ for $i\neq j$. Therefore

$$\|Z_{ii}\|^2_{L^2(\nu_p)}=\mathbb{E}\big[(p_i^2-\beta^{-1})^2\big]=2\beta^{-2},\qquad \|Z_{ij}\|^2_{L^2(\nu_p)}=\mathbb{E}[p_i^2p_j^2]=\beta^{-2}\quad (i\neq j),\tag{89}$$

and hence

$$c^2_{d,\beta}=d\,(2\beta^{-2})+d(d-1)\,\beta^{-2}=\frac{d(d+1)}{\beta^2},\qquad c_{d,\beta}=\frac{\sqrt{d(d+1)}}{\beta}.\tag{90}$$

Step 2: control of $\|\nabla_q^2\psi\|_{L^2(\nu)}$ by $\|(-Q_O)\psi\|_{L^2(\nu)}$ with explicit structural constants. Recall that the reversible overdamped generator acting on q-observables is

$$Q_O\psi=\beta^{-1}\Delta_q\psi-\nabla U\cdot\nabla_q\psi,\tag{91}$$

so that

$$\Delta_q\psi=\beta\big(Q_O\psi+\nabla U\cdot\nabla_q\psi\big).\tag{92}$$

Let $M_U:=\|\nabla U\|_{L^\infty(\mathbb{T}^d)}$.

(i) Hessian vs. Laplacian on the torus in $L^2(\nu)$. Write $d\nu(q)=\rho(q)\,dq$ with $\rho(q)=Z_q^{-1}e^{-\beta U(q)}$. Since ρ is smooth and strictly positive on the compact torus, we may set $\rho_{\min}:=\inf_{\mathbb{T}^d}\rho>0$ and $\rho_{\max}:=\sup_{\mathbb{T}^d}\rho<\infty$.

Fourier/Parseval step (in $L^2(dq)$). Write the Fourier series

$$\psi(q)=\sum_{k\in\mathbb{Z}^d}\hat\psi_k\,e^{ik\cdot q}.\tag{93}$$

Then

$$\partial_{q_iq_j}\psi(q)=\sum_{k\in\mathbb{Z}^d}(-k_ik_j)\,\hat\psi_k\,e^{ik\cdot q},\tag{94}$$

$$\Delta_q\psi(q)=\sum_{k\in\mathbb{Z}^d}(-|k|^2)\,\hat\psi_k\,e^{ik\cdot q}.\tag{95}$$

By Parseval's identity,

$$\|\nabla_q^2\psi\|^2_{L^2(dq)}=\sum_{i,j=1}^d\|\partial_{q_iq_j}\psi\|^2_{L^2(dq)}=\sum_{k\in\mathbb{Z}^d}\left(\sum_{i,j=1}^d k_i^2k_j^2\right)|\hat\psi_k|^2\tag{96}$$

$$=\sum_{k\in\mathbb{Z}^d}|k|^4|\hat\psi_k|^2\tag{97}$$

$$=\|\Delta_q\psi\|^2_{L^2(dq)}.\tag{98}$$

Hence $\|\nabla_q^2\psi\|_{L^2(dq)}=\|\Delta_q\psi\|_{L^2(dq)}$ for all periodic $\psi\in H^2(\mathbb{T}^d)$.

Norm-equivalence step (switching to $L^2(\nu)$). For any measurable f, the bounds $m\le\rho\le M$ imply

$$\rho_{\min}\,\|f\|^2_{L^2(dq)}\le\|f\|^2_{L^2(\nu)}:=\int_{\mathbb{T}^d}|f|^2\,d\nu=\int_{\mathbb{T}^d}|f|^2\,\rho\,dq\le\rho_{\max}\,\|f\|^2_{L^2(dq)}.\tag{99}$$

Applying this to $f=\nabla_q^2\psi$ and $f=\Delta_q\psi$ gives

$$\|\nabla_q^2\psi\|^2_{L^2(\nu)}\le\rho_{\max}\,\|\nabla_q^2\psi\|^2_{L^2(dq)}=\rho_{\max}\,\|\Delta_q\psi\|^2_{L^2(dq)}\le\frac{\rho_{\max}}{\rho_{\min}}\,\|\Delta_q\psi\|^2_{L^2(\nu)},\tag{100}$$

hence

$$\|\nabla_q^2\psi\|_{L^2(\nu)} \le \sqrt{\tfrac{\rho_{\max}}{\rho_{\min}}}\,\|\Delta_q\psi\|_{L^2(\nu)}. \tag{101}$$

(The factor $(\rho_{\max}/\rho_{\min})^{1/2}$ is structural, depending only on U and β.)

(ii) Bounding $\|\Delta_q\psi\|$ by $\|(-Q_O)\psi\|$. From the identity for $\Delta_q\psi$ and $\|\nabla U\cdot\nabla_q\psi\| \le M_U\,\|\nabla_q\psi\|$, we have

$$\|\Delta_q\psi\|_{L^2(\nu)} \le \beta\,\|(-Q_O)\psi\|_{L^2(\nu)} + \beta M_U\,\|\nabla_q\psi\|_{L^2(\nu)}. \tag{102}$$

To estimate the remaining gradient term with a structural constant, we use the Dirichlet form of Q_O. For mean-zero ψ (equivalently $\psi \in \ker(Q_O)^\perp$), the Poincaré inequality for ν gives a constant $C_{\mathrm{P}}(\nu) \in (0,\infty)$ such that

$$\|\psi\|_{L^2(\nu)}^2 \le C_{\mathrm{P}}(\nu)\,\|\nabla_q\psi\|_{L^2(\nu)}^2. \tag{103}$$

Moreover, since Q_O is reversible with invariant measure ν, we can compute the corresponding Dirichlet form explicitly. Writing $d\nu = \rho\,dq$ with $\rho \propto e^{-\beta U}$, we first rewrite Q_O in divergence form:

$$Q_O\psi = \beta^{-1}\Delta_q\psi - \nabla U\cdot\nabla_q\psi = \beta^{-1}\,\rho^{-1}\,\nabla_q\cdot\big(\rho\,\nabla_q\psi\big), \tag{104}$$

since $\nabla_q\rho = -\beta\rho\,\nabla U$. Therefore, using periodicity on $\mathbb{T}^d$ (so boundary terms vanish), integration by parts gives

$$\langle(-Q_O)\psi,\psi\rangle_{L^2(\nu)} = -\int_{\mathbb{T}^d}(Q_O\psi)\,\psi\,d\nu \qquad = -\beta^{-1}\int_{\mathbb{T}^d}\rho^{-1}\,\nabla_q\cdot(\rho\nabla_q\psi)\,\psi\,\rho\,dq \tag{105}$$

$$= -\beta^{-1}\int_{\mathbb{T}^d}\nabla_q\cdot(\rho\nabla_q\psi)\,\psi\,dq \tag{106}$$

$$= \beta^{-1}\int_{\mathbb{T}^d}\rho\,\nabla_q\psi\cdot\nabla_q\psi\,dq \tag{107}$$

$$= \beta^{-1}\,\|\nabla_q\psi\|_{L^2(\nu)}^2. \tag{108}$$

Combining this with Cauchy–Schwarz and (103),

$$\begin{aligned}\beta^{-1}\,\|\nabla_q\psi\|_{L^2(\nu)}^2 = \langle(-Q_O)\psi,\psi\rangle_{L^2(\nu)} &\le \|(-Q_O)\psi\|_{L^2(\nu)}\,\|\psi\|_{L^2(\nu)}\\ &\le \sqrt{C_{\mathrm{P}}(\nu)}\,\|(-Q_O)\psi\|_{L^2(\nu)}\,\|\nabla_q\psi\|_{L^2(\nu)},\end{aligned}$$

so that

$$\|\nabla_q\psi\|_{L^2(\nu)} \le \beta\sqrt{C_{\mathrm{P}}(\nu)}\,\|(-Q_O)\psi\|_{L^2(\nu)}. \tag{109}$$

Substituting (109) into (102) yields

$$\|\Delta_q\psi\|_{L^2(\nu)} \le \beta\big(1+\beta M_U\sqrt{C_{\mathrm{P}}(\nu)}\big)\,\|(-Q_O)\psi\|_{L^2(\nu)}. \tag{110}$$

Finally, combining with (101) gives a purely structural constant

$$C_{\mathrm{H}^2}(U,\beta,d) := \sqrt{\tfrac{\rho_{\max}}{\rho_{\min}}}\,\beta\big(1+\beta M_U\sqrt{C_{\mathrm{P}}(\nu)}\big) \tag{111}$$

such that

$$\|\nabla_q^2\psi\|_{L^2(\nu)} \le C_{\mathrm{H}^2}(U,\beta,d)\,\|(-Q_O)\psi\|_{L^2(\nu)}. \tag{112}$$

Conclusion. Combining (87) with the previous display yields

$$\big\|P_F\mathcal{V}_{\rm skew}P_F(-\mathcal{R}^+)g_\psi\big\|_{L^2(\mu_0)} \le \kappa_{\rm ell}\,\|(-Q_O)\psi\|_{L^2(\nu)}, \tag{113}$$

where we may take the explicit structural constant

$$\begin{aligned}\kappa_{\rm ell} := c_{d,\beta}\,C_{\mathrm{H}^2}(U,\beta,d) &= \frac{\sqrt{d(d+1)}}{\beta}\sqrt{\tfrac{\rho_{\max}}{\rho_{\min}}}\,\beta\big(1+\beta M_U\sqrt{C_{\rm P}(\nu)}\big)\\ &= \sqrt{d(d+1)}\sqrt{\tfrac{\rho_{\max}}{\rho_{\min}}}\,\big(1+\beta\|\nabla U\|_{L^\infty(\mathbb{T}^d)}\sqrt{C_{\rm P}(\nu)}\big).\end{aligned}$$

This verifies the flux regularity bound (24) with constants that are independent of treating $\lambda(Q_O)$ as a variable in the main theorems.

4(a)–4(b). Perturbation decomposition and energy bounds. We compute the $L^2(\mu_0)$-adjoint $\mathcal{V}_{\rm pert}^\dagger$ and split $\mathcal{V}_{\rm pert}$ into its skew and symmetric parts. Let $f,g\in\mathcal{C}_{\rm Lan}$. Write $d\mu_0(q,p)=Z^{-1}e^{-\beta U(q)}e^{-\frac{\beta}{2}|p|^2}\,dq\,dp$. Then, using that $F=F(q)$ is independent of p and integrating by parts in p (no boundary term since $f,g\in\mathcal{S}(\mathbb{R}^d)$ in p),

$$\langle\mathcal{V}_{\rm pert}f,g\rangle_{\mu_0} = \int_{\mathbb{T}^d} e^{-\beta U(q)}dq\int_{\mathbb{R}^d}\big(F(q)\cdot\nabla_p f(q,p)\big)\,\overline{g(q,p)}\,e^{-\frac{\beta}{2}|p|^2}\,dp \tag{114}$$

$$= -\int_{\mathbb{T}^d} e^{-\beta U(q)}dq\int_{\mathbb{R}^d} f(q,p)\,\nabla_p\cdot\Big(F(q)\,\overline{g(q,p)}\,e^{-\frac{\beta}{2}|p|^2}\Big)\,dp \tag{115}$$

$$= -\int_{\mathbb{T}^d} e^{-\beta U(q)}dq\int_{\mathbb{R}^d} f(q,p)\,\Big(F(q)\cdot\nabla_p\overline{g(q,p)}\Big)e^{-\frac{\beta}{2}|p|^2}\,dp \tag{116}$$

$$-\int_{\mathbb{T}^d} e^{-\beta U(q)}dq\int_{\mathbb{R}^d} f(q,p)\,\overline{g(q,p)}\,F(q)\cdot\nabla_p\Big(e^{-\frac{\beta}{2}|p|^2}\Big)\,dp. \tag{117}$$

Since $\nabla_p\big(e^{-\frac{\beta}{2}|p|^2}\big)=-\beta p\,e^{-\frac{\beta}{2}|p|^2}$, this becomes

$$\langle\mathcal{V}_{\rm pert}f,g\rangle_{\mu_0} = \int f\,\overline{\Big(-F\cdot\nabla_p g+\beta(F\cdot p)g\Big)}\,d\mu_0. \tag{118}$$

Hence the $L^2(\mu_0)$-adjoint is

$$\mathcal{V}_{\rm pert}^\dagger = -F\cdot\nabla_p+\beta(F\cdot p). \tag{119}$$

We then define the skew/symmetric parts in $L^2(\mu_0)$ by

$$\mathcal{K}_{\rm pert} := \tfrac12\big(\mathcal{V}_{\rm pert}-\mathcal{V}_{\rm pert}^\dagger\big) = F\cdot\nabla_p-\tfrac{\beta}{2}(F\cdot p),\qquad \mathcal{S}_{\rm pert} := \tfrac12\big(\mathcal{V}_{\rm pert}+\mathcal{V}_{\rm pert}^\dagger\big) = \tfrac{\beta}{2}(F\cdot p). \tag{120}$$

By definition one has $(\mathcal{K}_{\rm pert})^\dagger=-\mathcal{K}_{\rm pert}$ (hence $\Re\langle\mathcal{K}_{\rm pert}f,f\rangle_{\mu_0}=0$) and $(\mathcal{S}_{\rm pert})^\dagger=\mathcal{S}_{\rm pert}$.

Energy control of $\mathcal{K}_{\rm pert}$ (Assumption IV.1.4(a)). Since $F\in L^\infty(\mathbb{T}^d)$,

$$\|\mathcal{K}_{\rm pert}f\| \le \|F\|_{L^\infty}\,\|\nabla_p f\| + \tfrac{\beta}{2}\,\|F\|_{L^\infty}\,\||p|f\|. \tag{121}$$

For the Ornstein–Uhlenbeck generator on $L^2(\mu_0)$ one has $\langle f,-\mathcal{R}f\rangle_{\mu_0}=\beta^{-1}\|\nabla_p f\|^2$, hence $\|\nabla_p f\|\le\sqrt{\beta}\,\|(-\mathcal{R})^{1/2}f\|$. Moreover, one can bound the (unbounded) multiplier p by a Gaussian integration-by-parts estimate. Let $w(p):=e^{-\frac{\beta}{2}|p|^2}$. For each coordinate i and $f\in C_c^\infty(\mathbb{R}^d)$,

$$\int_{\mathbb{R}^d} p_i^2|f|^2\,w\,dp = -\frac{1}{\beta}\int_{\mathbb{R}^d} p_i|f|^2\,\partial_{p_i}w\,dp = \frac{1}{\beta}\int_{\mathbb{R}^d}\partial_{p_i}(p_i|f|^2)\,w\,dp \tag{122}$$

$$= \frac{1}{\beta}\int_{\mathbb{R}^d}|f|^2\,w\,dp + \frac{2}{\beta}\Re\int_{\mathbb{R}^d} p_i f\,\overline{\partial_{p_i}f}\,w\,dp. \tag{123}$$

We estimate the cross term by Young's inequality. For any $\varepsilon \in (0,1)$,

$$\frac{2}{\beta}\left|\Re\int_{\mathbb{R}^d} p_i f\,\overline{\partial_{p_i} f}\,w\,dp\right| \le \varepsilon\,\|p_i f\|^2_{L^2(wdp)} + \frac{1}{\varepsilon\beta^2}\,\|\partial_{p_i} f\|^2_{L^2(wdp)}. \tag{124}$$

Choosing $\varepsilon = \frac{1}{2}$ and inserting this into the previous inequality yields, for each $i \in \{1,\dots,d\}$,

$$\|p_i f\|^2_{L^2(wdp)} \le \frac{2}{\beta}\,\|f\|^2_{L^2(wdp)} + \frac{4}{\beta^2}\,\|\partial_{p_i} f\|^2_{L^2(wdp)}. \tag{125}$$

Summing over i gives

$$\||p|f\|^2_{L^2(wdp)} = \sum_{i=1}^{d}\|p_i f\|^2_{L^2(wdp)} \le \frac{2d}{\beta}\,\|f\|^2_{L^2(wdp)} + \frac{4}{\beta^2}\,\|\nabla_p f\|^2_{L^2(wdp)}. \tag{126}$$

Taking square roots and using $\sqrt{a+b} \le \sqrt{a} + \sqrt{b}$ yields the explicit multiplier bound

$$\||p|f\|_{L^2(\mu_0)} \le \sqrt{\tfrac{2d}{\beta}}\,\|f\|_{L^2(\mu_0)} + \frac{2}{\beta}\,\|\nabla_p f\|_{L^2(\mu_0)} =: C_0\,\|f\|_{L^2(\mu_0)} + C_1\,\|\nabla_p f\|_{L^2(\mu_0)}. \tag{127}$$

Combining this with $\|\nabla_p f\| \le \sqrt{\beta}\,\|(-\mathcal{R})^{1/2} f\|$ and $\|\mathcal{K}_{\mathrm{pert}} f\| \le \|F\|_{L^\infty}\,\|\nabla_p f\| + \frac{\beta}{2}\,\|F\|_{L^\infty}\,\||p|f\|$ (from the definition of $\mathcal{K}_{\mathrm{pert}}$), we obtain

$$\|\mathcal{K}_{\mathrm{pert}} f\| \le a_1\,\|(-\mathcal{R})^{1/2} f\| + b_1\,\|f\|, \tag{128}$$

for some explicit constants $a_1, b_1 \ge 0$ depending only on β, d and $\|F\|_{L^\infty}$.

Energy control of $\mathcal{S}_{\mathrm{pert}}$ (Assumption IV.1.4(b)). Recall

$$\mathcal{S}_{\mathrm{pert}} = \frac{\beta}{2}\,(F(q)\cdot p), \tag{129}$$

so by $|F\cdot p| \le \|F\|_{L^\infty}|p|$ and (127),

$$\|\mathcal{S}_{\mathrm{pert}} f\|_{L^2(\mu_0)} \le \frac{\beta}{2}\,\|F\|_{L^\infty}\,\||p|f\|_{L^2(\mu_0)} \le \frac{\beta}{2}\,\|F\|_{L^\infty}\left(C_0\,\|f\|_{L^2(\mu_0)} + C_1\,\|\nabla_p f\|_{L^2(\mu_0)}\right) \tag{130}$$

$$\le a_2\,\|(-\mathcal{R})^{1/2} f\|_{L^2(\mu_0)} + b_2\,\|f\|_{L^2(\mu_0)}, \tag{131}$$

for suitable constants $a_2, b_2 \ge 0$.[16] This verifies Assumption IV.1.4(b) in the natural hypocoercive functional setting. □

Which nonequilibrium forcing fields preserve kinetic acceleration? This question is answered precisely by Corollary V.2. The computation above shows that for any *bounded* forcing field $F \in L^\infty(\mathbb{T}^d;\mathbb{R}^d)$, the perturbation operator $\mathcal{V}_{\mathrm{pert}} = F(q)\cdot\nabla_p$ satisfies Assumption IV.1.4 with structural constants (a_1, b_1, a_2, b_2) that depend only on $(\beta, d, \|F\|_{L^\infty})$ and *not* on $\lambda(Q_O)$. Since $\Theta = \frac{1}{2}$, $\lambda_R = 1$, and $M_R = 1$ by the standard Ornstein–Uhlenbeck spectral theory (all independent of $\lambda(Q_O)$), the asymptotic-independence hypothesis in Corollary V.2 is verified in the bounded-barrier regime where $C_{\mathrm{P}}(\nu)$ and $\|\nabla U\|_{L^\infty}$ stay uniformly bounded.

Applying Corollary V.2, the optimal friction choice is $\gamma_{\mathrm{opt}} \asymp \sqrt{\lambda(Q_O)}$, and the admissibility threshold reads

$$\eta \le \eta^*(\gamma_{\mathrm{opt}}) \asymp C(\beta, d, \|F\|_{L^\infty})\cdot\sqrt{\lambda(Q_O)}, \tag{132}$$

where $C(\beta, d, \|F\|_{L^\infty})$ is an explicit constant independent of $\lambda(Q_O)$. Under this condition, the accelerated lower bound

$$\max_{\gamma}\,\Lambda_{\mathrm{hypo}}(\gamma) \asymp \sqrt{\lambda(Q_O)} \tag{133}$$

holds.

Note that[18] (working on $\mathbb{R}^d$) imposes additional regularity and growth conditions on U to ensure the Poincaré inequality and the elliptic/hypoelliptic estimates needed for a global L^2 theorem. In the present setting on $\mathbb{T}^d$, these growth issues are absent, and the structural conditions are verified using only boundedness of F together with the finiteness of $C_{\mathrm{P}}(\nu)$ and $\|\nabla U\|_{L^\infty(\mathbb{T}^d)}$.

Comparison with a sharp kinetic estimate. Despite the regularization required for the strict L^2 proof, the rate function $\Lambda_{\mathrm{hypo}}(\gamma)$ captures the fundamental scaling laws of kinetic theory. We compare it with the benchmark of Theorem 3 in[16].

Mapping the notation of[16] to ours (friction $\xi \leftrightarrow \gamma$, forcing $\tau \leftrightarrow \eta$), that result states: for sufficiently small perturbations, the convergence to the steady state h_η satisfies

$$\left\| e^{t\mathcal{L}^\dagger_{\gamma,\eta}} f - h_\eta \right\|_{L^2(\mu_0)} \leq C e^{-\overline{\lambda}_\delta \min(\gamma,\gamma^{-1})t} \|f - h_\eta\|_{L^2(\mu_0)}. \tag{134}$$

The rate scales as $\min(\gamma, \gamma^{-1})$, encapsulating the underdamped–overdamped crossover with a peak at $\gamma \sim 1$.

The scaling law (134) is an instance of the general rate function $\Lambda_{\mathrm{hypo}}(\gamma)$ established in Theorem V.1:

1. **The Overdamped Regime ($\gamma \to \infty$):** The benchmark (134) gives a rate of $\mathcal{O}(\gamma^{-1})$. This matches the asymptotic analysis in Section V A: the hypocoercive rate degenerates to the macroscopic spectral-gap scaling

$$\Lambda_{\mathrm{hypo}}(\gamma) \asymp \gamma^{-1}\lambda(Q_O). \tag{135}$$

   In the high-friction limit, convergence is thus governed by the slow effective diffusion process.

2. **The Underdamped Regime ($\gamma \to 0$):** The benchmark result scales as $\mathcal{O}(\gamma)$. Correspondingly, the denominator of $\Lambda_{\mathrm{hypo}}(\gamma)$ is dominated by constant structural terms as $\gamma \to 0$, yielding the linear dependence $\Lambda_{\mathrm{hypo}}(\gamma) \asymp \gamma$. This reflects the regime in which insufficient dissipation limits the rate of entropy production.

3. **Optimal Lift ($\gamma \sim 1$):** The function $\min(\gamma, \gamma^{-1})$ attains its global maximum at $\gamma = 1$. Our Corollary V.2 predicts an optimal friction of the form $\gamma_{\mathrm{opt}} \asymp \sqrt{\lambda(Q_O)}$. In the standard Langevin setting on the torus with unit mass and temperature considered in[16], the macroscopic operator on the slow variables has a spectral gap bounded away from 0 (and from ∞) by constants depending only on the model parameters; in particular, $\lambda(Q_O)$ is of order one. Therefore γ_{opt} is also of order one, consistent with the benchmark rate peaking at $\gamma \sim 1$.

   Note that the kinetic acceleration interpretation of $\gamma_{\mathrm{opt}} \asymp \sqrt{\lambda(Q_O)}$ is most significant in families of problems where $\lambda(Q_O) \downarrow 0$ (slow macroscopic mixing); when $\lambda(Q_O) = \mathcal{O}(1)$, the optimal lift still occurs at $\gamma = \mathcal{O}(1)$ but the improvement is at the level of constants.

The scaling $\min(\gamma, \gamma^{-1})$ can also be derived from specific commutator relations[16] and is fully captured by $\Lambda_{\mathrm{hypo}}(\gamma)$. Together, these observations confirm that the operator-modified Lyapunov framework is sharp: it recovers the optimal scaling for classical kinetic models while extending hypocoercive acceleration to general non-equilibrium quantum systems.

Numerical verification. We validate the lower- and upper-bound corollaries by reproducible spectral computations based on the Galerkin construction of[16]. We consider $d = 1$ with rotor potential $U(q) = U_0(1 - \cos q)$, inverse temperature $\beta = 100$, and mass $m = 1$. The full generator is discretized in the Fourier–Hermite basis $\psi_{nk}(q, p) = G_k(q)H_n(p)$, where $G_k(q) \propto e^{ikq+\beta U(q)/2}$ and $\{H_n\}$ are Hermite modes orthonormal in the Gaussian momentum measure. For each parameter set, the gap $\lambda(\mathcal{L})$ is extracted as the smallest positive real

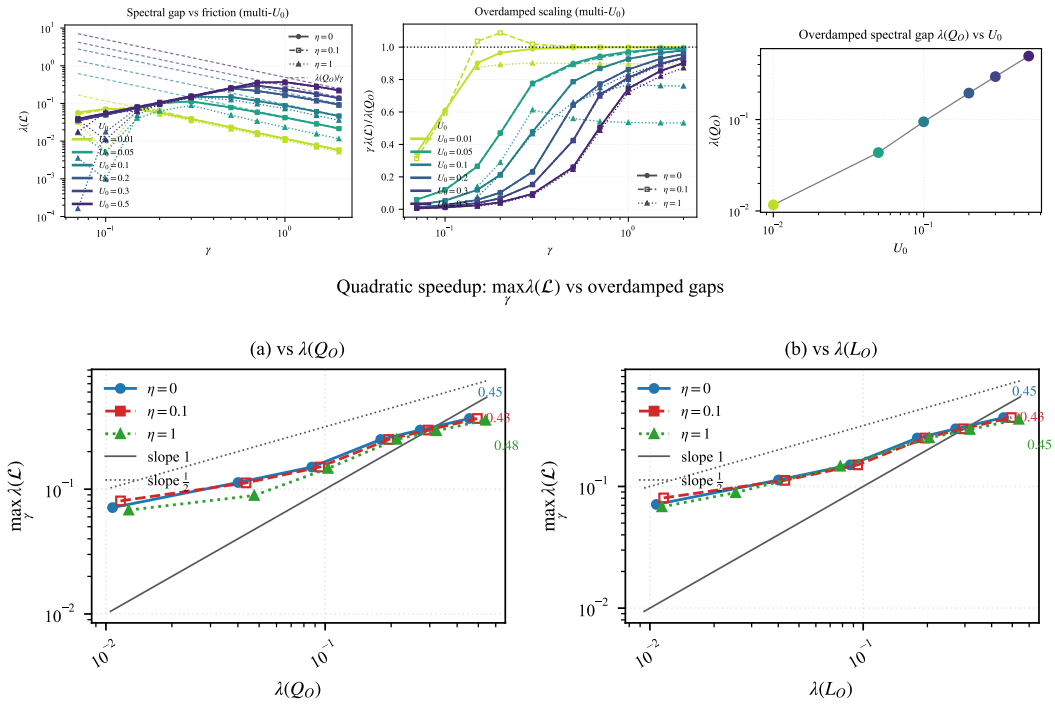

Figure 1. Numerical verification of Example VI A (Galerkin decay rate). *Left:* $\lambda(\mathcal{L})$ vs. γ (one curve per U_0) with the overdamped ceiling $\lambda(Q_O)/\gamma$ and $\gamma_{\rm opt} \sim \sqrt{\lambda(Q_O)}$. *Center:* scaled rate $\gamma\,\lambda(\mathcal{L})/\lambda(Q_O)$ approaching a constant for large γ. *Right:* $\lambda(Q_O)$ vs. U_0 (sub-Arrhenius growth at $\beta = 100$). *Bottom:* log-log speedup $\max_\gamma \lambda(\mathcal{L})$ vs. $\lambda(Q_O)$; slope ≈ 0.45 confirms the $\sqrt{\lambda(Q_O)}$ acceleration across nearly two decades. Parameters: $U_0 \in \{0.01, \ldots, 0.5\}$, $\beta = 100$, $m = 1$, $F = 0.05$, $K = 8$, $N = 16$, $K_{\rm ovd} = 32$, and $\eta \in \{0, 0.1, 1\}$.

part of $-\lambda$ over the eigenvalues of the resulting matrix representation of $\mathcal{L}$. For the slow effective operator, we compute $\lambda(Q_O)$ from the direct discretization $Q_O = -(1/\beta)D^\dagger D$ in Fourier space, with $D = \partial_q$ and $D^\dagger f = -\partial_q f + \beta U' f$.

We sweep

$$U_0 \in \{0.01, 0.05, 0.1, 0.2, 0.3, 0.5\},$$
$$\beta U_0 \in \{1, 5, 10, 20, 30, 50\},$$

with a constant nonequilibrium force $F = 0.05$. Truncations are $K = 8$ (Fourier) and $N = 16$ (Hermite) for $\mathcal{L}$, and $K_{\rm ovd} = 32$ for Q_O. Friction is scanned over $\gamma \in \{0.07, 0.1, 0.15, 0.2, 0.3, 0.5, 0.7, 1.0, 1.5, 2.0\}$ and forcing over $\eta \in \{0, 0.1, 1\}$. The computed macroscopic gaps $\lambda(Q_O) \in \{0.012, 0.044, 0.095, 0.195, 0.295, 0.495\}$ (one per U_0) grow roughly linearly with U_0 (at the chosen $\beta = 100$, the regime is sub-Arrhenius), spanning nearly two decades from the flat-torus lower bound $1/\beta = 0.01$ up to ≈ 0.5. The predicted optimal frictions $\gamma_{\rm opt} = \sqrt{\lambda(Q_O)} \in \{0.11, 0.21, 0.31, 0.44, 0.54, 0.70\}$ all fall within the sampled γ-grid.

The numerical data in Fig. 1 are fully consistent with the theoretical regime structure. For $\eta = 0$, one observes the underdamped branch $\lambda(\mathcal{L}) \approx \gamma/m$ at small friction and the overdamped ceiling $\lambda(\mathcal{L}) \approx \lambda(Q_O)/\gamma$ at large friction for every U_0. In post-processing, points flagged unstable (positive real-part spectrum) are excluded from fitted prefactors and slope estimates; reported exponents therefore correspond to the stable spectral branch. The speedup panel gives $\max_\gamma \lambda(\mathcal{L}) \propto \lambda(Q_O)^{0.45}$ for $\eta = 0$, in quantitative agreement with the predicted square-root law; the fitted exponents remain comparable for $\eta = 0.1$ (≈ 0.43) and $\eta = 1$ (≈ 0.48). At $\eta = 1$, the two smallest frictions ($\gamma \in \{0.07, 0.10\}$) fall outside the

perturbative stability window (positive real-part eigenvalues) and are therefore excluded from slope fitting; all remaining points ($\gamma \geq 0.15$) follow the predicted trend. Across the extended range $\lambda(Q_O) \in [0.012, 0.495]$ (nearly two decades), the kinetic acceleration remains robust throughout the sampled barrier heights.

As a consistency check of the discretization, we consider the free case $U_0 = 0$ (and $\eta = 0$). In this situation the generator is explicitly diagonalizable in the Fourier–Hermite basis, and the eigenvalues are known in closed form:[22]

$$\lambda_{n,k} = -\frac{n\gamma}{m} - \frac{k^2}{\beta\gamma}, \qquad n \in \mathbb{N}_0 := \{0, 1, 2, \ldots\},\ k \in \mathbb{Z},\ (n,k) \neq (0,0), \tag{136}$$

so the spectral gap equals

$$\min\Big(\frac{\gamma}{m}, \frac{1}{\beta\gamma}\Big), \tag{137}$$

(the first term is attained at $n = 1, k = 0$ and the second at $n = 0, |k| = 1$) with the crossover at $\gamma = \sqrt{m/\beta}$. For our choice $m = 1$, $\beta = 100$, the crossover is at $\gamma = 0.1$, and our Galerkin discretization recovers the formula to relative error $\lesssim 1.2 \times 10^{-12}$ across the tested friction values.

B. Example 2: A Finite-Dimensional Boundary-Driven Toy Model

We construct a finite-dimensional model in the spirit of boundary-driven and Zeno constructions, following Refs.[9,23,24]. The system satisfies Assumption IV.1 and has a generically non-self-adjoint effective slow generator $\mathcal{L}_O$, providing a tractable test case in which all operator-domain questions are vacuous yet non-normality of the effective generator persists.

Setup.
Hilbert space and reference state. Let the fast ancilla be a qubit $\mathfrak{h}_F \cong \mathbb{C}^2$ and the slow system be a qubit $\mathfrak{h}_S \cong \mathbb{C}^2$. We work on the observable space $\mathcal{H} = L^2(\sigma)$ associated with the product reference state

$$\sigma := \pi_F \otimes \pi_S, \qquad \pi_F := \tfrac{1}{2}\mathbf{1}_F, \qquad \pi_S := \tfrac{1}{2}\mathbf{1}_S. \tag{138}$$

Since σ is proportional to the identity, the KMS inner product $\langle X, Y\rangle_\sigma := \mathrm{tr}(\sigma^{1/2}X^\dagger\sigma^{1/2}Y)$ is proportional to the Hilbert–Schmidt inner product; we use this $L^2(\sigma)$ structure throughout. Because $\dim\mathcal{H} < \infty$ (here $\dim\mathcal{H} = 16$), all operators are bounded and $\mathrm{dom}(\mathcal{L}) = \mathrm{dom}(\mathcal{R}) = \mathrm{dom}(\mathcal{V}_{\mathrm{skew}}) = \mathrm{dom}(\mathcal{V}_{\mathrm{pert}}) = \mathcal{H}$; thus there are no core/closure subtleties in this example.

Slow/fast splitting. We take the slow subspace

$$\mathcal{H}_S := \mathbf{1}_F \otimes \mathcal{B}(\mathfrak{h}_S), \qquad P_S(X) := \mathbf{1}_F \otimes \mathrm{tr}_F(\pi_F X), \qquad P_F := \mathrm{id} - P_S. \tag{139}$$

Then P_S is the $L^2(\sigma)$-orthogonal projection (conditional expectation) onto $\mathcal{H}_S$.

Generator decomposition. We consider

$$\mathcal{L} = \mathcal{V}_{\mathrm{skew}} + \gamma\,\mathcal{R} + \eta\,\mathcal{V}_{\mathrm{pert}}. \tag{140}$$

1. **Fast dissipator ($\mathcal{R}$).** We fix an explicit primitive Lindbladian acting only on the ancilla,

$$\mathcal{R}_F X := \Gamma \sum_{j\in\{x,y,z\}} \mathcal{D}^\dagger[\sigma_j]X, \qquad \Gamma > 0, \tag{141}$$

and set $\mathcal{R} := \mathcal{R}_F \otimes \mathrm{id}_S$. Here σ_j are the Pauli matrices and, for any operator L, the Heisenberg dissipator is

$$\mathcal{D}^\dagger[L]X := L^\dagger X L - \tfrac{1}{2}\big(L^\dagger L\, X + X\, L^\dagger L\big). \tag{142}$$

Since each σ_j is unitary, $\mathcal{D}^\dagger[\sigma_j]X = \sigma_j X \sigma_j - X$. In particular, $\mathcal{R}_F$ (hence $\mathcal{R}$) is self-adjoint and non-positive on $L^2(\sigma)$, with $\ker(\mathcal{R}) = \mathcal{H}_S$. Moreover, for each Pauli matrix σ_j one has $\mathcal{R}_F(\sigma_j) = -4\Gamma\, \sigma_j$, so the spectral gap on the traceless (fast) subspace is $\lambda_R = 4\Gamma$.

2. **Hamiltonian transport ($\mathcal{V}_{\mathrm{skew}}$).** We choose the skew-adjoint transport generated by a Hamiltonian commutator,

$$\mathcal{V}_{\mathrm{skew}} X := -i[H_0, X], \qquad H_0 := g\, \sigma_z \otimes \sigma_z, \qquad g \in \mathbb{R}. \tag{143}$$

3. **Perturbation ($\mathcal{V}_{\mathrm{pert}}$).** To obtain an effective slow generator that is generically neither self-adjoint nor skew-adjoint, we choose a bounded centered perturbation

$$\mathcal{V}_{\mathrm{pert}} := \mathcal{L}_z \otimes \mathcal{G}_S, \qquad (\mathcal{L}_z Z_F) := \sigma_z Z_F, \tag{144}$$

where $\mathcal{G}_S$ is a fixed bounded generator on the slow qubit that is not self-adjoint in $L^2(\pi_S)$. For concreteness, take the (Heisenberg) amplitude-damping type map

$$\mathcal{G}_S(X) := \mathcal{D}^\dagger[\sigma_-]X, \qquad \sigma_- := \tfrac{1}{2}(\sigma_x - i\sigma_y), \qquad \sigma_+ := \sigma_-^\dagger = \tfrac{1}{2}(\sigma_x + i\sigma_y). \tag{145}$$

Since $\mathrm{tr}(\pi_F \sigma_z) = 0$, both couplings map $\mathcal{H}_S$ into $\mathcal{H}_F$. Indeed, for any $X_S \in \mathcal{B}(\mathfrak{h}_S)$,

$$\mathcal{V}_{\mathrm{skew}}(\mathbf{1}_F \otimes X_S) = -ig\, \sigma_z \otimes [\sigma_z, X_S], \qquad \mathcal{V}_{\mathrm{pert}}(\mathbf{1}_F \otimes X_S) = \sigma_z \otimes \mathcal{G}_S(X_S), \tag{146}$$

and hence $P_S \mathcal{V}_{\mathrm{skew}} P_S = 0$ and $P_S \mathcal{V}_{\mathrm{pert}} P_S = 0$.

Effective generator $\mathcal{L}_O$ (explicit formula). Let $X = \mathbf{1}_F \otimes X_S \in \mathcal{H}_S$.
Step 1 (first excursion to fast space).

$$(\mathcal{V}_{\mathrm{skew}} + \eta \mathcal{V}_{\mathrm{pert}})X = \sigma_z \otimes \Big(-ig\, [\sigma_z, X_S] + \eta\, \mathcal{G}_S(X_S)\Big) \in \mathcal{H}_F. \tag{147}$$

Step 2 (Poisson equation on the fast ancilla). In finite dimensions, the Moore–Penrose pseudoinverse can be written as

$$\mathcal{R}^+ := -\int_0^\infty \big(e^{s\mathcal{R}} - P_S\big)\, ds, \tag{148}$$

so that $\mathcal{R}\mathcal{R}^+ = \mathrm{id} - P_S = P_F$ and $P_S \mathcal{R}^+ = \mathcal{R}^+ P_S = 0$. In particular, since $\mathcal{R}_F(\sigma_z) = -\kappa\, \sigma_z$ with $\kappa := 4\Gamma$, we have $\mathcal{R}^+(\sigma_z \otimes Y) = \mathcal{R}_F^+(\sigma_z) \otimes Y = -(1/\kappa)\sigma_z \otimes Y$. Therefore

$$\mathcal{R}^+(\mathcal{V}_{\mathrm{skew}} + \eta \mathcal{V}_{\mathrm{pert}})X = -\frac{1}{\kappa}\sigma_z \otimes \Big(-ig\, [\sigma_z, X_S] + \eta\, \mathcal{G}_S(X_S)\Big). \tag{149}$$

Step 3 (return map and projection). Since $\mathcal{V}_{\mathrm{skew}}(\sigma_z \otimes Y) = -ig\, \mathbf{1}_F \otimes [\sigma_z, Y]$ and $\mathcal{V}_{\mathrm{pert}}(\sigma_z \otimes Y) = \mathbf{1}_F \otimes \mathcal{G}_S(Y)$, we compute

$$(\mathcal{V}_{\mathrm{skew}} + \eta \mathcal{V}_{\mathrm{pert}})\, \mathcal{R}^+ (\mathcal{V}_{\mathrm{skew}} + \eta \mathcal{V}_{\mathrm{pert}})X \tag{150}$$

$$= \frac{1}{\kappa}\mathbf{1}_F \otimes \Big(g^2\, [\sigma_z, [\sigma_z, X_S]] + i\eta g\, \big[\sigma_z, \mathcal{G}_S(X_S)\big] + i\eta g\, \mathcal{G}_S([\sigma_z, X_S]) - \eta^2\, \mathcal{G}_S^2(X_S)\Big). \tag{151}$$

Projecting with $P_S(\mathbf{1}_F \otimes Y) = \mathbf{1}_F \otimes Y$ yields

$$\mathcal{L}_O(\mathbf{1}_F \otimes X_S) = -\frac{1}{\kappa}\, \mathbf{1}_F \otimes \Big(g^2\, [\sigma_z, [\sigma_z, X_S]] \tag{152}$$

$$+ i\eta g\, \big[\sigma_z, \mathcal{G}_S(X_S)\big] \tag{153}$$

$$+ i\eta g\, \mathcal{G}_S([\sigma_z, X_S]) \tag{154}$$

$$- \eta^2\, \mathcal{G}_S^2(X_S)\Big). \tag{155}$$

The reference operator is the $\eta = 0$ part,

$$Q_O(\mathbf{1}_F \otimes X_S) = -\frac{g^2}{\kappa}\,\mathbf{1}_F \otimes \tag{156}$$

$$[\sigma_z, [\sigma_z, X_S]]. \tag{157}$$

Non-self-adjointness on the slow space. Since $\pi_S \propto \mathbf{1}_S$, the $L^2(\pi_S)$ inner product is proportional to the Hilbert–Schmidt inner product, and one checks that $(\mathcal{D}^\dagger[\sigma_-])^\dagger = \mathcal{D}^\dagger[\sigma_+] \neq \mathcal{D}^\dagger[\sigma_-]$. Thus $\mathcal{G}_S$ is not self-adjoint on $L^2(\pi_S)$. Consequently, the $\mathcal{O}(\eta)$ mixed operator

$$\mathcal{B} := ig\,\big(\,\mathrm{ad}_{\sigma_z} \circ \mathcal{G}_S + \mathcal{G}_S \circ \mathrm{ad}_{\sigma_z}\,\big), \qquad \mathrm{ad}_{\sigma_z}(X) := [\sigma_z, X], \tag{158}$$

is generically not self-adjoint either (unless additional commutation/self-adjointness relations hold), so $\mathcal{L}_O$ is generically non-self-adjoint for $\eta \neq 0$.

Verification of Assumption IV.1. All operators are bounded on the finite-dimensional space $\mathcal{H}$.

1. Assumption IV.1.1 holds for $\mathcal{R}$ by construction, with $\lambda_R = 4\Gamma$.

2. Assumption IV.1.2 holds since both $\mathcal{V}_{\mathrm{skew}}$ and $\mathcal{V}_{\mathrm{pert}}$ map $\mathcal{H}_S$ into $\mathcal{H}_F$.

3. Assumption IV.1.3 holds with $M_R := \|\mathcal{R}\|$. Moreover, in this toy model the flux regularity bound (24) can be checked explicitly. For $\psi = \mathbf{1}_F \otimes X_S \in \mathcal{H}_S$ one has $g_\psi = P_F \mathcal{V}_{\mathrm{skew}} P_S \psi = -ig\,\sigma_z \otimes [\sigma_z, X_S]$ and $(-\mathcal{R}^+)g_\psi = (1/\kappa)g_\psi$. Therefore

$$P_F \mathcal{V}_{\mathrm{skew}} P_F (-\mathcal{R}^+) g_\psi = 0, \tag{159}$$

$$(-Q_O)\psi = P_S \mathcal{V}_{\mathrm{skew}} P_F (-\mathcal{R}^+) g_\psi = \frac{g^2}{\kappa}\,\mathbf{1}_F \otimes [\sigma_z, [\sigma_z, X_S]]. \tag{160}$$

   Since the left-hand side is identically zero, (24) holds (e.g. with $\kappa_{\mathrm{ell}} := 1$).

4. For Assumption IV.1.4, decompose $\mathcal{V}_{\mathrm{pert}} = \mathcal{K}_{\mathrm{pert}} + \mathcal{S}_{\mathrm{pert}}$ by $\mathcal{K}_{\mathrm{pert}} := \frac{1}{2}(\mathcal{V}_{\mathrm{pert}} - \mathcal{V}_{\mathrm{pert}}^\dagger)$ and $\mathcal{S}_{\mathrm{pert}} := \frac{1}{2}(\mathcal{V}_{\mathrm{pert}} + \mathcal{V}_{\mathrm{pert}}^\dagger)$. Since $\mathcal{V}_{\mathrm{pert}}$ is bounded, the energy bounds (27)–(28) hold with $a_1 = a_2 = 0$ and $b_1 = \|\mathcal{K}_{\mathrm{pert}}\|$, $b_2 = \|\mathcal{S}_{\mathrm{pert}}\|$.

Numerical verification. We assemble the exact Liouville-space generator for the two-qubit model and compute its spectral gap on the mean-zero subspace. The scan uses $g \in \{0.3, 0.4, 0.5, 0.6, 0.8\}$, fixed bath damping $\Gamma = 0.8$ ($\kappa = 4\Gamma = 3.2$), $\gamma \in \{0.1, 0.2, 0.375, 0.5, 0.625, 0.75, 1.0, 2.0, 4.0, 8.0\}$, and $\eta \in \{0, 0.1, 0.15\}$. The additional points $\{0.375, 0.5, 0.625, 0.75\}$ are included to resolve the exact peak location $\gamma^* = g/\Gamma = 1.25g$.

The effective gap satisfies $\lambda(Q_O) = g^2/\Gamma = 1.25g^2$ exactly. At $\eta = 0$, the full generator has peak gap $2g$ at $\gamma^* = g/\Gamma$. This exact maximizer differs from $\sqrt{\lambda(Q_O)} = g/\sqrt{\Gamma}$ by a model-dependent constant factor; this is consistent with the theorem-level statement $\gamma_{\mathrm{opt}} \asymp \sqrt{\lambda(Q_O)}$, which is an order-scale prediction rather than an exact identity in each finite-dimensional realization.

Figure 2 shows the expected underdamped branch $\lambda(\mathcal{L}) = \kappa\gamma/2 = 1.6\gamma$, the overdamped ceiling $\lambda(Q_O)/\gamma = 1.25g^2/\gamma$, and the crossover peak at γ^* with machine-precision agreement. As in Example VI A, slope extraction is performed on stable points only. The speedup fit yields slope 0.500 for $\eta = 0$, and remains close for nonzero forcing (0.578 at $\eta = 0.1$, 0.605 at $\eta = 0.15$). For $\eta = 0.15$, only the smallest friction point ($\gamma = 0.1$) is unstable; all remaining points preserve the predicted scaling behavior.

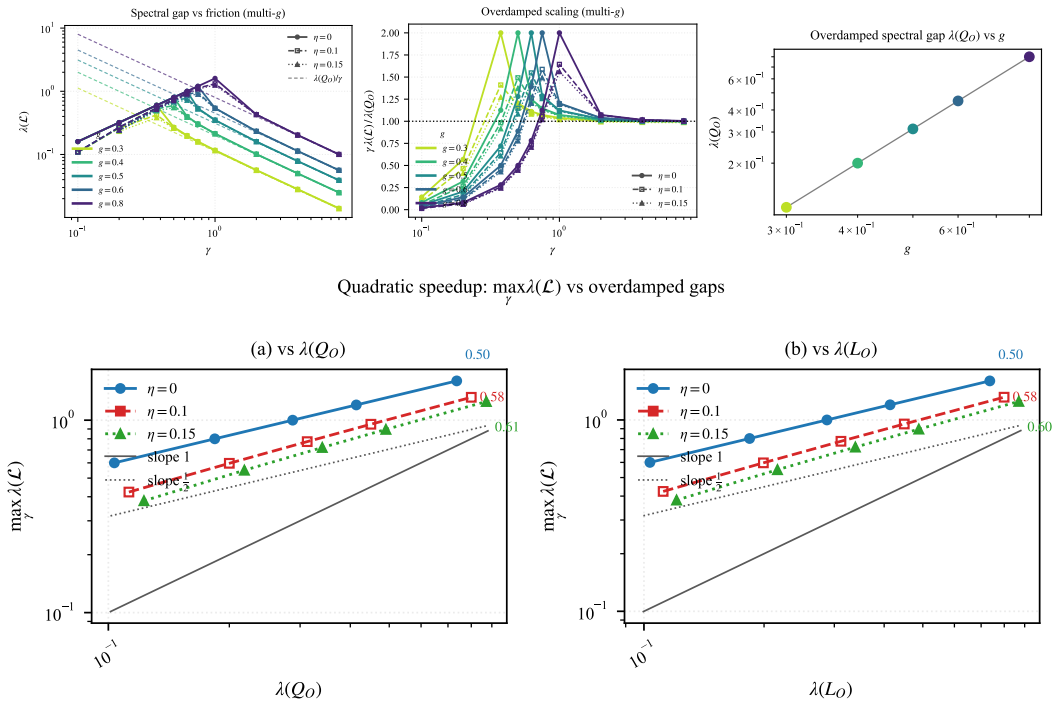

Figure 2. Numerical verification of Example VI B (spectral gap). *Left:* $\lambda(\mathcal{L})$ vs. γ (one curve per g) with the overdamped ceiling $\lambda(Q_O)/\gamma$ and the peak at $\gamma^* = g/\Gamma = 1.25g = \lambda(Q_O)/g$. *Center:* scaled gap $\gamma\,\lambda(\mathcal{L})/\lambda(Q_O)$ approaching a constant for large γ. *Right:* $\lambda(Q_O) = 1.25g^2$ vs. g (exact quadratic, confirmed numerically). *Bottom:* log-log speedup $\max_\gamma \lambda(\mathcal{L})$ vs. $\lambda(Q_O)$; slope $= 0.500$ (exact) for $\eta = 0$, rising to 0.578 and 0.605 for $\eta = 0.1$ and $\eta = 0.15$. Parameters: $\Gamma = 0.8$, $\kappa = 3.2$, $g \in \{0.3, 0.4, 0.5, 0.6, 0.8\}$, and $\eta \in \{0, 0.1, 0.15\}$.

VII. CONCLUSION

We have developed a quantitative hypocoercive framework for generators $\mathcal{L} = \gamma\mathcal{R} + \mathcal{V}_{\text{skew}} + \eta\mathcal{V}_{\text{pert}}$ on a slow–fast Hilbert decomposition. The analysis combines: (i) singular reduction to a slow operator (adiabatic elimination), (ii) a self-adjoint reference effective operator $Q_O = -P_S\mathcal{V}_{\text{skew}}\mathcal{R}^+\mathcal{V}_{\text{skew}}P_S$, and (iii) a resolvent-corrected Lyapunov functional that yields explicit decay inequalities.

The principal result is a finite-friction acceleration law. In the overdamped limit one recovers the diffusive ceiling of order γ^{-1}, while minimization of the explicit lower bound gives $\gamma_{\text{opt}} \asymp \sqrt{\lambda(Q_O)}$ and a decay rate of order $\sqrt{\lambda(Q_O)}$. The perturbative analysis is quantitative: permissible non-reversible forcing is controlled by an explicit threshold $\eta^*(\gamma)$, and at optimal friction the singular prefactors cancel so that $\eta^*(\gamma_{\text{opt}})$ shares the same square-root scaling as the accelerated rate.

The framework is validated on two model classes:

1. **Forced Langevin PDE:** the structural assumptions are verified on a natural core in $L^2(\mu_0)$, and numerics recover both the overdamped ceiling and the finite-friction speedup.[16]

2. **Boundary-driven finite-dimensional model:** all assumptions hold with explicit constants, and the effective slow operator is generically non-self-adjoint, extending the framework beyond reversible lifting settings.[9,19,23]

Limitations. The theory rests on structural inequalities (notably flux regularity and energy-norm perturbation bounds). In infinite-dimensional models these require model-

specific verification and may impose nontrivial regularity hypotheses; the predicted optimizer should therefore be interpreted as an intrinsic estimate within the validity regime of the reduced model, while external mechanisms may impose stricter operational constraints.[25]

Outlook. Natural extensions include multilevel lifting, sharper model-dependent estimates for Q_O, and finer control of non-normal effects beyond the symmetric reference reduction. Connections with linear response and dilation techniques may further improve explicit constants and robustness thresholds in concrete master-equation models.[26,27]

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Appendix A: Proofs for Preliminaries

This appendix contains the proofs of the spectral bounds and hypocoercivity characterizations stated in Section II. The arguments are carried out at Hilbert-space level with $\mathcal{H} = \mathbb{C}\mathbf{1} \oplus \mathcal{H}_0$. In the noncommutative realization, this corresponds to $\mathcal{H} = L^2(\sigma)$ and $\mathcal{H}_0 = \{X : \mathrm{tr}(\sigma X) = 0\}$.

1. Proof of Lemma II.3

We first clarify the geometric relationship between the fixed point space $\mathcal{F}(\mathcal{L}) = \mathbb{C}\mathbf{1}$ and the kernel of the symmetric part $\mathcal{S}$.

Proposition A.1 (Kernel Inclusion)**.** *For any dissipative generator with symmetric part* $\mathcal{S} := \frac{1}{2}(\mathcal{L} + \mathcal{L}^\dagger) \leq 0$*, one has*

$$\ker(\mathcal{L}) \subseteq \ker(\mathcal{S}). \tag{A1}$$

Proof. Let $X \in \ker(\mathcal{L})$. Then $\mathcal{L}(X) = 0$. Consequently,

$$\langle X, \mathcal{S}(X)\rangle = \Re\langle X, \mathcal{L}(X)\rangle = 0. \tag{A2}$$

Since $\mathcal{S}$ is a dissipative operator (negative semi-definite, $\mathcal{S} \leq 0$), the equality $\langle X, \mathcal{S}X\rangle = 0$ implies $\mathcal{S}X = 0$ (essentially due to the Cauchy-Schwarz inequality for the semi-inner product defined by $-\mathcal{S}$). Thus, $X \in \ker(\mathcal{S})$. □

Proof of Lemma II.3. We prove the coercivity characterization and show that, if $\ker(\mathcal{S}) \cap \mathcal{H}_0 \neq \{0\}$, then no exponential decay estimate can hold with prefactor $C = 1$.

Direction 1 (⇐): Assume $\ker(\mathcal{S}) \cap \mathcal{H}_0 = \{0\}$ and that $-\mathcal{S}|_{\mathcal{H}_0}$ has a positive spectral gap (in particular, this holds in finite dimensions). Then there exists $\lambda_S > 0$ such that

$$\langle X, \mathcal{S}X\rangle_\sigma \leq -\lambda_S \|X\|_\sigma^2, \qquad \forall X \in \mathcal{H}_0. \tag{A3}$$

For any $X_t = \mathcal{P}_t X_0 \in \mathcal{H}_0$, the evolution of the norm satisfies:

$$\frac{1}{2}\frac{d}{dt}\|X_t\|_\sigma^2 = \Re\langle X_t, \mathcal{L}X_t\rangle_\sigma = \langle X_t, \mathcal{S}X_t\rangle_\sigma \leq -\lambda_S \|X_t\|_\sigma^2. \tag{A4}$$

Integrating this differential inequality yields $\|X_t\|_\sigma \leq e^{-\lambda_S t}\|X_0\|_\sigma$. This is an exponential decay with prefactor $C = 1$ and rate λ_S. Thus, the system is coercive, not strictly hypocoercive.

Direction 2 (⇒): Assume the system is coercive ($C = 1$). By definition, there exists $\nu > 0$ such that $\|\mathcal{P}_t X\|_\sigma \leq e^{-\nu t}\|X\|_\sigma$ for all $X \in \mathcal{H}_0$. The Hille-Yosida theorem (or simple differentiation at $t = 0$) implies that the generator satisfies the dissipativity condition:

$$\Re\langle X, \mathcal{L}X\rangle_\sigma \leq -\nu \|X\|_\sigma^2, \quad \forall X \in \mathcal{H}_0. \tag{A5}$$

Substituting $\Re\langle X, \mathcal{L}X\rangle_\sigma = \langle X, \mathcal{S}X\rangle_\sigma$, we obtain:

$$\langle X, \mathcal{S}X\rangle_\sigma \leq -\nu \|X\|_\sigma^2. \tag{A6}$$

Now, suppose there exists a non-trivial vector $Y \in \ker(\mathcal{S}) \cap \mathcal{H}_0$. Then $\langle Y, \mathcal{S}Y\rangle_\sigma = 0$. However, the inequality above requires $\langle Y, \mathcal{S}Y\rangle_\sigma \le -\nu\|Y\|_\sigma^2 < 0$, a contradiction. Therefore, $\ker(\mathcal{S}) \cap \mathcal{H}_0 = \{0\}$, which implies $\ker(\mathcal{S}) = \mathbb{C}\mathbf{1} = \ker(\mathcal{L})$.

Finally, if $\ker(\mathcal{S}) \cap \mathcal{H}_0 \neq \{0\}$ then the above argument shows that coercivity ($C = 1$ with some positive rate) is impossible. In that case, any exponential decay estimate (when it holds) must necessarily come with some prefactor $C > 1$. □

Appendix B: Proof of Lemma IV.7

This appendix establishes the structural bounds of Lemma IV.7.

Proof. Recall the definition of the reference effective dynamics $Q_O = -P_S\mathcal{V}_{\text{skew}}\mathcal{R}^+\mathcal{V}_{\text{skew}}P_S$. By Assumption IV.1.1, the pseudoinverse is bounded on $\mathcal{H}_F$ with $\|\mathcal{R}^+\| \le \lambda_R^{-1}$, hence

$$\langle g, -\mathcal{R}^+ g\rangle \le \frac{1}{\lambda_R}\|g\|^2, \quad \forall g \in \mathcal{H}_F. \tag{B1}$$

The complementary (lower) control needed below is supplied by the flux conditioning assumption (22), which applies to fluxes of the form $g = g_\psi = P_F\mathcal{V}_{\text{skew}}P_S\psi$.

1. Proof of Properties 1 and 2 (Boundedness and Regularity)

Solver boundedness and forward regularity. Let $f \in \mathcal{H}$ and define $u := \mathcal{A}_\mu f \in \mathcal{H}_S$. The defining equation is $(\mu - Q_O)u = P_S\mathcal{V}_{\text{skew}}(-\mathcal{R}^+)P_F f$. Taking the inner product with u and using the skew-adjointness $\mathcal{V}_{\text{skew}}^\dagger = -\mathcal{V}_{\text{skew}}$:

$$\mu\|u\|^2 + \langle u, -Q_O u\rangle = \langle u, P_S\mathcal{V}_{\text{skew}}(-\mathcal{R}^+)P_F f\rangle = -\langle \mathcal{V}_{\text{skew}}u, (-\mathcal{R}^+)P_F f\rangle. \tag{B2}$$

Let $v := \mathcal{V}_{\text{skew}}u$. Since $P_S\mathcal{V}_{\text{skew}}P_S = 0$ (Centered Coupling), we strictly have $v \in \mathcal{H}_F$. Using the definition of Q_O and the pseudoinverse bound (B1):

$$\langle u, -Q_O u\rangle = \langle v, -\mathcal{R}^+ v\rangle = \|(-\mathcal{R}^+)^{1/2}v\|^2. \tag{B3}$$

Substituting this into the energy identity and applying Cauchy–Schwarz yields:

$$\mu\|u\|^2 + \|(-\mathcal{R}^+)^{1/2}v\|^2 \le \|(-\mathcal{R}^+)^{1/2}v\|\,\|(-\mathcal{R}^+)^{1/2}P_F f\|. \tag{B4}$$

Writing $a := \|(-\mathcal{R}^+)^{1/2}v\|$ and $b := \|(-\mathcal{R}^+)^{1/2}P_F f\|$, we have $\mu\|u\|^2 + a^2 \le ab$. Completing the square,

$$a^2 - ab = \left(a - \tfrac{1}{2}b\right)^2 - \tfrac{1}{4}b^2 \ge -\tfrac{1}{4}b^2, \tag{B5}$$

so $\mu\|u\|^2 \le \frac{1}{4}b^2$, hence

$$\|u\| \le \frac{1}{2}\,\mu^{-1/2}\,\|(-\mathcal{R}^+)^{1/2}P_F f\| \le \frac{1}{2\sqrt{\lambda_R}}\,\mu^{-1/2}\,\|f\|. \tag{B6}$$

Moreover, $a \le b$. In the bounded-operator setting one has $\text{dom}(Q_O) = \mathcal{H}_S$, so $u \in \text{dom}(Q_O)$. Then the vector $v = \mathcal{V}_{\text{skew}}u$ is a slow-induced flux of the form g_ψ with $\psi \in \text{dom}(Q_O)$; we may therefore apply the flux conditioning bound (22) to v to obtain

$$\|v\|^2 \le M_R\,\langle v, -\mathcal{R}^+ v\rangle = M_R a^2 \le M_R b^2 \le \frac{M_R}{\lambda_R}\,\|f\|^2, \tag{B7}$$

so $\|v\| \le \sqrt{M_R/\lambda_R}\,\|f\|$. This proves $\|\mathcal{A}_\mu\| \le k_{\mathcal{A}}\mu^{-1/2}$ and $\|\mathcal{V}_{\rm skew}\mathcal{A}_\mu\| \le k_{\rm reg}$, with

$$k_{\mathcal{A}} := \frac{1}{2\sqrt{\lambda_R}}, \qquad k_{\rm reg} := \sqrt{M_R/\lambda_R}. \tag{B8}$$

Dual regularity. We consider the operator $\mathcal{T} := \mathcal{A}_\mu \mathcal{V}_{\rm skew} P_F$. Its norm equals that of its adjoint $\mathcal{T}^\dagger$. We compute the adjoint of the solver:

$$\mathcal{A}_\mu^\dagger = \big((\mu - Q_O)^{-1} P_S \mathcal{V}_{\rm skew}(-\mathcal{R}^+) P_F\big)^\dagger = -P_F(-\mathcal{R}^+)\mathcal{V}_{\rm skew} P_S(\mu - Q_O)^{-1}. \tag{B9}$$

(Using $Q_O^\dagger = Q_O$ and $\mathcal{V}_{\rm skew}^\dagger = -\mathcal{V}_{\rm skew}$). Therefore:

$$\mathcal{T}^\dagger = P_F \mathcal{V}_{\rm skew}^\dagger \mathcal{A}_\mu^\dagger = P_F \mathcal{V}_{\rm skew} P_F(-\mathcal{R}^+)\mathcal{V}_{\rm skew} P_S(\mu - Q_O)^{-1}. \tag{B10}$$

Let $g \in \mathcal{H}_S$ and $\psi := (\mu - Q_O)^{-1} g$. Define the macroscopic flux

$$g_\psi := P_F \mathcal{V}_{\rm skew} P_S \psi \in \mathcal{H}_F. \tag{B11}$$

Then $\mathcal{T}^\dagger g = P_F \mathcal{V}_{\rm skew} P_F(-\mathcal{R}^+) g_\psi$. Invoking the **Flux Regularity** condition from Assumption IV.1.3:

$$\|P_F \mathcal{V}_{\rm skew} P_F(-\mathcal{R}^+) g_\psi\| \le \kappa_{\rm ell} \|(-Q_O)\psi\|. \tag{B12}$$

Substituting ψ back:

$$\|\mathcal{T}^\dagger g\| \le \kappa_{\rm ell} \|(-Q_O)(\mu - Q_O)^{-1} g\|. \tag{B13}$$

Since Q_O is self-adjoint and non-positive, the operator $(-Q_O)(\mu - Q_O)^{-1}$ is a spectral contraction ($\|\cdot\| \le 1$). Thus:

$$\|\mathcal{T}^\dagger g\| \le \kappa_{\rm ell} \|g\|. \tag{B14}$$

This proves $\|\mathcal{A}_\mu \mathcal{V}_{\rm skew} P_F\| \le \kappa_{\rm ell}$.

2. Proof of Property 3 (Perturbation Control)

We use the structural decomposition $\mathcal{V}_{\rm pert} = \mathcal{K}_{\rm pert} + \mathcal{S}_{\rm pert}$ from Assumption IV.1.4. We also recall the centered mapping property (40), which will be used later to sharpen the perturbation bookkeeping in the dissipation matrix. Recall the bound for the skew part (Assumption IV.1.4(a)):

$$\|\mathcal{K}_{\rm pert} f\| \le a_1 \|(-\mathcal{R})^{1/2} f\| + b_1 \|f\|, \qquad f \in {\rm dom}((-\mathcal{R})^{1/2}). \tag{B15}$$

For the symmetric part, we use the energy bound (Assumption IV.1.4(b)):

$$\|\mathcal{S}_{\rm pert} f\| \le a_2 \|(-\mathcal{R})^{1/2} f\| + b_2 \|f\|, \qquad f \in {\rm dom}((-\mathcal{R})^{1/2}). \tag{B16}$$

Reverse estimate ($\mathcal{V}_{\rm pert}^\dagger \mathcal{A}_\mu$). Let $f \in \mathcal{H}$ and $u := \mathcal{A}_\mu f$. The adjoint is $\mathcal{V}_{\rm pert}^\dagger = -\mathcal{K}_{\rm pert} + \mathcal{S}_{\rm pert}$. Applying the triangle inequality and the bounds above:

$$\|\mathcal{V}_{\rm pert}^\dagger u\| \le \|\mathcal{K}_{\rm pert} u\| + \|\mathcal{S}_{\rm pert} u\| \tag{B17}$$

$$\le b_1 \|u\| + b_2 \|u\| + (a_1 + a_2)\,\|(-\mathcal{R})^{1/2} u\|. \tag{B18}$$

Since $u \in \mathcal{H}_S = \ker(\mathcal{R})$ (by definition of $\mathcal{A}_\mu$), one has $(-\mathcal{R})^{1/2} u = 0$, hence

$$\|\mathcal{V}_{\rm pert}^\dagger u\| \le (b_1 + b_2)\,\|u\|. \tag{B19}$$

Substituting the Solver Bound ($\|u\| \le k_{\mathcal{A}}\mu^{-1/2}\|f\|$):

$$\|\mathcal{V}_{\text{pert}}^{\dagger}\mathcal{A}_{\mu}f\| \le (b_1+b_2)k_{\mathcal{A}}\mu^{-1/2}\,\|f\|. \tag{B20}$$

This establishes (37a) with $k_P(\mu) := (b_1+b_2)k_{\mathcal{A}}\mu^{-1/2}$.

Forward estimate ($\mathcal{A}_\mu\mathcal{V}_{\text{pert}}$). We aim to bound the operator norm of $\mathcal{W}_{\text{fwd}} := \mathcal{A}_\mu\mathcal{V}_{\text{pert}}$. By adjoints on Hilbert spaces, $\|\mathcal{W}_{\text{fwd}}\| = \|\mathcal{W}_{\text{fwd}}^{\dagger}\| = \|\mathcal{V}_{\text{pert}}^{\dagger}\mathcal{A}_{\mu}^{\dagger}\|$. Let $g \in \mathcal{H}_S$ be arbitrary with $\|g\| = 1$ and set $w := \mathcal{A}_{\mu}^{\dagger}g$. Using the bounds on $\mathcal{K}_{\text{pert}}$ and the energy bound on $\mathcal{S}_{\text{pert}}$,

$$\begin{aligned}\|\mathcal{V}_{\text{pert}}^{\dagger}w\| &\le \|\mathcal{K}_{\text{pert}}w\| + \|\mathcal{S}_{\text{pert}}w\| \\ &\le (b_1+b_2)\,\|w\| + (a_1+a_2)\,\|(-\mathcal{R})^{1/2}w\|.\end{aligned} \tag{B21}$$

We compute the norms on the right-hand side.

1. **Solver term ($\|w\|$):**

$$\|w\| = \|\mathcal{A}_{\mu}^{\dagger}g\| \le \|\mathcal{A}_{\mu}\|\,\|g\| \le k_{\mathcal{A}}\,\mu^{-1/2}. \tag{B22}$$

2. **Energy term ($\|(-\mathcal{R})^{1/2}w\|$):** Using $w = \mathcal{A}_{\mu}^{\dagger}g$ and the functional calculus for the non-positive self-adjoint operator Q_O,

$$\|(-\mathcal{R})^{1/2}w\| = \|(-\mathcal{R})^{1/2}\mathcal{A}_{\mu}^{\dagger}g\| = \|(-\mathcal{R})^{-1/2}P_F\mathcal{V}_{\text{skew}}P_S(\mu-Q_O)^{-1}g\|. \tag{B23}$$

Setting $\psi := (\mu - Q_O)^{-1}g \in \mathcal{H}_S$, we note the identity (for such slow-induced fluxes)

$$g_\psi := P_F\mathcal{V}_{\text{skew}}P_S\psi, \tag{B24}$$

$$\|(-\mathcal{R})^{-1/2}g_\psi\|^2 = \langle g_\psi, -\mathcal{R}^{+}g_\psi\rangle = -\langle\psi, Q_O\psi\rangle = \|(-Q_O)^{1/2}\psi\|^2, \tag{B25}$$

hence

$$\|(-\mathcal{R})^{1/2}w\| = \|(-Q_O)^{1/2}(\mu-Q_O)^{-1}g\| \le \frac{1}{2}\,\mu^{-1/2}\,\|g\|. \tag{B26}$$

(The last step follows from the scalar bound $\sup_{x\ge0}\frac{\sqrt{x}}{\mu+x} = \frac{1}{2\sqrt{\mu}}$.)

Substituting these estimates into (B21) yields

$$\|\mathcal{V}_{\text{pert}}^{\dagger}\mathcal{A}_{\mu}^{\dagger}g\| \le \Big((b_1+b_2)k_{\mathcal{A}} + \tfrac{a_1+a_2}{2}\Big)\,\mu^{-1/2}\,\|g\|. \tag{B27}$$

Since g is arbitrary, this establishes (37b) with $\tilde{k}_P(\mu) := \big((b_1+b_2)k_{\mathcal{A}} + \frac{a_1+a_2}{2}\big)\mu^{-1/2}$.

□

Appendix C: Proof of Theorem V.1

This appendix derives the dissipation matrix governing the hypocoercive convergence. The computation is based on the modified Lyapunov functional with resolvent shift $\mu > 0$ and mixing weight $\alpha > 0$:

$$\mathcal{E}_{\alpha,\mu}(f) := \frac{1}{2}\|f\|^2 + \alpha\Re\langle\mathcal{A}_\mu f, f\rangle, \tag{C1}$$

where $\mathcal{A}_\mu = (\mu - Q_O)^{-1}P_S\mathcal{V}_{\text{skew}}(-\mathcal{R}^{+})P_F$ is the resolvent corrector associated with the reference effective dynamics Q_O.

1. Preliminaries

We strictly utilize the structural estimates established in Lemma IV.7. We recall the constants derived therein, depending solely on the parameters from Assumption IV.1 $(M_R, \lambda_R, \kappa_{\text{ell}}, a_1, b_1, a_2, b_2)$. For convenience in the Lyapunov dissipation bounds below, we introduce the effective symmetric-perturbation coefficient

$$b_{2,\text{eff}} := b_2 + \frac{a_2}{2}(1 + \lambda_R), \tag{C2}$$

which absorbs both the slow-mode contribution $b_2 + \frac{a_2}{2}$ and the fast-mode λ_R-amplification $\frac{a_2\lambda_R}{2}$ that arises below when applying $\|(-\mathcal{R})^{1/2}f\|^2 \geq \lambda_R\|f_F\|^2$ to the symmetric perturbation energy.

- **Solver Bound:** $\|\mathcal{A}_\mu\| \leq k_{\mathcal{A}}\mu^{-1/2}$, with $k_{\mathcal{A}} := \frac{1}{2\sqrt{\lambda_R}}$.
- **Regularity:** $\|\mathcal{V}_{\text{skew}}\mathcal{A}_\mu\| \leq k_{\text{reg}}$ with $k_{\text{reg}} := \sqrt{M_R/\lambda_R}$, and $\|\mathcal{A}_\mu\mathcal{V}_{\text{skew}}P_F\| \leq \kappa_{\text{ell}}$.
- **Perturbation Control:**

$$\|\mathcal{V}_{\text{pert}}^\dagger\mathcal{A}_\mu\| \leq k_P(\mu) = (b_1 + b_2)k_{\mathcal{A}}\,\mu^{-1/2}, \tag{C3a}$$

$$\|\mathcal{A}_\mu\mathcal{V}_{\text{pert}}\| \leq \tilde{k}_P(\mu) = \Big((b_1 + b_2)k_{\mathcal{A}} + \tfrac{a_1+a_2}{2}\Big)\mu^{-1/2}. \tag{C3b}$$

We denote the spectral gap of the self-adjoint operator $-Q_O$ by $\lambda(Q_O)$.

2. Derivation of Differential Estimates

Standing regularity for the Lyapunov computation. In the bounded-operator setting, for any $f_0 \in \mathcal{H}$ the solution $f_t = e^{t\mathcal{L}}f_0$ is continuously differentiable and satisfies $\partial_t f_t = \mathcal{L}f_t$. In unbounded settings, the computations below are performed on a common invariant core $\mathcal{C} \subset \text{dom}(\mathcal{L}) \cap \text{dom}(\mathcal{A}_\mu)$ and extended to the energy domain by density.

Lemma C.1 (Energy identity for the fast dissipator)**.** *Assume $\mathcal{R}$ is self-adjoint and non-positive on $\mathcal{H}$. Then for all $f \in \text{dom}(\mathcal{R})$,*

$$\Re\langle\mathcal{R}f, f\rangle = \langle\mathcal{R}f, f\rangle = -\|(-\mathcal{R})^{1/2}f\|^2. \tag{C4}$$

In particular, if $f = f_S + f_F$ with $f_S \in \ker(\mathcal{R})$ and $f_F \in \mathcal{H}_F$, then

$$\Re\langle\mathcal{R}f, f\rangle = \Re\langle\mathcal{R}f_F, f_F\rangle. \tag{C5}$$

Proof. Since $\mathcal{R}$ is self-adjoint and non-positive, $-\mathcal{R} \geq 0$ admits a functional calculus square root $(-\mathcal{R})^{1/2}$ and $\langle\mathcal{R}f, f\rangle = -\langle(-\mathcal{R})f, f\rangle = -\|(-\mathcal{R})^{1/2}f\|^2$. Moreover $\mathcal{R}f_S = 0$ and $\mathcal{R}f = \mathcal{R}f_F$. □

Let f_t satisfy $\partial_t f_t = \mathcal{L}f_t$. By differentiating (42) along the flow and using that $\mathcal{A}_\mu$ is time-independent, we obtain

$$\frac{d}{dt}\,\mathcal{E}_{\alpha,\mu}(f_t) = \Re\langle\partial_t f_t, f_t\rangle + \alpha\,\Re\langle\mathcal{A}_\mu\partial_t f_t, f_t\rangle + \alpha\,\Re\langle\mathcal{A}_\mu f_t, \partial_t f_t\rangle \tag{C6}$$

$$= \Re\langle\mathcal{L}f_t, f_t\rangle + \alpha\,\Re\langle\mathcal{A}_\mu\mathcal{L}f_t, f_t\rangle + \alpha\,\Re\langle\mathcal{A}_\mu f_t, \mathcal{L}f_t\rangle. \tag{C7}$$

We now split the right-hand side into the "plain" dissipation coming from the generator itself and the hypocoercive correction generated by the cross term:

$$\frac{d}{dt}\,\mathcal{E}_{\alpha,\mu}(f_t) = \mathcal{D}_I(f_t) + \mathcal{D}_{\text{mod}}(f_t), \tag{C8}$$

where we define

$$\mathcal{D}_I(f) := \Re\langle \mathcal{L}f, f\rangle, \tag{C9}$$
$$\mathcal{D}_{\mathrm{mod}}(f) := \alpha\,\Re\langle \mathcal{A}_\mu \mathcal{L}f, f\rangle + \alpha\,\Re\langle \mathcal{A}_\mu f, \mathcal{L}f\rangle. \tag{C10}$$

Expanding $\mathcal{L} = \gamma\mathcal{R} + \mathcal{V}_{\mathrm{skew}} + \eta\mathcal{V}_{\mathrm{pert}}$ in $\mathcal{D}_I$ then yields the decomposition displayed below.

Natural dissipation ($\mathcal{D}_I$).

$$\mathcal{D}_I = \gamma\Re\langle \mathcal{R}f, f\rangle + \Re\langle \mathcal{V}_{\mathrm{skew}} f, f\rangle + \eta\Re\langle \mathcal{V}_{\mathrm{pert}} f, f\rangle. \tag{C11}$$

Using Assumption IV.1.1 ($\mathcal{R}$ coercivity) and Assumption IV.1.4 (Perturbation decomposition $\mathcal{V}_{\mathrm{pert}} = \mathcal{K}_{\mathrm{pert}} + \mathcal{S}_{\mathrm{pert}}$):

- $\gamma\Re\langle \mathcal{R}f, f\rangle \le -\gamma\lambda_R \|f_F\|^2$.
- $\Re\langle \mathcal{V}_{\mathrm{skew}} f, f\rangle = 0$ and $\Re\langle \mathcal{K}_{\mathrm{pert}} f, f\rangle = 0$ (Skew-adjointness).
- For the symmetric part, using (28) and Cauchy–Schwarz,

$$\eta\Re\langle \mathcal{S}_{\mathrm{pert}} f, f\rangle \le \eta\|\mathcal{S}_{\mathrm{pert}} f\|\,\|f\| \le \eta\Big(b_2\|f\|^2 + a_2\,\|(-\mathcal{R})^{1/2} f\|\,\|f\|\Big) \tag{C12}$$
$$\le \eta\Big(\big(b_2 + \tfrac{a_2}{2}\big)\|f\|^2 + \tfrac{a_2}{2}\,\|(-\mathcal{R})^{1/2} f\|^2\Big). \tag{C13}$$

Combining with the dissipator contribution, note that by Lemma C.1, $\Re\langle \mathcal{R}f, f\rangle = \Re\langle \mathcal{R}f_F, f_F\rangle = -\|(-\mathcal{R})^{1/2} f\|^2$. Hence

$$\gamma\Re\langle \mathcal{R}f, f\rangle + \eta\Re\langle \mathcal{S}_{\mathrm{pert}} f, f\rangle \le -\big(\gamma - \tfrac{\eta a_2}{2}\big)\,\|(-\mathcal{R})^{1/2} f\|^2 + \eta\big(b_2 + \tfrac{a_2}{2}\big)\,\|f\|^2 \tag{C14}$$
$$\le -\big(\gamma\lambda_R - \eta b_{2,\mathrm{eff}}\big)\,\|f_F\|^2 + \eta\big(b_2 + \tfrac{a_2}{2}\big)\,\|f_S\|^2, \tag{C15}$$

where we used $\|(-\mathcal{R})^{1/2} f\|^2 \ge \lambda_R\|f_F\|^2$, the decomposition $\|f\|^2 = \|f_S\|^2 + \|f_F\|^2$, and the fact that $\frac{a_2\lambda_R}{2} + (b_2 + \frac{a_2}{2}) = b_{2,\mathrm{eff}}$ by definition. The fast mode carries the full effective coefficient $b_{2,\mathrm{eff}}$, while the slow mode retains only the smaller $b_2 + \frac{a_2}{2}$ (since $f_S \in \ker(\mathcal{R})$ does not benefit from the λ_R-amplification).

Thus:

$$\mathcal{D}_I \le -\big(\gamma\lambda_R - \eta b_{2,\mathrm{eff}}\big)\,\|f_F\|^2 + \eta\big(b_2 + \tfrac{a_2}{2}\big)\,\|f_S\|^2. \tag{C16}$$

Hypocoercive correction ($\mathcal{D}_{\mathrm{mod}}$). Recall

$$\mathcal{D}_{\mathrm{mod}}(f) = \alpha\,\Re\langle \mathcal{A}_\mu \mathcal{L}f, f\rangle + \alpha\,\Re\langle \mathcal{A}_\mu f, \mathcal{L}f\rangle. \tag{C17}$$

Using $\mathcal{L} = \gamma\mathcal{R} + \mathcal{V}_{\mathrm{skew}} + \eta\mathcal{V}_{\mathrm{pert}}$ and linearity, we regroup $\mathcal{D}_{\mathrm{mod}}$ according to which part of the generator appears:

$$\mathcal{D}_{\mathrm{mod}}(f) = T_{\mathcal{R}}(f) + T_{\mathcal{V}}(f) + T_{\mathcal{P}}(f), \tag{C18}$$

where

$$T_{\mathcal{R}}(f) := \alpha\gamma\,\Re\langle \mathcal{A}_\mu \mathcal{R}f, f\rangle + \alpha\gamma\,\Re\langle \mathcal{A}_\mu f, \mathcal{R}f\rangle, \tag{C19}$$
$$T_{\mathcal{V}}(f) := \alpha\,\Re\langle \mathcal{A}_\mu \mathcal{V}_{\mathrm{skew}} f, f\rangle + \alpha\,\Re\langle \mathcal{A}_\mu f, \mathcal{V}_{\mathrm{skew}} f\rangle, \tag{C20}$$
$$T_{\mathcal{P}}(f) := \alpha\eta\,\Re\langle \mathcal{A}_\mu \mathcal{V}_{\mathrm{pert}} f, f\rangle + \alpha\eta\,\Re\langle \mathcal{A}_\mu f, \mathcal{V}_{\mathrm{pert}} f\rangle. \tag{C21}$$

a. Interaction with the fast dissipator. We start from the regrouped definition

$$T_{\mathcal{R}}(f) = \alpha\gamma\,\Re\langle \mathcal{A}_\mu \mathcal{R} f, f\rangle + \alpha\gamma\,\Re\langle \mathcal{A}_\mu f, \mathcal{R} f\rangle. \tag{C22}$$

Since $\mathcal{R}f = \mathcal{R}f_F$ and $\mathcal{A}_\mu f \in \mathcal{H}_S = \ker(\mathcal{R})$, the second term vanishes by self-adjointness of $\mathcal{R}$:

$$\Re\langle \mathcal{A}_\mu f, \mathcal{R} f\rangle = \Re\langle \mathcal{R}\mathcal{A}_\mu f, f\rangle = 0. \tag{C23}$$

Moreover $\mathcal{A}_\mu \mathcal{R} f_F \in \mathcal{H}_S$ is orthogonal to $f_F \in \mathcal{H}_F$, hence

$$T_{\mathcal{R}}(f) = \alpha\gamma\,\Re\langle \mathcal{A}_\mu \mathcal{R} f_F, f_S\rangle. \tag{C24}$$

Using $\mathcal{A}_\mu = (\mu - Q_O)^{-1} P_S \mathcal{V}_{\text{skew}}(-\mathcal{R}^+) P_F$ and the pseudoinverse identity $\mathcal{R}^+\mathcal{R} = P_F$ on $\text{dom}(\mathcal{R})$, we obtain

$$\mathcal{A}_\mu \mathcal{R} f_F = (\mu - Q_O)^{-1} P_S \mathcal{V}_{\text{skew}}(-\mathcal{R}^+)\mathcal{R} f_F = -(\mu - Q_O)^{-1} P_S \mathcal{V}_{\text{skew}} f_F. \tag{C25}$$

Therefore, by skew-adjointness of $\mathcal{V}_{\text{skew}}$,

$$|T_{\mathcal{R}}| = \alpha\gamma\left|\Re\langle (\mu - Q_O)^{-1} P_S \mathcal{V}_{\text{skew}} f_F, f_S\rangle\right| = \alpha\gamma\left|\Re\langle f_F, \mathcal{V}_{\text{skew}}(\mu - Q_O)^{-1} f_S\rangle\right| \tag{C26}$$

$$\leq \alpha\gamma\,\|f_F\|\,\|\mathcal{V}_{\text{skew}}(\mu - Q_O)^{-1} f_S\|. \tag{C27}$$

To bound $\|\mathcal{V}_{\text{skew}}(\mu - Q_O)^{-1} f_S\|$, set $\psi := (\mu - Q_O)^{-1} f_S \in \text{dom}(Q_O) \subset \mathcal{H}_S$ and denote its induced microscopic flux

$$g_\psi := P_F \mathcal{V}_{\text{skew}} P_S \psi \in \mathcal{H}_F. \tag{C28}$$

Since $\psi \in \mathcal{H}_S$, the centered-coupling assumption gives $P_S \mathcal{V}_{\text{skew}} \psi = 0$, hence $g_\psi = \mathcal{V}_{\text{skew}}\psi = \mathcal{V}_{\text{skew}}(\mu - Q_O)^{-1} f_S$. Applying the flux conditioning bound (22) yields

$$\|\mathcal{V}_{\text{skew}}(\mu - Q_O)^{-1} f_S\|^2 = \|g_\psi\|^2 \leq M_R\,\langle g_\psi, -\mathcal{R}^+ g_\psi\rangle. \tag{C29}$$

The "extra Q_O" comes from rewriting the right-hand side using the definition $Q_O := -P_S \mathcal{V}_{\text{skew}} \mathcal{R}^+ \mathcal{V}_{\text{skew}} P_S$: indeed, for $\psi \in \mathcal{H}_S$,

$$\langle g_\psi, -\mathcal{R}^+ g_\psi\rangle = \langle \mathcal{V}_{\text{skew}}\psi, (-\mathcal{R}^+)\mathcal{V}_{\text{skew}}\psi\rangle = \langle \psi, (-Q_O)\psi\rangle = \|(-Q_O)^{1/2}\psi\|^2. \tag{C30}$$

Therefore

$$\|\mathcal{V}_{\text{skew}}(\mu - Q_O)^{-1} f_S\| \leq \sqrt{M_R}\,\|(-Q_O)^{1/2}(\mu - Q_O)^{-1} f_S\|. \tag{C31}$$

Finally, by functional calculus for the self-adjoint operator $-Q_O \geq 0$,

$$\|(-Q_O)^{1/2}(\mu - Q_O)^{-1}\| = \sup_{x\geq 0} \frac{\sqrt{x}}{\mu + x} = \frac{1}{2\sqrt{\mu}}, \tag{C32}$$

which implies

$$\|\mathcal{V}_{\text{skew}}(\mu - Q_O)^{-1} f_S\| \leq \frac{\sqrt{M_R}}{2}\,\mu^{-1/2}\,\|f_S\|. \tag{C33}$$

Hence

$$|T_{\mathcal{R}}| \leq \alpha\gamma\,\frac{\sqrt{M_R}}{2}\,\mu^{-1/2}\,\|f_F\|\,\|f_S\|. \tag{C34}$$

b. Interaction with transport. Starting from the regrouped definition

$$T_{\mathcal{V}}(f) = \alpha\,\Re\langle \mathcal{A}_\mu \mathcal{V}_{\text{skew}} f, f\rangle + \alpha\,\Re\langle \mathcal{A}_\mu f, \mathcal{V}_{\text{skew}} f\rangle, \tag{C35}$$

Set $\mathcal{V} := \mathcal{V}_{\text{skew}}$ for this paragraph. we decompose $f = f_S + f_F$ and use that $\mathcal{A}_\mu$ maps into the slow space and annihilates the slow component: $\mathcal{A}_\mu = \mathcal{A}_\mu P_F$ (since P_F is on the right in (32)). Hence $\mathcal{A}_\mu f = \mathcal{A}_\mu f_F$. Expanding in the *first* bracket gives

$$\Re\langle \mathcal{A}_\mu \mathcal{V} f, f\rangle = \Re\langle \mathcal{A}_\mu \mathcal{V} f_S, f_S\rangle + \Re\langle \mathcal{A}_\mu \mathcal{V} f_F, f_S\rangle \tag{C36}$$

$$+ \Re\langle \mathcal{A}_\mu \mathcal{V} f_S, f_F\rangle + \Re\langle \mathcal{A}_\mu \mathcal{V} f_F, f_F\rangle. \tag{C37}$$

The last term vanishes because $\mathcal{A}_\mu \mathcal{V} f_F \in \mathcal{H}_S$ is orthogonal to $f_F \in \mathcal{H}_F$. Moreover, since $\mathcal{A}_\mu \mathcal{V} f_S \in \mathcal{H}_S$, we have $\Re\langle \mathcal{A}_\mu \mathcal{V} f_S, f_F\rangle = 0$. Therefore

$$\Re\langle \mathcal{A}_\mu \mathcal{V} f, f\rangle = \Re\langle \mathcal{A}_\mu \mathcal{V} f_S, f_S\rangle + \Re\langle \mathcal{A}_\mu \mathcal{V} f_F, f_S\rangle. \tag{C38}$$

In the *second* bracket we use $\mathcal{A}_\mu f = \mathcal{A}_\mu f_F$ and expand only $\mathcal{V} f$:

$$\Re\langle \mathcal{A}_\mu f, \mathcal{V} f\rangle = \Re\langle \mathcal{A}_\mu f_F, \mathcal{V} f_S\rangle + \Re\langle \mathcal{A}_\mu f_F, \mathcal{V} f_F\rangle. \tag{C39}$$

Putting the pieces together, we obtain the explicit decomposition

$$T_{\mathcal{V}} = \alpha\,\Re\langle \mathcal{A}_\mu \mathcal{V}_{\text{skew}} f_S, f_S\rangle + \alpha\,\Re\langle \mathcal{A}_\mu \mathcal{V}_{\text{skew}} f_F, f_S\rangle \tag{C40}$$

$$+ \alpha\,\Re\langle \mathcal{A}_\mu f_F, \mathcal{V}_{\text{skew}} f_S\rangle + \alpha\,\Re\langle \mathcal{A}_\mu f_F, \mathcal{V}_{\text{skew}} f_F\rangle. \tag{C41}$$

- **Main Coercivity:** Since $f_S \in \mathcal{H}_S$ and $P_S \mathcal{V}_{\text{skew}} P_S = 0$, we have $P_F \mathcal{V}_{\text{skew}} f_S = \mathcal{V}_{\text{skew}} f_S$. By the definition of $\mathcal{A}_\mu$ and Q_O this yields

$$\alpha\Re\langle \mathcal{A}_\mu \mathcal{V}_{\text{skew}} f_S, f_S\rangle = \alpha\langle (\mu - Q_O)^{-1} Q_O f_S, f_S\rangle = -\alpha\,\langle (\mu - Q_O)^{-1}(-Q_O) f_S, f_S\rangle \tag{C42}$$

$$\leq -\alpha\,\frac{\lambda(Q_O)}{\mu + \lambda(Q_O)}\,\|f_S\|^2. \tag{C43}$$

- **Cross Terms:** Using Regularity estimates:

$$\alpha\Re\langle \mathcal{A}_\mu \mathcal{V}_{\text{skew}} f_F, f_S\rangle + \alpha\Re\langle \mathcal{A}_\mu f_F, \mathcal{V}_{\text{skew}} f_S\rangle \tag{C44}$$

$$\leq \alpha(\|\mathcal{A}_\mu \mathcal{V}_{\text{skew}} P_F\| + \|\mathcal{V}_{\text{skew}} \mathcal{A}_\mu\|)\|f_F\|\|f_S\| \tag{C45}$$

$$\leq \alpha(\kappa_{\text{ell}} + k_{\text{reg}})\|f_F\|\|f_S\|. \tag{C46}$$

- **Fast–fast remainder.** Using $\mathcal{V}_{\text{skew}}^\dagger = -\mathcal{V}_{\text{skew}}$ and Cauchy–Schwarz,

$$\big|\alpha\Re\langle \mathcal{A}_\mu f_F, \mathcal{V}_{\text{skew}} f_F\rangle\big| = \alpha\big|\Re\langle -\mathcal{V}_{\text{skew}} \mathcal{A}_\mu f_F, f_F\rangle\big| \leq \alpha\,\|\mathcal{V}_{\text{skew}} \mathcal{A}_\mu\|\,\|f_F\|^2 \leq \alpha\,k_{\text{reg}}\,\|f_F\|^2. \tag{C47}$$

c. Interaction with perturbation.

$$T_{\mathcal{P}} = \alpha\eta\Re\langle \mathcal{V}_{\text{pert}}^\dagger \mathcal{A}_\mu f, f\rangle + \alpha\eta\Re\langle \mathcal{A}_\mu \mathcal{V}_{\text{pert}} f, f\rangle. \tag{C48}$$

We estimate each term using Lemma IV.7 and the centered mapping property (40).

- **Reverse term.** Since $\mathcal{A}_\mu f \in \mathcal{H}_S$ and $\mathcal{V}_{\text{pert}}^\dagger \mathcal{H}_S \subset \mathcal{H}_F$, we have $\langle \mathcal{V}_{\text{pert}}^\dagger \mathcal{A}_\mu f, f\rangle = \langle \mathcal{V}_{\text{pert}}^\dagger \mathcal{A}_\mu f, f_F\rangle$. Therefore,

$$\big|\alpha\eta\Re\langle \mathcal{V}_{\text{pert}}^\dagger \mathcal{A}_\mu f, f\rangle\big| \leq \alpha\eta\,\|\mathcal{V}_{\text{pert}}^\dagger \mathcal{A}_\mu f\|\,\|f_F\| \leq \alpha\eta\,k_P(\mu)\,\|f\|\,\|f_F\| \tag{C49}$$

$$\leq \alpha\eta\,k_P(\mu)\,\big(\|f_F\|^2 + \|f_F\|\,\|f_S\|\big). \tag{C50}$$

- **Forward term.** Since $\mathcal{A}_\mu \mathcal{H} \subset \mathcal{H}_S$, we have $\langle \mathcal{A}_\mu \mathcal{V}_{\text{pert}} f, f\rangle = \langle \mathcal{A}_\mu \mathcal{V}_{\text{pert}} f, f_S\rangle$. Therefore,

$$\big|\alpha\eta\Re\langle \mathcal{A}_\mu \mathcal{V}_{\text{pert}} f, f\rangle\big| \leq \alpha\eta\,\|\mathcal{A}_\mu \mathcal{V}_{\text{pert}} f\|\,\|f_S\| \leq \alpha\eta\,\tilde{k}_P(\mu)\,\|f\|\,\|f_S\| \tag{C51}$$

$$\leq \alpha\eta\,\tilde{k}_P(\mu)\,\big(\|f_S\|^2 + \|f_F\|\,\|f_S\|\big). \tag{C52}$$

3. Dissipation Matrix Construction

Summing all estimates and using Cauchy–Schwarz bounds of the form $|\langle Xf_F, f_S\rangle| \leq \|X\|\,\|f_F\|\,\|f_S\|$, we obtain a closed differential inequality of the form

$$\frac{d}{dt}\,\mathcal{E}_{\alpha,\mu}(f_t) \leq -\mathbf{v}(t)^T\mathbf{M}\,\mathbf{v}(t), \qquad \mathbf{v}(t) := \begin{pmatrix}\|P_F f_t\| \\ \|P_S f_t\|\end{pmatrix}. \tag{C53}$$

Here $\mathbf{M}$ is a real symmetric 2×2 matrix whose entries depend only on the structural constants and the chosen parameters (α,μ). The entries of $\mathbf{M}$ are explicitly decomposed by powers of μ below.

Fast-Fast Block (M_{11}): From Natural Dissipation, the fast–fast transport remainder, and Reverse Perturbation.

$$\begin{aligned} M_{11} &= \big(\gamma\lambda_R - \eta b_{2,\mathrm{eff}}\big) - \alpha k_{\mathrm{reg}} - \alpha\eta k_P(\mu) \\ &= \gamma\lambda_R - \eta b_{2,\mathrm{eff}} - \alpha k_{\mathrm{reg}} - \alpha\eta(b_1+b_2)k_{\mathcal{A}}\,\mu^{-1/2}. \end{aligned} \tag{C54}$$

Slow-Slow Block (M_{22}): From Main Coercivity, Natural Dissipation (gain), and both Perturbations.

$$\begin{aligned} M_{22} &= \alpha\frac{\lambda(Q_O)}{\mu+\lambda(Q_O)} - \eta\big(b_2 + \tfrac{a_2}{2}\big) - \alpha\eta\,\tilde{k}_P(\mu) \\ &= \alpha\frac{\lambda(Q_O)}{\mu+\lambda(Q_O)} - \eta\big(b_2 + \tfrac{a_2}{2}\big) - \alpha\eta\left[\Big((b_1+b_2)k_{\mathcal{A}} + \tfrac{a_1+a_2}{2}\Big)\mu^{-1/2}\right]. \end{aligned} \tag{C55}$$

Cross Block (M_{12}): From Mixing, Transport Cross terms, and Forward Perturbation.

$$\begin{aligned} M_{12} &= -\frac{1}{2}\alpha\left[\gamma\frac{\sqrt{M_R}}{2}\,\mu^{-1/2} + (k_{\mathrm{reg}}+\kappa_{\mathrm{ell}}) + \eta\big(k_P(\mu) + \tilde{k}_P(\mu)\big)\right] \\ &= -\frac{1}{2}\alpha\left[k_{\mathrm{reg}} + \kappa_{\mathrm{ell}} + \left(\gamma\frac{\sqrt{M_R}}{2} + \eta\Big(2(b_1+b_2)k_{\mathcal{A}} + \tfrac{a_1+a_2}{2}\Big)\right)\mu^{-1/2}\right]. \end{aligned} \tag{C56}$$

This explicit form allows for direct asymptotic analysis by choosing $\mu \sim \lambda(Q_O)$, confirming the stability for small η.

4. Optimization of the Decay Rate

We now derive the explicit convergence rate by simultaneously optimizing the resolvent shift μ and the mixing weight α. We consider the unperturbed regime ($\eta = 0$) to establish the dominant hypocoercive rate.

Let $\mathbf{M}_0(\alpha,\mu)$ denote the dissipation matrix for $\eta = 0$. Since $\mathbf{M}_0$ is a 2×2 symmetric positive definite matrix (verified below), its smallest eigenvalue $\lambda_{\min}$ satisfies the rigorous algebraic bound:

$$\lambda_{\min} \geq \frac{\det(\mathbf{M}_0)}{\mathrm{Tr}(\mathbf{M}_0)} = \frac{M_{11}^0 M_{22}^0 - (M_{12}^0)^2}{M_{11}^0 + M_{22}^0}. \tag{C57}$$

The entries of $\mathbf{M}_0$ are:

$$M_{11}^0 = \gamma\lambda_R - \alpha k_{\mathrm{reg}}, \tag{C58}$$

$$M_{22}^0 = \alpha\mathcal{Y}(\mu), \quad \text{where } \mathcal{Y}(\mu) := \frac{\lambda(Q_O)}{\mu+\lambda(Q_O)}, \tag{C59}$$

$$M_{12}^0 = -\frac{1}{2}\alpha\mathcal{Z}(\mu,\gamma), \quad \text{where } \mathcal{Z}(\mu,\gamma) := \gamma\frac{\sqrt{M_R}}{2}\,\mu^{-1/2} + (k_{\mathrm{reg}}+\kappa_{\mathrm{ell}}). \tag{C60}$$

Choice of the resolvent shift μ. To maximize the structural efficiency of the rate, we select μ to align with the macroscopic spectral gap. A simple way to see the relevant μ-dependence is to look at the large-friction leading balance in the unperturbed matrix: the cross term contributes $\mathcal{Z}(\mu,\gamma) \sim \gamma\Theta\,\mu^{-1/2}$, while the slow coercivity contributes $\mathcal{Y}(\mu) = \frac{\lambda(Q_O)}{\mu+\lambda(Q_O)}$. In the overdamped-dominated part of the optimized bound one therefore obtains (up to γ-independent structural factors)

$$\Lambda_{\text{hypo}}(\gamma) \;\sim\; \frac{\gamma\,\mathcal{Y}(\mu)^2}{\gamma^2\,\mu^{-1}} \;=\; \frac{1}{\gamma}\,\mu\left(\frac{\lambda(Q_O)}{\mu+\lambda(Q_O)}\right)^2 \;=\; \frac{\lambda(Q_O)^2}{\gamma}\,\frac{\mu}{(\mu+\lambda(Q_O))^2}. \tag{C61}$$

Thus the "effective diffusion strength" is governed by the scalar function $\mu(\mu+\lambda(Q_O))^{-2}$, whose maximizer is $\mu = \lambda(Q_O)$. Therefore we fix:

$$\mu_* := \lambda(Q_O). \tag{C62}$$

At this value, the structural coefficients simplify to:

$$\mathcal{Y}_* := \mathcal{Y}(\mu_*) = \frac{1}{2}, \tag{C63}$$

$$\mathcal{Z}_*(\gamma) := \mathcal{Z}(\mu_*,\gamma) = \gamma\Theta\lambda(Q_O)^{-1/2} + K_{\text{geom}}, \tag{C64}$$

where $\Theta := \frac{\sqrt{M_R}}{2}$ and $K_{\text{geom}} := k_{\text{reg}} + \kappa_{\text{ell}}$.

Optimization of the mixing weight α. We aim to maximize the determinant in the numerator of (C57), which is the dominant factor for the rate. The determinant as a function of α is:

$$D(\alpha) := M_{11}^0 M_{22}^0 - (M_{12}^0)^2 = \gamma\lambda_R\mathcal{Y}_*\alpha - \Big(\mathcal{Y}_* k_{\text{reg}} + \frac{1}{4}\mathcal{Z}_*(\gamma)^2\Big)\alpha^2. \tag{C65}$$

This is a concave parabola maximizing at:

$$\alpha_* := \frac{\gamma\lambda_R\mathcal{Y}_*}{2\left(\mathcal{Y}_* k_{\text{reg}} + \frac{1}{4}\mathcal{Z}_*(\gamma)^2\right)} = \frac{2\gamma\lambda_R\mathcal{Y}_*}{\mathcal{Z}_*(\gamma)^2 + 4\mathcal{Y}_* k_{\text{reg}}}. \tag{C66}$$

At this optimal weight, the maximum determinant is:

$$D(\alpha_*) = \frac{(\gamma\lambda_R\mathcal{Y}_*)^2}{\mathcal{Z}_*(\gamma)^2 + 4\mathcal{Y}_* k_{\text{reg}}}. \tag{C67}$$

General lower bound. We substitute α_* into the full lower bound expression (C57). First, we compute an *upper bound* on the trace at the optimal weight. The exact value is $\mathrm{Tr}(\mathbf{M}_0)|_{\alpha_*} = (\gamma\lambda_R - \alpha_* k_{\text{reg}}) + \alpha_*\mathcal{Y}_*$, which contains the strictly negative term $-\alpha_* k_{\text{reg}}$. Since the trace appears in the *denominator* of the lower bound $\lambda_{\min} \geq \det(\mathbf{M}_0)/\,\mathrm{Tr}(\mathbf{M}_0)$, bounding the trace from above yields a valid (conservative) lower bound:

$$\mathrm{Tr}(\mathbf{M}_0)|_{\alpha_*} = \big(\gamma\lambda_R - \alpha_* k_{\text{reg}}\big) + \alpha_*\mathcal{Y}_* \leq \gamma\lambda_R + \alpha_*\mathcal{Y}_* = \gamma\lambda_R\left(1 + \frac{2\mathcal{Y}_*^2}{\mathcal{Z}_*(\gamma)^2 + 4\mathcal{Y}_* k_{\text{reg}}}\right). \tag{C68}$$

The inequality $\leq$ holds because $-\alpha_* k_{\text{reg}} \leq 0$. Dropping this negative term inflates the denominator, yielding a weaker but *rigorous* lower bound on $\lambda_{\min}$ (and hence on Λ_{hypo}). The convergence rate Λ is bounded by the ratio $D(\alpha_*)/\,\mathrm{Tr}(\mathbf{M}_0)|_{\alpha_*}$:

$$\Lambda \geq \frac{\frac{(\gamma\lambda_R\mathcal{Y}_*)^2}{\mathcal{Z}_*(\gamma)^2+4\mathcal{Y}_* k_{\text{reg}}}}{\gamma\lambda_R\left(1+\frac{2\mathcal{Y}_*^2}{\mathcal{Z}_*(\gamma)^2+4\mathcal{Y}_* k_{\text{reg}}}\right)} = \frac{\gamma\lambda_R\mathcal{Y}_*^2}{\mathcal{Z}_*(\gamma)^2 + 4\mathcal{Y}_* k_{\text{reg}} + 2\mathcal{Y}_*^2}. \tag{C69}$$

Substituting the explicit forms of $\mathcal{Y}_*$ and $\mathcal{Z}_*(\gamma)$:

$$\Lambda_{\text{hypo}}(\gamma) := \frac{\frac{1}{4}\,\gamma\,\lambda_R}{\left(\gamma\Theta\lambda(Q_O)^{-1/2} + K_{\text{geom}}\right)^2 + 2k_{\text{reg}} + \frac{1}{2}}. \tag{C70}$$

Using $k_{\text{reg}} = \sqrt{M_R/\lambda_R}$, we may rewrite

$$\Theta = \frac{\sqrt{M_R}}{2}, \qquad K_{\text{geom}} = k_{\text{reg}} + \kappa_{\text{ell}} = \sqrt{M_R/\lambda_R} + \kappa_{\text{ell}}, \tag{C71}$$

so that

$$\Lambda_{\text{hypo}}(\gamma) = \frac{\frac{1}{4}\,\gamma\,\lambda_R}{\left[\frac{\sqrt{M_R}}{2}\,\lambda(Q_O)^{-1/2}\,\gamma + \left(\sqrt{M_R/\lambda_R} + \kappa_{\text{ell}}\right)\right]^2 + 2\sqrt{M_R/\lambda_R} + \frac{1}{2}}. \tag{C72}$$

This result provides a rigorous, explicit lower bound valid for all $\gamma > 0$ (provided η is small).

Asymptotic scaling analysis. Equation (C70) reveals the precise scaling behavior across regimes without relying on ad-hoc approximations.

- **Overdamped Limit ($\gamma \to \infty$):** The denominator is dominated by the $\mathcal{O}(\gamma^2)$ term.

$$\lim_{\gamma\to\infty} \gamma\Lambda_{\text{hypo}}(\gamma) = \frac{\lambda_R\,\mathcal{Y}_*^2\,\lambda(Q_O)}{\Theta^2} = \lambda(Q_O)\,\frac{\lambda_R}{M_R}. \tag{C73}$$

  Thus, $\Lambda_{\text{hypo}}(\gamma) = \lambda(Q_O)\,\frac{\lambda_R}{M_R}\,\gamma^{-1} + \mathcal{O}(\gamma^{-2})$. In particular, $\Lambda_{\text{hypo}}(\gamma) \asymp \lambda(Q_O)\,\gamma^{-1}$.

- **Underdamped/Intermediate Regime:** For small or order-one γ, the denominator is controlled by the γ-independent structural constants K_{geom}, $2k_{\text{reg}}$, and $\frac{1}{2}$. These terms originate from the slow–fast coupling through $\mathcal{V}_{\text{skew}}$ (captured by the solver bounds and regularity constants) and therefore prevent the estimate from degenerating even though $\mathcal{R}$ does not directly dissipate on $\mathcal{H}_S$. In other words, the transport mixes slow directions into dissipative fast directions, yielding a strictly positive hypocoercive lower bound in the small/intermediate-friction regime.

We now establish the stability condition by computing the operator norm of the perturbation matrix $\mathbf{E}(\gamma)$ as a function of γ, tracking the dependence on the macroscopic spectral gap $\mu_* = \lambda(Q_O)$.

Explicit norm of the perturbation matrix. Recalling that $\mathbf{M} = \mathbf{M}_0 - \eta\mathbf{E}$, where $\mathbf{M}_0$ is the unperturbed ($\eta = 0$) matrix with entries $M_{11}^0 = \gamma\lambda_R - \alpha k_{\text{reg}}$ and $M_{22}^0 = \alpha\mathcal{Y}(\mu)$, the perturbation matrix $\mathbf{E}$ has entries read off from the full $\mathbf{M}$ computation:

$$E_{11} = b_{2,\text{eff}} + \alpha_*\,k_P(\mu_*), \tag{C74}$$

$$E_{22} = \left(b_2 + \tfrac{a_2}{2}\right) + \alpha_*\,\tilde{k}_P(\mu_*), \tag{C75}$$

$$E_{12} = \frac{1}{2}\,\alpha_*\left(k_P(\mu_*) + \tilde{k}_P(\mu_*)\right). \tag{C76}$$

Here E_{11} contains the full effective coefficient $b_{2,\text{eff}} = b_2 + \frac{a_2}{2}(1 + \lambda_R)$, while E_{22} contains only the slow-mode coefficient $b_2 + \frac{a_2}{2}$; the extra $\frac{a_2\lambda_R}{2}$ absorbed into $b_{2,\text{eff}}$ arises in the fast block from the bound $\|(-\mathcal{R})^{1/2}f\|^2 \geq \lambda_R\|f_F\|^2$ applied to the symmetric perturbation in $\mathcal{D}_I$ (see (C16)). Since $\mathbf{E}$ is symmetric, we bound its spectral norm by the maximal absolute row sum (Gershgorin):

$$\|\mathbf{E}(\gamma)\| \leq \max\{E_{11} + |E_{12}|,\, E_{22} + |E_{12}|\}. \tag{C77}$$

Since $b_{2,\text{eff}} \geq b_2 + \frac{a_2}{2}$ (as $\lambda_R \geq 0$), we have $E_{11} \geq E_{22}$, so the maximum is attained by the first row. To obtain a uniform closed-form upper bound, we additionally replace k_P with $\tilde{k}_P$ (since $\tilde{k}_P \geq k_P$) in E_{11}:

$$\|\mathbf{E}(\gamma)\| \leq b_{2,\text{eff}} + \alpha_*(\gamma)\left[\tfrac{1}{2}\,\tilde{k}_P(\mu_*) + \tfrac{3}{2}\,\tilde{k}_P(\mu_*)\right] = b_{2,\text{eff}} + 2\alpha_*(\gamma)\,\tilde{k}_P(\mu_*). \tag{C78}$$

Recall from Lemma IV.7 that the bounds k_P and $\tilde{k}_P$ depend on $\mu_*^{-1/2}$. We define the *strictly constant* structural coefficients (independent of μ and γ):

$$C_{\text{sing}} := 2(b_1 + b_2)k_{\mathcal{A}} + \frac{3}{4}(a_1 + a_2). \tag{C79}$$

Indeed, $2\tilde{k}_P(\mu_*) = 2\big((b_1 + b_2)k_{\mathcal{A}} + \frac{a_1+a_2}{2}\big)\mu_*^{-1/2}$; using $\frac{1}{2}k_P + \frac{3}{2}\tilde{k}_P = (2(b_1 + b_2)k_{\mathcal{A}} + \frac{3(a_1+a_2)}{4})\mu^{-1/2} = C_{\text{sing}}\mu^{-1/2}$ (tighter than $2\tilde{k}_P$), the bound (C78) refines to

$$\|\mathbf{E}(\gamma)\| \leq b_{2,\text{eff}} + \alpha_*(\gamma)\, C_{\text{sing}}\, \mu_*^{-1/2}. \tag{C80}$$

Substituting the explicit optimal mixing weight $\alpha_*(\gamma) = \frac{2\gamma\lambda_R\mathcal{Y}_*}{\mathcal{Z}_*(\gamma)^2+4\mathcal{Y}_*k_{\text{reg}}}$ derived above, the bound becomes:

$$\|\mathbf{E}(\gamma)\| \leq b_{2,\text{eff}} + \frac{2\gamma\lambda_R\mathcal{Y}_*}{\mathcal{Z}_*(\gamma)^2 + 4\mathcal{Y}_*k_{\text{reg}}}\, C_{\text{sing}}\, \lambda(Q_O)^{-1/2}. \tag{C81}$$

This expression is now fully explicit, revealing the interplay between the mixing weight (vanishing at large γ) and the macroscopic singularity of the solver.

Friction-dependent threshold. The stability condition $\lambda_{\min}(\mathbf{M}) > 0$ requires that the perturbative spectral shift does not exceed the unperturbed gap. A sufficient condition is $\eta\|\mathbf{E}(\gamma)\| \leq \frac{1}{2}\Lambda_{\text{hypo}}(\gamma)$. This defines the explicit stability threshold function $\eta^*(\gamma)$:

$$\eta^*(\gamma) := \frac{\Lambda_{\text{hypo}}(\gamma)}{2\|\mathbf{E}(\gamma)\|}. \tag{C82}$$

Substituting the explicit form of $\Lambda_{\text{hypo}}(\gamma)$ from Eq. (C70) and the perturbation bound derived above:

$$\eta^*(\gamma) = \frac{\frac{\gamma\lambda_R\mathcal{Y}_*^2}{\mathcal{Z}_*(\gamma)^2+4\mathcal{Y}_*k_{\text{reg}}+2\mathcal{Y}_*^2}}{2b_{2,\text{eff}} + \frac{4\gamma\lambda_R\mathcal{Y}_*}{\mathcal{Z}_*(\gamma)^2+4\mathcal{Y}_*k_{\text{reg}}}\, C_{\text{sing}}\lambda(Q_O)^{-1/2}}. \tag{C83}$$

Multiplying numerator and denominator by $\mathcal{Z}_*(\gamma)^2 + 4\mathcal{Y}_*k_{\text{reg}}$ simplifies the expression to

$$\eta^*(\gamma) = \frac{\gamma\lambda_R\mathcal{Y}_*^2}{2b_{2,\text{eff}}\big(\mathcal{Z}_*(\gamma)^2 + 4\mathcal{Y}_*k_{\text{reg}}\big) + 4\gamma\lambda_R\mathcal{Y}_*\, C_{\text{sing}}\lambda(Q_O)^{-1/2}} \tag{C84}$$

$$\times\, \frac{\mathcal{Z}_*(\gamma)^2 + 4\mathcal{Y}_*k_{\text{reg}}}{\mathcal{Z}_*(\gamma)^2 + 4\mathcal{Y}_*k_{\text{reg}} + 2\mathcal{Y}_*^2}. \tag{C85}$$

Recalling $\mathcal{Y}_* = \frac{1}{2}$ and rewriting yields the more explicit form

$$\eta^*(\gamma) = \frac{\frac{1}{4}\,\gamma\lambda_R}{2b_{2,\text{eff}}\big(\mathcal{Z}_*(\gamma)^2 + 2k_{\text{reg}}\big) + 2\gamma\lambda_R\, C_{\text{sing}}\, \lambda(Q_O)^{-1/2}} \times \frac{\mathcal{Z}_*(\gamma)^2 + 2k_{\text{reg}}}{\mathcal{Z}_*(\gamma)^2 + 2k_{\text{reg}} + \frac{1}{2}}. \tag{C86}$$

Asymptotic analysis of stability. We analyze the scaling of $\eta^*(\gamma)$ in the limiting regimes. Note that $\mathcal{Z}_*(\gamma) \asymp \gamma$.

- **Overdamped Limit ($\gamma \to \infty$):** The numerator scales as $\mathcal{O}(\gamma)$. In the denominator, the term $2b_{2,\text{eff}}\mathcal{Z}_*(\gamma)^2$ scales as $\mathcal{O}(\gamma^2)$, while the skew-term scales as $\mathcal{O}(\gamma)$.

- If $b_{2,\text{eff}} > 0$ (symmetric perturbation present): The $\mathcal{O}(\gamma^2)$ term dominates the denominator. Thus, $\eta^*(\gamma) \asymp \gamma^{-1}$.
- If $b_{2,\text{eff}} = 0$ (purely skew perturbation, i.e. $b_2 = a_2 = 0$): The denominator is dominated by the skew term $\mathcal{O}(\gamma)$. Thus, $\eta^*(\gamma) \asymp \mathcal{O}(1)$.

- **Optimal Lift ($\gamma \sim \sqrt{\lambda(Q_O)}$):** At the optimal friction γ_{opt} (will be derived the optimization section later), we have $\mathcal{Z}_*(\gamma) \asymp \mathcal{O}(1)$. Consider the singular term in the denominator:

$$\gamma_{\text{opt}} \cdot C_{\text{sing}} \lambda(Q_O)^{-1/2} \asymp \sqrt{\lambda(Q_O)} \cdot \lambda(Q_O)^{-1/2} \asymp \mathcal{O}(1). \tag{C87}$$

  Crucially, the singularity cancels out exactly. This implies that at the optimal lift, the stability threshold is well-behaved:

$$\eta^*(\gamma_{\text{opt}}) \asymp \sqrt{\lambda(Q_O)}. \tag{C88}$$

This explicit derivation confirms that the stability estimates are robust and structurally consistent with the macroscopic limit.

From coercivity to global decay. Having established the positivity of the dissipation matrix, we convert the algebraic estimate into a global semigroup decay bound. This step involves explicit algebraic substitution to derive the final decay rate, ensuring that the singular scaling of the perturbation is correctly propagated.

Uniform norm equivalence. Recall the distortion parameter $\delta(\gamma)$ defined by

$$\delta(\gamma) := 2\alpha_*(\gamma)\|\mathcal{A}_{\mu_*}\|. \tag{C89}$$

Using $\|\mathcal{A}_{\mu_*}\| \le k_{\mathcal{A}}\, \mu_*^{-1/2}$ with $k_{\mathcal{A}} = (2\sqrt{\lambda_R})^{-1}$ and the explicit weight $\alpha_*(\gamma) = \frac{\gamma\lambda_R}{\mathcal{Z}_*(\gamma)^2 + 2k_{\text{reg}}}$, we obtain the crude but uniform estimate

$$\delta(\gamma) \le 2\,\frac{\gamma\lambda_R}{\mathcal{Z}_*(\gamma)^2}\, k_{\mathcal{A}}\, \mu_*^{-1/2} = \frac{\gamma\sqrt{\lambda_R}\,\mu_*^{-1/2}}{\left(\gamma\Theta\mu_*^{-1/2} + K_{\text{geom}}\right)^2}. \tag{C90}$$

Setting $x := \gamma\Theta\mu_*^{-1/2}$ gives

$$\delta(\gamma) \le \frac{\sqrt{\lambda_R}}{\Theta}\,\frac{x}{(x + K_{\text{geom}})^2} \le \frac{\sqrt{\lambda_R}}{4\,\Theta\,K_{\text{geom}}} =: \delta_{\max}. \tag{C91}$$

Since $K_{\text{geom}} \ge \sqrt{M_R/\lambda_R}$ and $\Theta = \sqrt{M_R}/2$, we obtain $\delta_{\max} \le \frac{\lambda_R}{2M_R}$. By Assumption IV.1.3, the flux conditioning constant M_R is monotone (any larger value also satisfies the assumption); we may therefore take $M_R \ge \lambda_R$ without loss of generality, which gives $\delta_{\max} \le \frac{1}{2}$. (Choosing $M_R > \lambda_R$ can only *increase* $\delta_{\max} = \lambda_R/(2M_R)$ relative to its natural value $\lambda_R/(2M_R^\natural)$ where $M_R^\natural$ is the smallest valid constant; when $M_R^\natural \ge \lambda_R$ already, no enlargement is needed. The assumption $M_R \ge \lambda_R$ is verified in both examples of Section VI.) This implies that the condition number of the norm equivalence is uniformly bounded for all γ:

$$C_\gamma := \sqrt{\frac{1+\delta(\gamma)}{1-\delta(\gamma)}} \le \sqrt{\frac{1+1/2}{1-1/2}} \tag{C92}$$

$$= \sqrt{3}. \tag{C93}$$

Thus, we have the uniform equivalence bounds:

$$\frac{1}{4}\|f\|^2 \le \frac{1}{2}(1-\delta(\gamma))\|f\|^2 \le \mathcal{E}_{\alpha_*,\mu_*}(f) \le \frac{1}{2}(1+\delta(\gamma))\|f\|^2 \le \frac{3}{4}\|f\|^2. \tag{C94}$$

Explicit calculation of the decay rate. Set

$$c(\gamma,\eta) := \Lambda_{\rm hypo}(\gamma) - \eta\,\|\mathbf{E}(\gamma)\|. \tag{C95}$$

The algebraic dissipation estimate reads

$$\frac{d}{dt}\,\mathcal{E}_{\alpha_*,\mu_*}(f_t) \le -c(\gamma,\eta)\,\|f_t\|^2. \tag{C96}$$

Next we use the norm equivalence from Proposition IV.9 in the convenient form

$$\frac{1}{2}(1-\delta(\gamma))\,\|f\|^2 \le \mathcal{E}_{\alpha_*,\mu_*}(f) \le \frac{1}{2}(1+\delta(\gamma))\,\|f\|^2, \tag{C97}$$

where $\delta(\gamma) := 2\alpha_*(\gamma)\,\|\mathcal{A}_{\mu_*}\| \in (0,1)$. The upper bound implies

$$\|f\|^2 \ge \frac{2}{1+\delta(\gamma)}\,\mathcal{E}_{\alpha_*,\mu_*}(f). \tag{C98}$$

Substituting this into the dissipation inequality gives a closed differential inequality for the Lyapunov functional:

$$\frac{d}{dt}\,\mathcal{E}_{\alpha_*,\mu_*}(f_t) \le -\frac{2c(\gamma,\eta)}{1+\delta(\gamma)}\,\mathcal{E}_{\alpha_*,\mu_*}(f_t). \tag{C99}$$

By Grönwall's inequality,

$$\mathcal{E}_{\alpha_*,\mu_*}(f_t) \le \exp\Big(-\frac{2c(\gamma,\eta)}{1+\delta(\gamma)}\,t\Big)\,\mathcal{E}_{\alpha_*,\mu_*}(f_0). \tag{C100}$$

Finally, converting back to the L^2 norm using the two-sided equivalence yields

$$\|f_t\|^2 \le \frac{2}{1-\delta(\gamma)}\,\mathcal{E}_{\alpha_*,\mu_*}(f_t) \le \frac{1+\delta(\gamma)}{1-\delta(\gamma)}\,e^{-\frac{2c(\gamma,\eta)}{1+\delta(\gamma)}t}\,\|f_0\|^2. \tag{C101}$$

Therefore

$$\|f_t\| \le \sqrt{\frac{1+\delta(\gamma)}{1-\delta(\gamma)}}\,e^{-\lambda_\eta t}\,\|f_0\|, \qquad \lambda_\eta := \frac{c(\gamma,\eta)}{1+\delta(\gamma)}. \tag{C102}$$

We now substitute the explicit expressions derived in the previous steps. For the unperturbed rate, using $\mathcal{Y}_* = \frac{1}{2}$:

$$\Lambda_{\rm hypo}(\gamma) = \frac{\frac{1}{4}\lambda_R\,\gamma}{\mathcal{Z}_*(\gamma)^2 + 2k_{\rm reg} + \frac{1}{2}}, \quad \text{with } \mathcal{Z}_*(\gamma) = \gamma\Theta\lambda(Q_O)^{-1/2} + K_{\rm geom}. \tag{C103}$$

For the perturbation norm, we use the sharp bound (C80) at $\mu_* = \lambda(Q_O)$ to handle the macroscopic singularity:

$$\|\mathbf{E}(\gamma)\| \le b_{2,\rm eff} + \underbrace{\frac{\gamma\lambda_R}{\mathcal{Z}_*(\gamma)^2 + 2k_{\rm reg}}}_{\alpha_*(\gamma)}\,C_{\rm sing}\,\lambda(Q_O)^{-1/2}. \tag{C104}$$

Substituting these into the expression for λ_η and using the uniform bound $1+\delta(\gamma) \le 3/2$:

$$\lambda_\eta \ge \frac{2}{3}\left[\frac{\frac{1}{4}\lambda_R\,\gamma}{\mathcal{Z}_*(\gamma)^2 + 2k_{\rm reg} + \frac{1}{2}} - \eta\left(b_{2,\rm eff} + \frac{\gamma\lambda_R}{\mathcal{Z}_*(\gamma)^2 + 2k_{\rm reg}}\,C_{\rm sing}\,\lambda(Q_O)^{-1/2}\right)\right]. \tag{C105}$$

This expression constitutes the rigorous lower bound. It explicitly demonstrates that the term driving the instability ($C_{\rm sing}\lambda(Q_O)^{-1/2}$) is multiplied by the mixing weight factor $\frac{\gamma}{\mathcal{Z}_*^2}$, which scales as $\sqrt{\lambda(Q_O)}$ at the optimal lift, ensuring the boundedness of the perturbation.

Finally, converting back to the L^2 norm via Grönwall's lemma yields the global estimate:

$$\|e^{t\mathcal{L}}f\|_{\mathcal{H}} \le \sqrt{3}\,e^{-\lambda_\eta t}\|f\|_{\mathcal{H}}. \tag{C106}$$

This completes the proof.

5. Optimization over Friction γ: The Kinetic Acceleration

We finally determine the optimal friction coefficient γ_{opt} that maximizes the hypocoercive convergence rate. This optimization reveals the kinetic (hypocoercive) acceleration phenomenon, where the rate is improved from the diffusive scaling $\mathcal{O}(\lambda(Q_O))$ to the kinetic scaling $\mathcal{O}(\sqrt{\lambda(Q_O)})$.

Recall the explicit lower bound derived in Eq. (C70):

$$\Lambda_{\text{hypo}}(\gamma) = \frac{C_{\text{num}}\gamma}{(C_1\gamma + C_0)^2 + C_{\text{tr}}}, \tag{C107}$$

where the structural constants (independent of γ) are defined as:

$$C_{\text{num}} := \frac{1}{4}\lambda_R, \tag{C108}$$

$$C_1 := \Theta\lambda(Q_O)^{-1/2} = \frac{\sqrt{M_R}}{2}\lambda(Q_O)^{-1/2}, \tag{C109}$$

$$C_0 := K_{\text{geom}} = \sqrt{M_R/\lambda_R} + \kappa_{\text{ell}}, \tag{C110}$$

$$C_{\text{tr}} := 2k_{\text{reg}} + \frac{1}{2} = 2\sqrt{M_R/\lambda_R} + \frac{1}{2}. \tag{C111}$$

Optimal friction coefficient. The function $f(\gamma) = \frac{\gamma}{(C_1\gamma+C_0)^2+C_{\text{tr}}}$ attains its global maximum on $(0,\infty)$ when the derivative vanishes. Computing the critical point:

$$\frac{d}{d\gamma}\log f(\gamma) = \frac{1}{\gamma} - \frac{2C_1(C_1\gamma + C_0)}{(C_1\gamma + C_0)^2 + C_{\text{tr}}} = 0. \tag{C112}$$

This implies the balance condition $(C_1\gamma + C_0)^2 + C_{\text{tr}} = 2C_1\gamma(C_1\gamma + C_0)$. Simplifying this quadratic equation yields $C_1^2\gamma^2 = C_0^2 + C_{\text{tr}}$. Thus, the optimal friction is explicitly:

$$\gamma_{\text{opt}} = \frac{\sqrt{C_0^2 + C_{\text{tr}}}}{C_1} = \frac{\sqrt{K_{\text{geom}}^2 + 2k_{\text{reg}} + \frac{1}{2}}}{\Theta}\sqrt{\lambda(Q_O)}. \tag{C113}$$

We define the structural prefactor $C_{\text{lift}} := \Theta^{-1}\sqrt{K_{\text{geom}}^2 + 2k_{\text{reg}} + \frac{1}{2}}$. The optimal lift scales as the square root of the macroscopic spectral gap:

$$\gamma_{\text{opt}} = C_{\text{lift}}\sqrt{\lambda(Q_O)}. \tag{C114}$$

We evaluate the maximal rate $\Lambda_{\max} := \Lambda_{\text{hypo}}(\gamma_{\text{opt}})$. Substituting the optimality condition $C_1\gamma_{\text{opt}} = \sqrt{C_0^2 + C_{\text{tr}}}$ into the denominator of (C107):

$$\text{Denom} = (\sqrt{C_0^2 + C_{\text{tr}}} + C_0)^2 + C_{\text{tr}} = 2\left(C_0^2 + C_{\text{tr}} + C_0\sqrt{C_0^2 + C_{\text{tr}}}\right). \tag{C115}$$

The numerator is $C_{\text{num}}\gamma_{\text{opt}} = C_{\text{num}}C_1^{-1}\sqrt{C_0^2 + C_{\text{tr}}}$. Thus:

$$\Lambda_{\max} = \frac{C_{\text{num}}}{2C_1(\sqrt{C_0^2 + C_{\text{tr}}} + C_0)}. \tag{C116}$$

Substituting the scaling $C_1 = \Theta\lambda(Q_O)^{-1/2}$, we isolate the dependency on $\lambda(Q_O)$:

$$\Lambda_{\max} = \left[\frac{C_{\text{num}}}{2\Theta(\sqrt{C_0^2 + C_{\text{tr}}} + C_0)}\right]\sqrt{\lambda(Q_O)}. \tag{C117}$$

This explicitly confirms the kinetic (hypocoercive) acceleration:

$$\Lambda_{\max} \asymp \sqrt{\lambda(Q_O)} \quad \text{vs.} \quad \Lambda_{\text{overdamped}} \asymp \lambda(Q_O). \tag{C118}$$

Stability threshold at optimal lift. The admissible perturbation range $\eta^*(\gamma_{\text{opt}})$ at the accelerated regime is determined as follows. From (C83), the threshold depends on $\|\mathbf{E}(\gamma)\|$. The dominant term involves the mixing weight $\alpha_*(\gamma)$. Evaluating at γ_{opt}:

$$\alpha_*(\gamma_{\text{opt}}) = \frac{2\gamma_{\text{opt}}\lambda_R\mathcal{Y}_*}{\mathcal{Z}_*(\gamma_{\text{opt}})^2 + 4\mathcal{Y}_* k_{\text{reg}}}. \tag{C119}$$

Using the scaling $\gamma_{\text{opt}} \asymp \sqrt{\lambda(Q_O)}$:

- Numerator: $\gamma_{\text{opt}}\lambda_R \asymp \sqrt{\lambda(Q_O)}$.
- Denominator: $\mathcal{Z}_*(\gamma_{\text{opt}}) = C_1\gamma_{\text{opt}} + C_0 = \sqrt{C_0^2 + C_{\text{tr}}} + C_0 = \text{const}$ (independent of $\lambda(Q_O)$).

Thus, $\alpha_*(\gamma_{\text{opt}}) \asymp \sqrt{\lambda(Q_O)}$. We now apply the perturbation bound (C78), which accounts for the macroscopic singularity:

$$\|\mathbf{E}(\gamma_{\text{opt}})\| \le b_{2,\text{eff}} + \alpha_*(\gamma_{\text{opt}})\, C_{\text{sing}}\, \lambda(Q_O)^{-1/2}. \tag{C120}$$

The key observation is the cancellation of the singularity by the mixing weight:

$$\alpha_*(\gamma_{\text{opt}}) \cdot \lambda(Q_O)^{-1/2} \asymp \sqrt{\lambda(Q_O)} \cdot \frac{1}{\sqrt{\lambda(Q_O)}} \asymp \mathcal{O}(1). \tag{C121}$$

Consequently, the perturbation matrix norm is uniformly bounded at the optimal lift:

$$\|\mathbf{E}(\gamma_{\text{opt}})\| \le b_{2,\text{eff}} + K_{\text{robust}}, \quad \text{where } K_{\text{robust}} \asymp \mathcal{O}(1). \tag{C122}$$

The stability threshold $\eta^* = \Lambda_{\max}/(2\|\mathbf{E}\|)$ therefore scales as:

$$\eta^*(\gamma_{\text{opt}}) \asymp \frac{\sqrt{\lambda(Q_O)}}{b_{2,\text{eff}} + K_{\text{robust}}} \asymp \sqrt{\lambda(Q_O)}. \tag{C123}$$

This proves that the accelerated convergence retains an $\mathcal{O}(\sqrt{\lambda(Q_O)})$ threshold even in the presence of singular macroscopic interactions.

Appendix D: Upper bound via singular value gap

This appendix proves Theorem V.4.

1. Exponential decay implies a singular value-gap bound

For a (possibly unbounded) closed operator T on $\mathcal{H}_0$, define its injectivity modulus

$$s(T) := \inf_{f \in \text{dom}(T)\cap\mathcal{H}_0\setminus\{0\}} \frac{\|Tf\|}{\|f\|}. \tag{D1}$$

Lemma D.1 (From exponential decay to a singular value-gap bound)**.** *Assume* $\ker(\mathcal{L}) \cap \mathcal{H}_0 = \{0\}$. *If there exist constants* $C \ge 1$ *and* $\nu > 0$ *such that*

$$\|e^{t\mathcal{L}} f\| \le C\, e^{-\nu t}\, \|f\|, \qquad \forall t \ge 0,\ \forall f \in \mathcal{H}_0, \tag{D2}$$

then 0 *belongs to the resolvent set of* $\mathcal{L}|_{\mathcal{H}_0}$ *and*

$$\nu \le C\, s(\mathcal{L}). \tag{D3}$$

Proof. For $f \in \mathcal{H}_0$, define

$$u := -\int_0^\infty e^{t\mathcal{L}} f \, dt. \tag{D4}$$

The assumed exponential bound $\|e^{t\mathcal{L}} f\| \le Ce^{-\nu t}\|f\|$ implies that the Bochner integral converges in $\mathcal{H}_0$ and

$$\|u\| \le \int_0^\infty \|e^{t\mathcal{L}} f\| \, dt \le \int_0^\infty Ce^{-\nu t} \, dt \, \|f\| = \frac{C}{\nu} \, \|f\|. \tag{D5}$$

Moreover, $u \in \mathrm{dom}(\mathcal{L})$ and differentiating under the integral gives

$$\mathcal{L}u = -\int_0^\infty \mathcal{L}e^{t\mathcal{L}} f \, dt = -\int_0^\infty \partial_t\big(e^{t\mathcal{L}} f\big) \, dt = f. \tag{D6}$$

Thus $(\mathcal{L}|_{\mathcal{H}_0})^{-1}$ exists and satisfies $\|(\mathcal{L}|_{\mathcal{H}_0})^{-1}\| \le C/\nu$, hence $s(\mathcal{L}) = \|(\mathcal{L}|_{\mathcal{H}_0})^{-1}\|^{-1} \ge \nu/C$, i.e. $\nu \le C\, s(\mathcal{L})$. □

Lemma D.2 (Relaxation time is bounded below by the singular value gap)**.** *Assume* $\ker(\mathcal{L}) \cap \mathcal{H}_0 = \{0\}$. *Assume there exist constants* $C \ge 1$ *and* $\nu > 0$ *such that*

$$\|e^{t\mathcal{L}} f\| \le C\, e^{-\nu t} \, \|f\|, \qquad \forall t \ge 0, \ \forall f \in \mathcal{H}_0. \tag{D7}$$

Define the L^2 *relaxation time*

$$t_{\mathrm{rel}}(\mathcal{L}) := \inf\Big\{ t \ge 0 \,;\ \|e^{t\mathcal{L}} f\| \le e^{-1} \, \|f\| \text{ for all } f \in \mathcal{H}_0 \Big\}. \tag{D8}$$

Then

$$t_{\mathrm{rel}}(\mathcal{L}) \ge \frac{1}{2C\, s(\mathcal{L})}. \tag{D9}$$

This L^2 *bound is a continuous-time analogue of discrete-time relaxation estimates based on singular values for non-reversible Markov chains; compare*[21].

Proof. By the assumed exponential bound $\|e^{t\mathcal{L}}\| \le Ce^{-\nu t}$, for any $r \ge 0$ we have $\|e^{r\mathcal{L}}\| \le C$. By the definition of $t_{\mathrm{rel}}(\mathcal{L})$, for all $t \ge 0$ we can write $t = n\, t_{\mathrm{rel}}(\mathcal{L}) + r$ with $n \in \mathbb{N}_0$ and $r \in [0, t_{\mathrm{rel}}(\mathcal{L}))$, and obtain from the semigroup property and $\|e^{t_{\mathrm{rel}}(\mathcal{L})\,\mathcal{L}}\| \le e^{-1}$:

$$\|e^{t\mathcal{L}}\| \le \|e^{r\mathcal{L}}\| \, \|e^{t_{\mathrm{rel}}(\mathcal{L})\,\mathcal{L}}\|^n \le C\, e^{-n}.$$

As in the proof of Lemma D.1, the inverse on $\mathcal{H}_0$ is given by $(\mathcal{L}|_{\mathcal{H}_0})^{-1} = -\int_0^\infty e^{t\mathcal{L}} \, dt$, hence

$$\|(\mathcal{L}|_{\mathcal{H}_0})^{-1}\| \le \int_0^\infty \|e^{t\mathcal{L}}\| \, dt \le \sum_{n=0}^\infty \int_0^{t_{\mathrm{rel}}(\mathcal{L})} C\, e^{-n} \, dr = \frac{C\, t_{\mathrm{rel}}(\mathcal{L})}{1 - e^{-1}} \le 2C\, t_{\mathrm{rel}}(\mathcal{L}).$$

Therefore $s(\mathcal{L}) = \|(\mathcal{L}|_{\mathcal{H}_0})^{-1}\|^{-1} \ge (2C\, t_{\mathrm{rel}}(\mathcal{L}))^{-1}$. □

2. Test function and completion of the proof

Proof of Theorem V.4. Let $\phi \in \mathrm{dom}(Q_O) \cap \mathcal{H}_S \cap \mathcal{H}_0 \cap \mathcal{K}^\perp$ be a normalized eigenfunction of Q_O with

$$Q_O \phi = -\lambda(Q_O)\, \phi, \qquad \|\phi\| = 1. \tag{D10}$$

Define the first-order corrector

$$\psi := \mathcal{R}^+\mathcal{V}_{\text{skew}}\phi \in \mathcal{H}_F, \tag{D11}$$

and the test function $f_\gamma := \phi - \gamma^{-1}\psi$. Since $\phi \in \mathcal{H}_S$ and $\psi \in \mathcal{H}_F$ we have $\langle \phi, \psi \rangle = 0$ and therefore

$$\|f_\gamma\|^2 = \|\phi\|^2 + \gamma^{-2}\|\psi\|^2 \geq 1. \tag{D12}$$

Using $\mathcal{R}\phi = 0$ and $\mathcal{R}\mathcal{R}^+ = P_F$ on $\mathcal{H}$, together with centered coupling $\mathcal{V}_{\text{skew}}\phi \in \mathcal{H}_F$, we obtain

$$\mathcal{R}\psi = \mathcal{R}\mathcal{R}^+\mathcal{V}_{\text{skew}}\phi = P_F\mathcal{V}_{\text{skew}}\phi = \mathcal{V}_{\text{skew}}\phi. \tag{D13}$$

Consequently,

$$\mathcal{L}f_\gamma = \big(\gamma\mathcal{R} + \mathcal{V}_{\text{skew}} + \eta\mathcal{V}_{\text{pert}}\big)\big(\phi - \gamma^{-1}\psi\big) \tag{D14}$$

$$= \underbrace{\mathcal{V}_{\text{skew}}\phi - \mathcal{R}\psi}_{=0} - \gamma^{-1}\mathcal{V}_{\text{skew}}\psi \;+\; \eta\mathcal{V}_{\text{pert}}f_\gamma. \tag{D15}$$

We decompose the remaining transport term into slow/fast components:

$$\mathcal{V}_{\text{skew}}\psi = P_S\mathcal{V}_{\text{skew}}\psi + P_F\mathcal{V}_{\text{skew}}\psi. \tag{D16}$$

For the slow part, using $Q_O := -P_S\mathcal{V}_{\text{skew}}\mathcal{R}^+\mathcal{V}_{\text{skew}}P_S$ and $P_S\phi = \phi$, we have

$$P_S\mathcal{V}_{\text{skew}}\psi = P_S\mathcal{V}_{\text{skew}}\mathcal{R}^+\mathcal{V}_{\text{skew}}\phi = -Q_O\phi = \lambda(Q_O)\,\phi. \tag{D17}$$

For the fast part, apply the flux regularity estimate (24) with the slow input ϕ. Writing $g_\phi := P_F\mathcal{V}_{\text{skew}}P_S\phi = \mathcal{V}_{\text{skew}}\phi$ and noting that $(-\mathcal{R}^+)g_\phi = -\psi$ on $\mathcal{H}_F$, we obtain

$$\|P_F\mathcal{V}_{\text{skew}}\psi\| = \|P_F\mathcal{V}_{\text{skew}}P_F\psi\| = \|P_F\mathcal{V}_{\text{skew}}P_F(-\mathcal{R}^+)g_\phi\| \leq \kappa_{\text{ell}}\,\|(-Q_O)\phi\| = \kappa_{\text{ell}}\,\lambda(Q_O). \tag{D18}$$

Combining the slow and fast contributions yields

$$\|\mathcal{V}_{\text{skew}}\psi\| \leq (1 + \kappa_{\text{ell}})\,\lambda(Q_O). \tag{D19}$$

Hence

$$\|\mathcal{L}f_\gamma\| \leq \frac{(1 + \kappa_{\text{ell}})\,\lambda(Q_O)}{\gamma} + \eta\,\|\mathcal{V}_{\text{pert}}f_\gamma\|. \tag{D20}$$

To bound the perturbation, use Assumption IV.1.4 to estimate

$$\|\mathcal{V}_{\text{pert}}f\| \leq \|\mathcal{K}_{\text{pert}}f\| + \|\mathcal{S}_{\text{pert}}f\| \tag{D21}$$

$$\leq (a_1 + a_2)\,\|(-\mathcal{R})^{1/2}f\| + (b_1 + b_2)\,\|f\|, \qquad f \in \text{dom}((-\mathcal{R})^{1/2}). \tag{D22}$$

Since $\phi \in \ker(\mathcal{R})$ and $f_\gamma = \phi - \gamma^{-1}\psi$, we have

$$\|(-\mathcal{R})^{1/2}f_\gamma\| = \gamma^{-1}\,\|(-\mathcal{R})^{1/2}\psi\|. \tag{D23}$$

Finally, using $\psi = \mathcal{R}^+\mathcal{V}_{\text{skew}}\phi$ and the definition of Q_O,

$$\|(-\mathcal{R})^{1/2}\psi\|^2 = \langle(-\mathcal{R})\psi, \psi\rangle = \langle\mathcal{V}_{\text{skew}}\phi, (-\mathcal{R}^+)\mathcal{V}_{\text{skew}}\phi\rangle = -\langle Q_O\phi, \phi\rangle = \lambda(Q_O). \tag{D24}$$

Therefore $\|(-\mathcal{R})^{1/2}f_\gamma\| = \sqrt{\lambda(Q_O)}/\gamma$, and

$$\|\mathcal{V}_{\text{pert}}f_\gamma\| \leq (b_1 + b_2)\,\|f_\gamma\| + (a_1 + a_2)\,\frac{\sqrt{\lambda(Q_O)}}{\gamma}. \tag{D25}$$

Dividing by $\|f_\gamma\| \geq 1$ gives

$$\frac{\|\mathcal{L}f_\gamma\|}{\|f_\gamma\|} \leq \frac{(1 + \kappa_{\text{ell}})\,\lambda(Q_O)}{\gamma} + \eta\,(b_1 + b_2) + \eta\,(a_1 + a_2)\,\frac{\sqrt{\lambda(Q_O)}}{\gamma}. \tag{D26}$$

By definition of $s(\mathcal{L})$, this implies $s(\mathcal{L}) \leq \|\mathcal{L}f_\gamma\|/\|f_\gamma\|$. Combining with Lemma D.1 yields the claimed bound on ν. □